Set 1: Mirror reflections: what a reflection must show

Local outcome LOGIC.SPATIAL.01: Decide whether an image is consistent with a reflection in a horizontal or vertical mirror line

Puzzle set · genre: reflection and symmetry · MP7, MP6

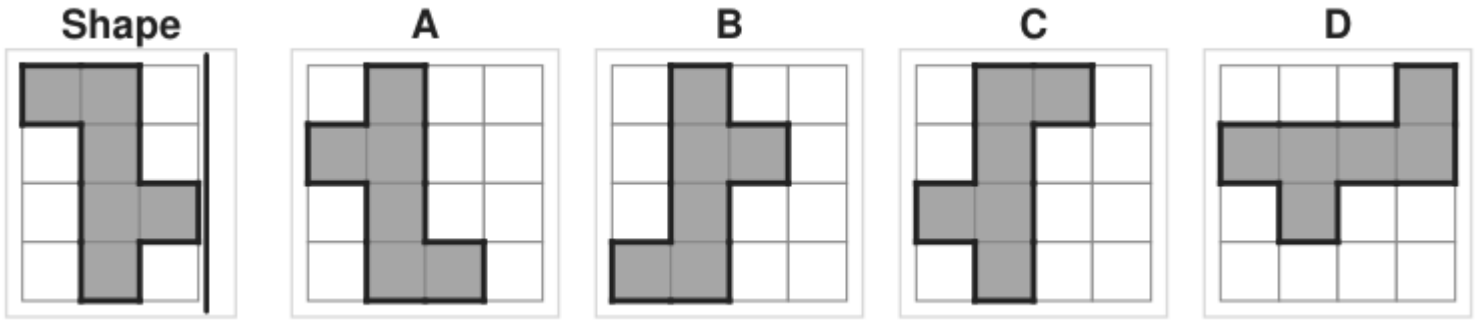
MAIN: Decide what a mirror reflection must show

BACKWARD 1: Strategy: reflect one cell at a time

FORWARD 1: ADVANCED APPLICATION: Combine reflections and read mirrored clocks and words

FORWARD 2: EXPERT CHALLENGE: Construct and count symmetric figures

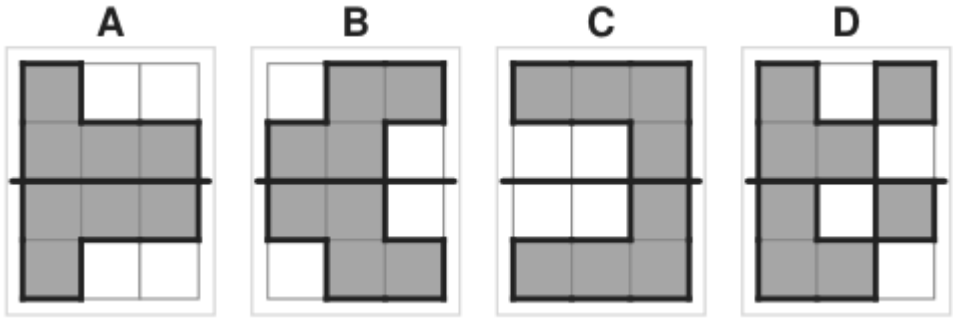
The shaded shape is reflected in the dashed mirror line on its right. Which figure shows its mirror image? (Only the shape matters, not where it sits in its box.)



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Each figure shows a shape above a dashed water line and, below the line, what is supposed to be its reflection in the water. In which figure can the lower shape NOT be the reflection of the upper one?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Words are written left to right in block capitals and seen in a left-right mirror (a vertical mirror line). In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same in the mirror?

A. TOOT
B. ZOOM
C. NOON
D. BIB

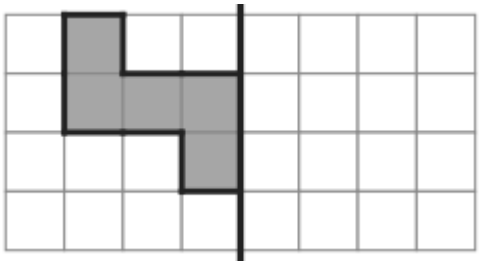
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A word is written in block capitals and seen reflected in calm water below it (a horizontal mirror line). In the block capitals used here, exactly these letters look the same in a top-bottom mirror: B C D E H I K O X. The letter order stays the same in this reflection. Which word looks exactly the same in the water?

A. BEAD
B. CODE
C. CORE
D. BOXES

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The grid has 4 rows and 8 columns. The dashed mirror line runs between columns 4 and 5. The shaded cells are in row 1 column 2, row 2 columns 2, 3 and 4, and row 3 column 4. Mark the cells of their mirror image. (Cells are named row,column; row 1 is the top row and column 1 the left column.)



Mark cells on the 4 × 8 grid (on screen: click each cell; cells are named row,column).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A grid has 10 columns. A vertical mirror line lies between columns 5 and 6. A shaded cell is in column 2. In which column is its mirror image?

Column: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A grid has 8 rows. A horizontal mirror line lies between rows 4 and 5. A shaded cell is in row 1. In which row is its mirror image?

Row: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A reflection in a vertical mirror line keeps every cell in its own row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A cell is 3 columns to the left of a vertical mirror line. Where is its mirror image?

A. 6 columns to the right of the line
B. on the mirror line itself
C. 3 columns to the left of the line
D. 3 columns to the right of the line

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A row of 6 cells has a mirror line between cells 3 and 4. Cell 1 is shaded. Mark the cell of its mirror image. (The grid is 1 row by 6 columns; cells are named row,column.)

Mark cells on the 1 × 6 grid (on screen: click each cell; cells are named row,column).

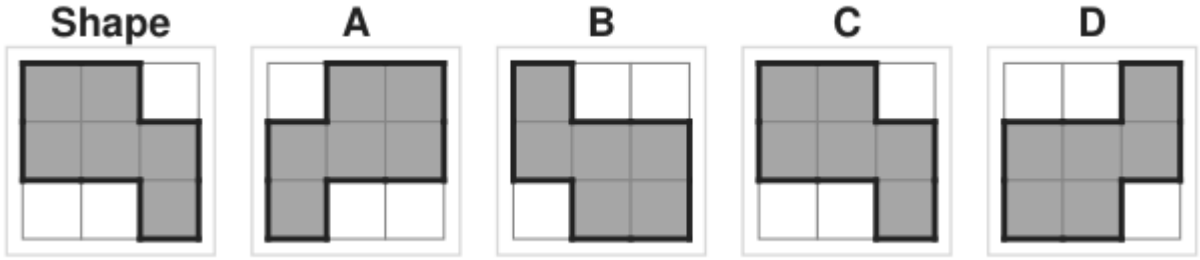
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A clock face with no numbers shows 2:20. You see it in a left-right mirror. What time does the mirrored clock appear to show?

A. 10:20
B. 2:40
C. 9:40
D. 3:40

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The shape is reflected in a vertical mirror line and then the result is reflected in a horizontal mirror line. Which figure shows the final result? (Only the shape matters, not where it sits.)



A. Figure A
B. Figure B
C. Figure C
D. Figure D

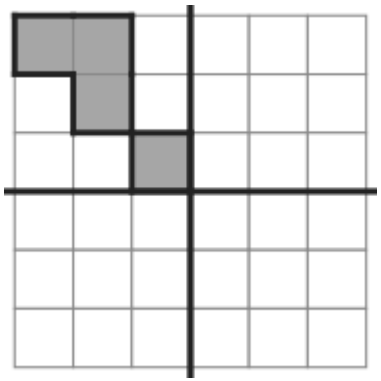
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A word is written DOWN the page, one block capital under another, and seen in a left-right mirror. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same?

A. MATH
B. MAZE
C. BATH
D. MASK

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A 6 by 6 grid has two dashed mirror lines: a vertical one between columns 3 and 4 and a horizontal one between rows 3 and 4. The figure must look the same in BOTH mirrors. Four cells are already shaded in the top-left quarter: (1,1), (1,2), (2,2) and (3,3). Mark every shaded cell of the completed figure, including the four given cells.



Mark cells on the 6 × 6 grid (on screen: click each cell; cells are named row,column).

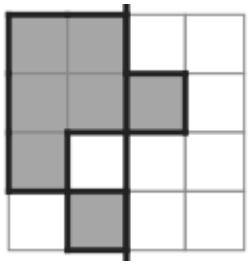
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A figure looks the same after a reflection in a vertical line through its center and after a reflection in a horizontal line through its center. Then it also looks the same after a half-turn about its center.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The figure is on a 4 by 4 grid with a dashed vertical line between columns 2 and 3. Shaded cells can be added but not removed. What is the least number of cells you must add so that the figure looks the same in a mirror along the dashed line?



Cells to add: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Strings of 3 letters are made from the letters A, H, M, O and T (letters may repeat). Each is written left to right in block capitals. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. How many of these strings look exactly the same in a left-right mirror?

Number of strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Two vertical mirror lines are 3 cells apart: line 1 on the left, line 2 on the right. A shape is reflected in line 1 and the image is then reflected in line 2. Where does the final image end up?

A. a mirror image of the shape, 6 cells to the right
B. back exactly where the shape started
C. the original shape moved 3 cells to the right, not flipped
D. the original shape moved 6 cells to the right, not flipped

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

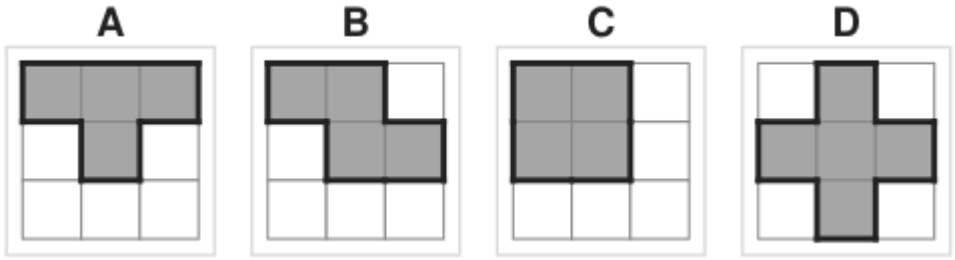
A row of 12 cells is numbered 1 to 12. A mirror line lies between cells 6 and 7. Give two cell numbers a and b, with a less than b, that are mirror images of each other.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Count only mirror lines that run vertically, horizontally or diagonally through the center of the shape (the middle of the smallest rectangle around it). Which figure has exactly ONE mirror line?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Set 2: Read exactly what is stated

Local outcome LOGIC.READING.01: Read negations, double negations, units and relationships exactly as stated

Puzzle set · genre: precise reading · MP6, MP1

MAIN: Read negations, units and relationships exactly as stated
BACKWARD 1: Strategy: rewrite each statement in plain form
FORWARD 1: ADVANCED APPLICATION: Read layered negations, units and family clues
FORWARD 2: EXPERT CHALLENGE: Precise reading in multi-step puzzles

Three students each make one statement about themself, and every statement is true. Dee: "It is not true that I finished the race." Eli: "I did not fail to finish the race." Fay: "I never said that I finished the race." Who definitely finished the race?

A. Eli and Fay
B. Eli only
C. Dee and Fay
D. Dee only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q1
Logic, Set 2: Read exactly what is stated · MAIN · Q1/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Shoelaces are sold only in packs of 2 laces for $3 a pack. Each sneaker needs exactly one lace. A club needs laces for 12 sneakers and also buys 4 whistles at $5 each. What is the least the club can spend, in dollars?

Dollars: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q2
Logic, Set 2: Read exactly what is stated · MAIN · Q2/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

In the Ortiz family, each son has exactly one brother, and each daughter has exactly two brothers and three sisters. How many children are in the family?

Children: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q3
Logic, Set 2: Read exactly what is stated · MAIN · Q3/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

A box holds only red pens and blue pens: at least 3 red pens, at most 2 blue pens, and exactly 6 pens in all. Which statement MUST be true?

A. There are exactly 2 blue pens.
B. There are exactly 3 red pens.
C. There are at least 4 red pens.
D. There are no blue pens.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q4
Logic, Set 2: Read exactly what is stated · MAIN · Q4/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

On this team, April, June and May are the names of three players (not months). A coach writes: "June scored after April and before May." Put the players in the order in which they scored, first to last.

Put these in order:

A April
J June
M May

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q5
Logic, Set 2: Read exactly what is stated · MAIN · Q5/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Treat "unhappy" as meaning exactly "not happy". What does "Rui is not unhappy" say?

A. Rui is unhappy.
B. Nothing at all about Rui.
C. Rui is sad and tired.
D. Rui is happy.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q1
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

A drawer holds 18 single socks that make complete matching pairs. How many pairs is that?

Pairs: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q2
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

"At most 3 people may enter" allows exactly 3 people to enter.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q3
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

"Not all of the doors are locked." Which statement means the same?

A. At least one door is unlocked.
B. No door is locked.
C. Every door is unlocked.
D. Exactly one door is unlocked.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q4
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Lin has 3 brothers. Each of her brothers has exactly one sister. How many children are in the family?

Children: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q5
Logic, Set 2: Read exactly what is stated · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Every statement below is true. Which person is going to the game? Gus: "It is not true that I am not going." Hal: "I am not not staying home." Ivy: "It is false that I won't go."

A. Hal and Ivy
B. Gus and Ivy
C. Gus only
D. all three

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q1
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Eggs are sold in half-dozen boxes (6 eggs) for $2 a box. A recipe uses 4 eggs per cake, and a baker makes 9 cakes. What is the least she can spend on eggs, in dollars?

Dollars: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q2
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

In the Novak family, each son has exactly 3 sisters and 2 brothers. Each daughter has exactly 2 sisters. How many children are in the family?

Children: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q3
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

A bag holds only red and blue marbles: at least twice as many red as blue, at least 3 blue, and at most 10 marbles in all. Which statement MUST be true?

A. There are exactly 10 marbles.
B. There are exactly 6 red marbles.
C. There are exactly 3 blue marbles.
D. There are more blue marbles than red ones.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q4
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

"Only members may vote" means that everyone who votes is a member.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q5
Logic, Set 2: Read exactly what is stated · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Four statements: (1) "It is not the case that 7 is not odd." (2) "It is false that 10 is not even." (3) "It is not true that 9 is not a multiple of 4." (4) "5 is not unequal to 5." How many of the four statements are true?

Number true: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q1
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

On a quiz show, Monday, Tuesday, Wednesday and Friday are the first names of four contestants. The host says: "Wednesday answered before Monday. Friday answered right after Tuesday. Monday did not answer last." Put the contestants in answering order, first to last.

Put these in order:

Mo Monday
Tu Tuesday
We Wednesday
Fr Friday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q2
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

A bag holds only red and blue marbles: the number of red marbles is at least twice the number of blue, there are at least 3 blue marbles, and there are at most 10 marbles in all. Give a possible number of red marbles r and blue marbles b.

red r = _____

blue b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q3
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

In a group of brothers and sisters, every girl has exactly as many brothers as sisters, and there are 3 boys. How many girls are there?

Girls: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q4
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

"No student who did not study passed the test." Which statement follows?

A. Every student who passed studied.
B. Some students who studied failed.
C. Every student who studied passed.
D. No student passed.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q5
Logic, Set 2: Read exactly what is stated · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.READING.01 (local) · MP6, MP1

Set 3: Statements, premises and conclusions

Local outcome LOGIC.ARGUMENT.01: Identify statements, premises, conclusions and unstated assumptions in an argument

Formal logic set · genre: formal logic · MP3

- MAIN: Identify statements, premises, conclusions and assumptions
- BACKWARD 1: Which sentences are statements?
- BACKWARD 2: Find the conclusion by its signal words
- FORWARD 1: ADVANCED APPLICATION: Arguments with several premises and hidden structure
- FORWARD 2: EXPERT CHALLENGE: Evaluate structure and missing links

"The trail is closed whenever it rains. It is raining now. So the trail is closed." Which sentence is the conclusion?

A. The trail is closed whenever it rains.
B. Rain is likely tomorrow.
C. It is raining now.
D. The trail is closed.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q1
Logic, Set 3: Statements, premises and conclusions · MAIN · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which of these is a statement (a sentence that is either true or false)?

A. Please close the window.
B. Every square has four sides.
C. Is 15 a prime number?
D. What a great game!

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q2
Logic, Set 3: Statements, premises and conclusions · MAIN · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Since every ticket holder may enter, and Zoe holds a ticket, Zoe may enter." Which sentence is a premise?

A. Only Zoe may enter.
B. Zoe may enter.
C. Zoe holds a ticket.
D. Zoe bought the ticket online.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q3
Logic, Set 3: Statements, premises and conclusions · MAIN · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Ravi will be on time, because the bus left on schedule." Which unstated assumption does this argument need?

A. The bus is red.
B. Ravi was late yesterday.
C. Some buses are late.
D. Ravi is on that bus.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q4
Logic, Set 3: Statements, premises and conclusions · MAIN · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"All of our club's robots passed inspection. Unit 7 is one of our club's robots. The room is cold. Therefore Unit 7 passed inspection." Which sentence plays no part in the reasoning?

A. The room is cold.
B. Unit 7 is one of our club's robots.
C. All of our club's robots passed inspection.
D. Therefore Unit 7 passed inspection.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q5
Logic, Set 3: Statements, premises and conclusions · MAIN · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Close the door." is a statement.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q1
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"7 is greater than 9." is a statement.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q2
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which is NOT a statement?

A. Nobody scored.
B. Who won the game?
C. Our team won.
D. The game ended in a tie.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q3
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"x + 2 = 5" with no value of x given is a statement.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q4
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which sentence is a statement?

A. Visit Paris.
B. Is Paris far?
C. Paris is in Spain.
D. Let's go to Paris!

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q5
Logic, Set 3: Statements, premises and conclusions · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which word usually introduces a conclusion?

A. given that
B. therefore
C. because
D. since

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q1
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Which word usually introduces a reason (a premise)?

A. thus
B. consequently
C. because
D. so

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q2
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"The lake is frozen, so we can skate." What is the conclusion?

A. Skating is fun.
B. The lake is frozen.
C. It is winter.
D. We can skate.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q3
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"We should leave now, since the show starts at 7." What is the conclusion?

A. We should leave now.
B. We are late.
C. The show is long.
D. The show starts at 7.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q4
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

In "Because the battery is dead, the car will not start," the conclusion is "the battery is dead."

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B2-Q5
Logic, Set 3: Statements, premises and conclusions · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"The museum must be open today. After all, it is open every weekday except Monday, and today is Thursday." Which sentence is the conclusion?

A. The museum must be open today.
B. It is open every weekday except Monday.
C. The museum is closed on Monday.
D. Today is Thursday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q1
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Put these sentences in the order of a clear argument: the general rule first, then the fact, then the conclusion.

Put these in order:

A Therefore, 91 is not prime.

B Any whole number greater than 1 with a factor other than 1 and itself is not prime.

C 91 = 7 x 13, so 91 has the factor 7.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q2
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Coach Lee will pick Maya, because Coach Lee always picks the fastest runner." Which assumption is needed?

A. Maya ran yesterday.
B. Coach Lee likes Maya.
C. Maya wants to be picked.
D. Maya is the fastest runner.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q3
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

An argument can have a true conclusion even when one of its premises is false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q4
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"If the alarm is set, it rings at 6. It did not ring at 6. Hence it was not set." How many premises does this argument have?

A. 0
B. 2
C. 3
D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q5
Logic, Set 3: Statements, premises and conclusions · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"All sprinters on the team train daily. So Kim trains daily." Which added premise makes the conclusion follow for certain?

A. Kim trains often.
B. Kim is on the team.
C. Kim is a sprinter on the team.
D. Some sprinters train daily.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q1
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every number divisible by 6 is even. So 84 is even." Which added premise makes this valid?

A. 84 is greater than 6.
B. 84 is divisible by 4.
C. Some even numbers are divisible by 6.
D. 84 is divisible by 6.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q2
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

In a valid argument, adding an extra true premise can make the conclusion no longer follow.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q3
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Arrange these sentences into an argument that ends with its conclusion: chain the two rules.

Put these in order:

A If the dam opens, the river rises.
B If the river rises, the ferry stops.
C The dam opens.
D So the ferry stops.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q4
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Our new ads worked: sales rose this month." Which assumption is the argument making?

A. Sales rose this month.
B. Nothing other than the ads caused most of the rise in sales.
C. Sales will rise next month.
D. The ads were expensive.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q5
Logic, Set 3: Statements, premises and conclusions · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

Set 4: Eliminate candidates clue by clue

Local outcome LOGIC.CONSTRAINT.01: Find the only candidate left after applying every clue to a finite list

Puzzle set · genre: elimination · MP1, MP6

MAIN: Find the only candidate left after every clue

BACKWARD 1: Strategy: cross out, one clue at a time

FORWARD 1: ADVANCED APPLICATION: Eliminate with clues about neighbors and positions

FORWARD 2: EXPERT CHALLENGE: Which clues are really needed?

Which month is it? Its name has fewer than 8 letters. Its name contains the letter R. It is the 1st, 3rd, 5th, 7th, 9th or 11th month. Its name does not start with J.

A. March
B. May
C. January
D. September

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q1
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q1/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Which day of the week is it? Its name has no letter E. It comes after Wednesday in the week that starts on Monday. Its name has exactly 6 letters. Its name contains the letter U.

A. Friday
B. Thursday
C. Sunday
D. Saturday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q2
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q2/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A whole number from 1 to 50 leaves remainder 3 when divided by 7. It is greater than 20. It is even. It does not leave remainder 4 when divided by 5. What is the number?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q3
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q3/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Three friends, Ana, Ben and Cai, each have exactly one pet: a cat, a dog or a fish, all different. Ana does not have the dog. Ben has neither the cat nor the dog. Which pet does each friend have?

Choose one column for each row. Each column is used once.

	cat	dog	fish
Ana			
Ben			
Cai			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q4
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q4/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A locker number is one of 12, 15, 18, 21, 24, 27. It is even. It is not a multiple of 4. Which locker is it?

A. 24
B. 27
C. 12
D. 18

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q5
Logic, Set 4: Eliminate candidates clue by clue · MAIN · Q5/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A number from 1 to 10 is even, greater than 7 and less than 10. What is it?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q1
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A day of the week (Monday to Sunday) starts with T and has more than 7 letters. Which day?

A. Saturday
B. Thursday
C. Tuesday
D. Wednesday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q2
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

How many months have a name with exactly 4 letters?

Months: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q3
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

If two candidates are left after every clue, the clues determine the answer.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q4
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Candidates: 2, 3, 4, 5, 6. Clue 1: the number is prime. Clue 2: the number is odd. How many candidates are left?

A. 0
B. 3
C. 2
D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q5
Logic, Set 4: Eliminate candidates clue by clue · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Which month is it? The month before it has a name longer than 5 letters. The month after it does not end in "ber". The month after it contains the letter U. Its own name contains the letter A.

A. January
B. May
C. February
D. December

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q1
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

In the week Monday to Sunday, exactly one day has a name whose day before and day after both contain the letter T.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q2
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A whole number from 1 to 100 is a perfect square, is odd, has two digits, and its digits add to 13. What is it?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q3
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Four runners, Dev, Eva, Finn and Gia, finished 1st to 4th, no ties. Eva finished right after Dev. Gia finished before Finn. Finn finished before Dev. Assign each runner a place.

Choose one column for each row. Each column is used once.

	1st	2nd	3rd	4th
Dev				
Eva				
Finn				
Gia				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q4
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Candidates are the whole numbers 1 to 30. Clues: A: the number is odd. B: it is a multiple of 3. C: it is greater than 10. D: it is a multiple of 5. Which number fits all four clues?

A. 30
B. 15
C. 5
D. 25

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q5
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Candidates are the whole numbers 1 to 30, with four clues: A: odd; B: a multiple of 3; C: greater than 10; D: a multiple of 5. Only 15 fits all four. Which ONE clue can be removed so that exactly one number is still left?

A. C (greater than 10)
B. D (multiple of 5)
C. A (odd)
D. B (multiple of 3)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q1
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

A number from 1 to 30 is a multiple of 4. What is the least number of extra yes/no questions of the form "Is it greater than k?" that always identify it?

Questions: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q2
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Adding a new clue to a puzzle can never increase the number of candidates left.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q3
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Candidates are the whole numbers 1 to 20. Write a number n that survives BOTH clues: n is prime, and n leaves remainder 1 when divided by 4.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q4
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Six boxes are numbered 1 to 6. One holds a prize. Which single clue, added alone, would leave exactly one box?

A. "The box number is odd."
B. "The box number is a multiple of 3."
C. "The box number is less than 4."
D. "The box number is prime and even."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q5
Logic, Set 4: Eliminate candidates clue by clue · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONSTRAINT.01 (local) · MP1, MP6

Set 5: Negation and truth values

Local outcome LOGIC.CONNECTIVE.01: Form the negation of a statement and find truth values of simple statements

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Form negations and find truth values

BACKWARD 1: Truth values of simple statements

BACKWARD 2: Negation is not the same as the opposite

FORWARD 1: ADVANCED APPLICATION: Negate ranges, counts and double negatives

FORWARD 2: EXPERT CHALLENGE: Negation inside conditions and counts

What is the negation of "Sam is taller than Ana"?

A. Ana is taller than Sam.
B. Sam and Ana are the same height.
C. Sam is not taller than Ana.
D. Sam is shorter than Ana.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q1
Logic, Set 5: Negation and truth values · MAIN · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

x stands for a real number. What is the negation of "x > 5"?

A. $x \leq 5$

B. $x < 5$

C. $x \geq 5$

D. $x < -5$

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q2
Logic, Set 5: Negation and truth values · MAIN · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

P is the statement "48 is a multiple of 9." The negation of P is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q3
Logic, Set 5: Negation and truth values · MAIN · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Give a number x that makes the NEGATION of the statement "$x \leq 4$" true.

$x =$ _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q4
Logic, Set 5: Negation and truth values · MAIN · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Which pair of statements are negations of each other (in every situation, exactly one of the two is true)?

A. "The cup is full." / "The cup is empty."

B. "Every student passed." / "Every student failed."

C. "The coin landed heads up." / "The coin did not land heads up."

D. "x > 3" / "x < 3"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q5
Logic, Set 5: Negation and truth values · MAIN · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

"17 is a prime number" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q1
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

"3 x 7 = 24" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q2
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

"Every even number is a multiple of 4" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q3
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

Which statement is false?

A. 0.5 equals one half.
B. 0.5 + 0.25 = 0.75.
C. 0.75 is greater than 0.5.
D. 0.5 is greater than 0.75.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q4
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

"10 is not an even number" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q5
Logic, Set 5: Negation and truth values · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP6, MP3

What is the negation of "The door is open"?

A. The door is locked.
B. The door is not open.
C. The door is wide open.
D. The window is open.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q1
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

"The water is cold" and "The water is hot" are negations of each other.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q2
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

What is the negation of "Tom won the race"?

A. Tom did not win the race.
B. Someone else won.
C. Tom lost the race.
D. Tom came last.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q3
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

If a statement is true, its negation is also true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q4
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

n is a whole number. What is the negation of "n is even"?

A. n is 0.
B. n is negative.
C. n is odd.
D. n is prime.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B2-Q5
Logic, Set 5: Negation and truth values · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

x stands for a real number. What is the negation of "3 < x < 7" (x is between 3 and 7)?

A. $x < 3$ or $x > 7$

B. $3 \geq x \geq 7$

C. $x \leq 3$ and $x \geq 7$

D. $x \leq 3$ or $x \geq 7$

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q1
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Exactly 9 lamps are in a room. What is the negation of "At least 5 lamps are on"?

A. At most 5 lamps are on.
B. At most 4 lamps are on.
C. No lamps are on.
D. At least 4 lamps are off.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q2
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Complete the truth table: the columns are "not P" and "not (not P)".

Fill in T or F in every empty cell.

P	not P	not (not P)
T		
F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q3
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

How many of these are true? (1) "It is not true that 8 is odd." (2) "It is false that 0 is not a whole number." (3) "It is not the case that 2 + 2 is not 4." (4) "It is not true that 5 is not less than 3."

Number true: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q4
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

The negation of "Mia always walks to school" is "Mia never walks to school".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q5
Logic, Set 5: Negation and truth values · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Give a number x that makes the negation of "x < 2 or x > 5" true.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q1
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

There are 6 lights. What is the negation of "At most 2 lights are on"?

A. At least 2 lights are on.
B. At most 2 lights are off.
C. At least 3 lights are on.
D. Exactly 3 lights are on.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q2
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Ana says: "It is false that it is not true that the test is not on Friday." What is Ana saying?

A. Nothing about the test.
B. The test might be on Friday.
C. The test is on Friday.
D. The test is not on Friday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q3
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

For every statement P, exactly one of P and "not P" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q4
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Statements about a whole number n from 1 to 10: A: "n is even." B: "n is not greater than 6." C: "It is not true that n is a multiple of 3." For how many values of n are all three statements true?

Values: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q5
Logic, Set 5: Negation and truth values · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

Set 6: Crack a code with a stated key

Local outcome LOGIC.CODE.01: Decode and encode with a stated symbol or value key, and infer a key from examples within a stated family

Puzzle set · genre: symbol codes · MP7, MP2

- MAIN: Decode and encode with a stated key, and find a key within a stated family
- BACKWARD 1: Strategy: write the key as a table
- FORWARD 1: ADVANCED APPLICATION: Codes with structure: shapes of letters and combined rules
- FORWARD 2: EXPERT CHALLENGE: Find a key and prove when none exists

In a code, each letter has the value of its position in the alphabet (A = 1, B = 2, ..., Z = 26), and a word's value is the sum of its letters' values. For example, CAB = 3 + 1 + 2 = 6. What is the value of BEAD?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q1
Logic, Set 6: Crack a code with a stated key · MAIN · Q1/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

In a code, every letter is replaced by the letter 3 places later in the alphabet (A becomes D, B becomes E, and X, Y, Z wrap around to A, B, C). A coded word reads FDW. What was the original word?

A. CAW
B. CAT
C. DBX
D. IGZ

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q2
Logic, Set 6: Crack a code with a stated key · MAIN · Q2/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code adds the same whole number k to every digit and keeps only the last digit of each sum (so 8 + 3 gives 1). The code turns 2025 into 4247. What does it turn 1990 into?

Code: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q3
Logic, Set 6: Crack a code with a stated key · MAIN · Q3/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Each symbol stands for a whole number. Star + Star = 10. Star + Circle = 8. Circle + Square = 9. What number is Square?

Square: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q4
Logic, Set 6: Crack a code with a stated key · MAIN · Q4/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Two 3-letter words were interleaved: first letter of word 1, first letter of word 2, second letter of word 1, and so on. The result is CDAORT. Which two words were interleaved?

A. CAR and DOT
B. CAD and ORT
C. COD and ART
D. CAT and DOR

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q5
Logic, Set 6: Crack a code with a stated key · MAIN · Q5/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code replaces each letter by its position in the alphabet (A = 1, ..., Z = 26) and adds them. What is the value of DOG?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q1
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code moves each letter 1 place later (A -> B, ..., Z -> A). What is BOX in code?

A. CPY
B. DQZ
C. CPX
D. ANW

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q2
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code replaces each digit d by 9 - d. What is the code for 306?

Code: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q3
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

If a code moves every letter 2 places later, then decoding moves every letter 2 places earlier.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q4
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Triangle + Triangle + Triangle = 12. What number is Triangle?

Triangle: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q5
Logic, Set 6: Crack a code with a stated key · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. What is the value of BREAD?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q1
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. Which word has value 0?

A. SHIRT
B. FLIES
C. MAZE
D. TOWEL

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q2
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code first writes a word backward and then moves each letter 1 place later in the alphabet. What is the code for MAP?

Code: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q3
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Each symbol stands for a whole number. Moon + Moon + Sun = 13. Sun + Sun = 6. Moon + Star = 9. What number is Star?

Star: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q4
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code moves every letter 13 places later in the alphabet (wrapping around). Applying this code twice gives back the original word.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q5
Logic, Set 6: Crack a code with a stated key · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code adds 3 to each digit of a 4-digit number and keeps only the last digit of each sum (so 8 becomes 1). Is there a 4-digit number whose code is the same number written backward?

A. No: but only because the first digit cannot be 0.

B. Yes: 1234 works.

C. No: the middle digits would need b + 6 to end in the same digit as b, which never happens.

D. Yes: any number whose digits are all equal works.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q1
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code moves every letter the same number of places later in the alphabet (wrapping around). It turns GO into MU. What does it turn LOGIC into?

Code: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q2
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Each symbol stands for a whole number from 1 to 9. Square + Square = Circle, and Circle + Square = 12. Find Square s and Circle c.

Square s = _____

Circle c = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q3
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

In this code, a letter's value is the number of closed spaces inside it in plain block capitals: A, D, O, P, Q and R have 1, B has 2, and every other letter has 0. A word's value is the sum of its letters' values. Using only the letters B, O and T (each as often as you like), what is the greatest value of a 4-letter string?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q4
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

A code sends each letter to the letter at the same distance from the other end of the alphabet (A <-> Z, B <-> Y, C <-> X, ...). Which word is coded as itself?

A. WIZARD

B. MOON

C. NOON

D. No word: no letter is coded as itself.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q5
Logic, Set 6: Crack a code with a stated key · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CODE.01 (local) · MP7, MP2

Set 7: Order from relational clues

Local outcome LOGIC.ORDER.01: Build an ordering that satisfies relational clues (before, after, between, next to)

Puzzle set · genre: ordering · MP1, MP2

MAIN: Build an ordering that satisfies relational clues

BACKWARD 1: Strategy: place the most constrained item first

FORWARD 1: ADVANCED APPLICATION: Orderings with "next to" and "not next to"

FORWARD 2: EXPERT CHALLENGE: Count and compare orderings

Four books stand on a shelf. The atlas is left of the cookbook. The diary is right of the cookbook. The biography is at the far left. Order the books from left to right.

Put these in order:

A atlas
B biography
C cookbook
D diary

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q1
Logic, Set 7: Order from relational clues · MAIN · Q1/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five friends arrived at a party one at a time. Gil arrived before Hana but after Ivo. Jo arrived last. Kai arrived right after Ivo. Order the arrivals, first to last.

Put these in order:

G Gil
H Hana
I Ivo
J Jo
K Kai

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q2
Logic, Set 7: Order from relational clues · MAIN · Q2/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five runners finished a race with no ties. Uma beat Vic. Wes finished right after Uma. Xia beat Uma. Yan finished last. Who finished in the middle (3rd)?

A. Uma
B. Vic
C. Wes
D. Xia

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q3
Logic, Set 7: Order from relational clues · MAIN · Q3/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Four flags hang on a pole, from top to bottom. The green flag is somewhere between the red and the blue flags. The red flag is not at the top. The yellow flag is directly below the blue flag. Order the flags from top to bottom.

Put these in order:

R red
G green
B blue
Y yellow

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q4
Logic, Set 7: Order from relational clues · MAIN · Q4/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Seven students stand in a line. Lee is 2nd from the left. Max is 3rd from the right. How many students stand between Lee and Max?

Students between: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q5
Logic, Set 7: Order from relational clues · MAIN · Q5/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Three cups stand in a row. The red cup is between the other two. The blue cup is on the right. Order them from left to right.

Put these in order:

R red
B blue
W white

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q1
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

In a line of 9 people, Ola is 4th from the front. What is her position counted from the back?

Position from the back: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q2
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

"Bea is right after Cy" means no one is between Cy and Bea.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q3
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Ava is taller than Ben, and Ben is taller than Cal. Who is shortest?

A. Ben
B. Cal
C. Ava
D. It cannot be known.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q4
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Four racers: Dot finished first. Eli beat Fay. Gus finished last. Order them, first to last.

Put these in order:

D Dot
E Eli
F Fay
G Gus

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q5
Logic, Set 7: Order from relational clues · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five houses stand in a row, numbered 1 to 5 from left to right. The bakery is next to the bank. The café is at one end. The dentist is directly right of the café. The florist is not next to the dentist. The bank is in house 4. Order the businesses from left to right.

Put these in order:

K bakery
N bank
C café
D dentist
F florist

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q1
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Six people, A to F, sit in seats 1 to 6 in a row. C is in seat 1 and D is in seat 6. B sits right after A, and E sits right after B. F does not sit next to C. Which seat is F in?

Seat: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q2
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five students line up. Pat is in front of Quinn. Rae is right behind Sol. Tia is first. Quinn is last. Which statement must be true?

A. Rae is 4th.

B. Pat is 2nd.

C. Sol is right behind Pat.

D. Tia is directly in front of Pat or of Sol.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q3
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Four friends sit in seats 1 to 4 in a row. Ari is not next to Bo. Cy sits in seat 2. Bo is in a higher seat number than Cy. Di is not in seat 4. Give each friend's seat.

Choose one column for each row. Each column is used once.

	1	2	3	4
Ari				
Bo				
Cy				
Di				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q4
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

If five people stand in a line and three of them must stand next to each other in a fixed order, there are exactly 6 possible lines.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q5
Logic, Set 7: Order from relational clues · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Five students, P, Q, R, S and T, stand in a line. P is somewhere in front of Q, and R is somewhere in front of S. How many different lines are possible?

Lines: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q1
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Six relay swimmers, A to F, swim in order. B swims right after E. D swims before A. C swims last. F swims first. A swims right after B. Give the swimming order.

Put these in order:

A A
B B
C C
D D
E E
F F

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q2
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Four people, W, X, Y and Z, stand in a line. Clues: (1) W is not first. (2) X is right after W. (3) Y is last. Which clue could be removed and still leave exactly one possible line?

A. No clue can be removed.

B. Clue 3

C. Clue 1

D. Clue 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q3
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Seven people stand in a line, numbered 1 to 7 from the front. Ana is in position a and Ben is in position b. Ana is in front of Ben, exactly two people stand between them, and Ben is not last. Give one possible pair of positions.

Ana a = _____

Ben b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q4
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

With the clues "J is before K" and "K is before L" for three people in a line, the order is fully determined.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q5
Logic, Set 7: Order from relational clues · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.ORDER.01 (local) · MP1, MP2

Set 8: And, inclusive or, exclusive or

Local outcome LOGIC.CONNECTIVE.02: Find truth values of conjunctions and of inclusive and exclusive disjunctions

Formal logic set · genre: formal logic · MP6

- MAIN: Truth values of and, inclusive or, exclusive or
- BACKWARD 1: Truth values of the parts
- BACKWARD 2: Inclusive or exclusive?
- FORWARD 1: ADVANCED APPLICATION: Compound statements with three parts
- FORWARD 2: EXPERT CHALLENGE: Design statements with given truth values

"7 is odd and 7 is greater than 10" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q1
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"7 is odd or 7 is greater than 10" is true (with "or" meaning at least one).

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q2
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

A sign says: "Each meal comes with soup or salad, but not both." Which order follows the sign?

A. a meal with salad and no soup
B. a meal with soup and salad
C. a meal with two salads and a soup
D. a meal with neither soup nor salad

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q3
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Give a whole number n that makes "n is even or n is greater than 20" true but "n is even and n is greater than 20" false.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q4
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

P is true and Q is false. Which statement is FALSE?

A. P and not Q
B. P exclusive-or Q
C. P or Q
D. P and Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q5
Logic, Set 8: And, inclusive or, exclusive or · MAIN · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"12 is a multiple of 5" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q1
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

"-3 < 1" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q2
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

"A triangle has 4 sides" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q3
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

"0 is even" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q4
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

Which part of "9 is odd and 9 is prime" is false?

A. both parts
B. 9 is prime
C. 9 is odd
D. neither part

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q5
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6

With inclusive "or", "P or Q" is false when both P and Q are true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q1
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

With exclusive "or", "P or Q" is true when both P and Q are true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q2
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

A rule says "You may bring a pencil or a pen (or both)." Which kind of "or" is this?

A. neither kind

B. it cannot be told

C. inclusive

D. exclusive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q3
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"The light is either on or off" describes two cases that can never both happen. Which "or" fits best?

A. "and"
B. exclusive
C. inclusive
D. neither

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q4
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

When P is false and Q is false, exclusive "P or Q" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B2-Q5
Logic, Set 8: And, inclusive or, exclusive or · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"(3 > 2 and 5 > 9) or 4 is even" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q1
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"3 > 2 and (5 > 9 or 4 is odd)" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q2
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

For how many whole numbers n from 1 to 20 is "n is a multiple of 3 exclusive-or n is a multiple of 4" true?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q3
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Complete the truth table for "P exclusive-or Q" and for "P or Q" (inclusive).

Fill in T or F in every empty cell.

P	Q	P exclusive-or Q	P or Q
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q4
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Statement S: "P or Q" (inclusive). S is false. What can be concluded?

A. P is false and Q is false.
B. P is true and Q is false.
C. P is false or Q is false.
D. Exactly one of P and Q is false.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q5
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

P, Q and R are statements. In how many of the 8 combinations of truth values is "(P or Q) and not R" true?

Combinations: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q1
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Give whole numbers a and b with "a > b exclusive-or a + b is even" true and both numbers between 1 and 9.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q2
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

"P exclusive-or Q" has the same truth value as "(P or Q) and not (P and Q)" in every row of a truth table.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q3
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

Three lamps A, B and C are each on or off. Exactly one of these statements is true: (1) "A is on or B is on." (2) "C is on." (3) "A is on exclusive-or C is on." Which lamps are on?

A. A only
B. A and B
C. none
D. B only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q4
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

For how many whole numbers n from 1 to 30 is "n is odd and (n is a multiple of 3 or n is a multiple of 5)" true?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q5
Logic, Set 8: And, inclusive or, exclusive or · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6

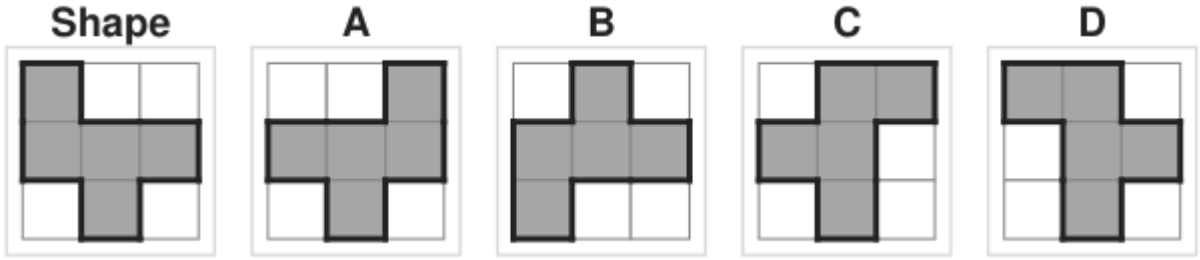
Set 9: Rotation or reflection?

Local outcome LOGIC.SPATIAL.02: Distinguish rotations from reflections of a figure on a grid

Puzzle set · genre: rotation and reflection · MP7

- MAIN: Distinguish rotations from reflections
- BACKWARD 1: Strategy: follow one marked corner
- FORWARD 1: ADVANCED APPLICATION: Combine turns and flips
- FORWARD 2: EXPERT CHALLENGE: Symmetry under turns and flips

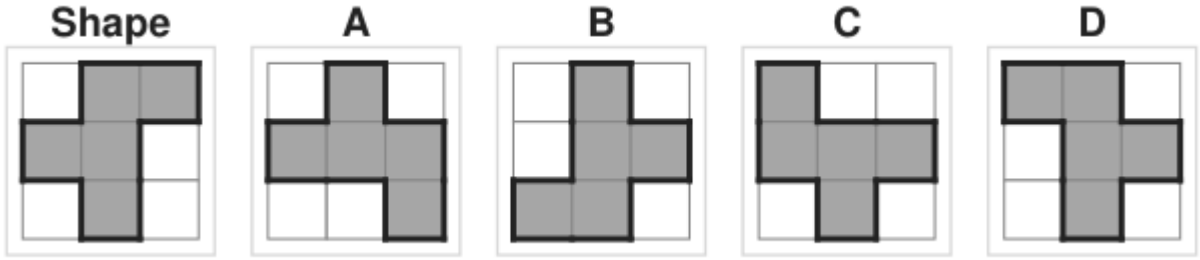
Which figure shows the shape TURNED (rotated in the page), not flipped over? (Only the shape matters, not where it sits in its box.)



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q1
Logic, Set 9: Rotation or reflection? · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

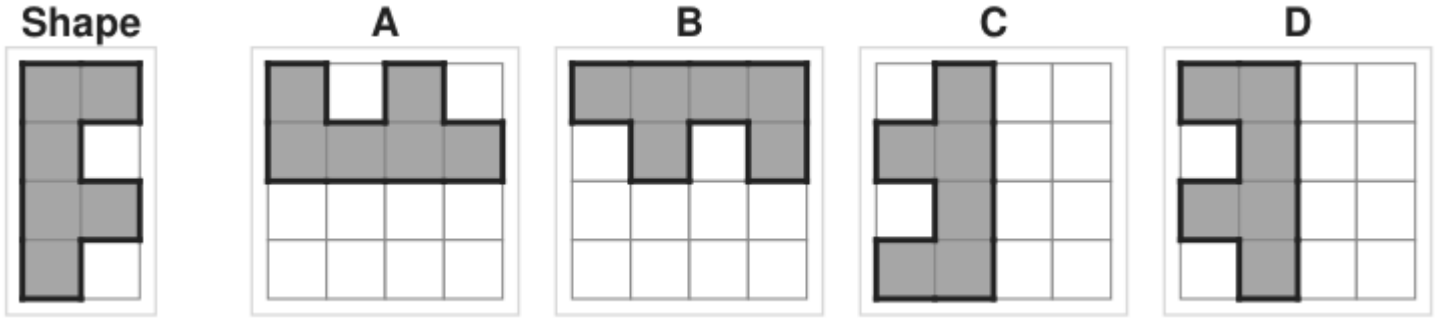
Three of the figures are the shape turned (rotated). One is the shape flipped over (a mirror image). Which figure is the flipped one?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q2
Logic, Set 9: Rotation or reflection? · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

The shape is turned a quarter-turn CLOCKWISE (90 degrees, like a clock hand moving from 12 to 3). Which figure shows the result?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

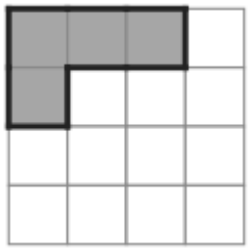
REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q3
Logic, Set 9: Rotation or reflection? · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Capital letters are turned upside down by a half-turn (180 degrees in the page). In the block capitals used here, exactly these letters look the same after a half-turn: H I N O S X Z. Which letter looks the same after the half-turn?

A. A
B. M
C. N
D. E

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q4
Logic, Set 9: Rotation or reflection? · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A 4 by 4 grid has shaded cells at (1,1), (1,2), (1,3) and (2,1), written as row,column with row 1 at the top. The whole grid is turned a quarter-turn clockwise about its center. Mark the cells where the shaded cells end up.



Mark cells on the 4 × 4 grid (on screen: click each cell; cells are named row,column).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q5
Logic, Set 9: Rotation or reflection? · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A quarter-turn clockwise followed by another quarter-turn clockwise is the same as a half-turn.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q1
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

An arrow points up. It is turned a quarter-turn clockwise. Which way does it point now?

A. right
B. up
C. left
D. down

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q2
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

An arrow points right. It is reflected in a vertical mirror line. Which way does it point now?

A. down
B. left
C. right
D. up

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q3
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Flipping a shape over twice in the same mirror line brings it back to where it started.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q4
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

How many quarter-turns clockwise bring a shape back to its starting position for the first time, if the shape has no symmetry at all?

Quarter-turns: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q5
Logic, Set 9: Rotation or reflection? · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Reflecting a shape in a vertical line and then in a horizontal line gives the same result as a half-turn.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q1
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A shape is turned a quarter-turn clockwise three times. Which single move gives the same result?

A. a reflection
B. a quarter-turn clockwise
C. a quarter-turn counterclockwise
D. a half-turn

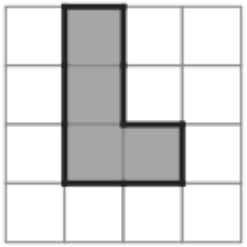
REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q2
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A word written in capital letters is turned a half-turn (180 degrees). In the block capitals used here, exactly these letters look the same after a half-turn: H I N O S X Z. A half-turn also reverses the order of the letters. Which word looks exactly the same after the turn?

A. NOW
B. MOM
C. ZOO
D. SOS

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q3
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A 4 by 4 grid has shaded cells at (1,2), (2,2), (3,2) and (3,3), written as row,column with row 1 at the top. The grid is given a half-turn about its center. Mark where the shaded cells end up.



Mark cells on the 4 × 4 grid (on screen: click each cell; cells are named row,column).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q4
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

A shape has no symmetry. How many different-looking positions can it be put in using turns AND flips (on a grid, counting each distinct picture once)?

Positions: _____

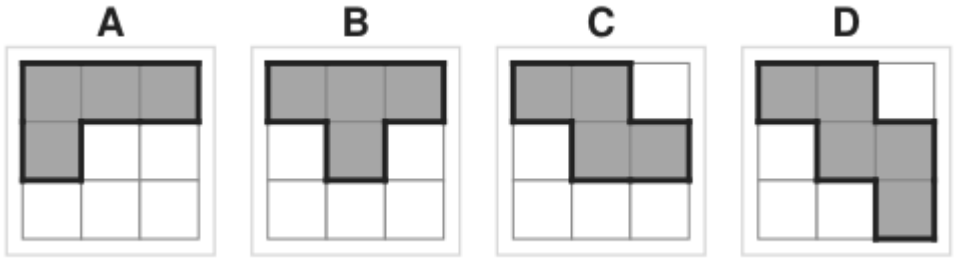
REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q5
Logic, Set 9: Rotation or reflection? · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Of the 8 moves of a square grid figure (4 turns: 0, 90, 180, 270 degrees; and 4 flips: vertical, horizontal and both diagonals), how many leave a plus-shaped figure of 5 cells (a center cell with one cell on each side) unchanged?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q1
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

For which figure can the mirror image be produced by turning the figure (without flipping it over)? Exactly two of the four figures have this property; choose the answer that names both.



A. B and D
B. A and C
C. A and B
D. C and D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q2
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

An S-shaped figure of 4 cells (two cells in a row, and two more cells one row down, shifted one place left) is turned and flipped. Which statement is true?

A. No move other than doing nothing leaves it unchanged.

B. A quarter-turn leaves it unchanged.

C. A half-turn leaves it unchanged, but no flip does.

D. A flip in a vertical line leaves it unchanged.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q3
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

If a figure looks the same after a quarter-turn, then it also looks the same after a half-turn.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q4
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

How many of the 7 shapes made of 4 grid cells joined edge to edge (counting a shape and its turns as the same, but its mirror image as different) look the same after a half-turn?

Shapes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q5
Logic, Set 9: Rotation or reflection? · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.02 (local) · MP7

Set 10: Deduction and induction

Local outcome LOGIC.ARGUMENT.02: Tell deductive conclusions (forced by the premises) from inductive ones (supported by examples)

Formal logic set · genre: formal logic · MP3, MP8

- MAIN: Tell deductive conclusions from inductive ones
- BACKWARD 1: Premises and conclusions
- BACKWARD 2: Certain or only likely?
- FORWARD 1: ADVANCED APPLICATION: Patterns that break
- FORWARD 2: EXPERT CHALLENGE: From pattern to proof

Which argument is DEDUCTIVE (the conclusion must be true if the premises are true)?

A. Every swan I have seen is white. So all swans are white.

B. Every prime number greater than 2 is odd. 31 is a prime number greater than 2. So 31 is odd.

C. Most students like recess. So Ana likes recess.

D. The last 20 buses were on time. So the next bus will be on time.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q1
Logic, Set 10: Deduction and induction · MAIN · Q1/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

"1 + 3 = 4, 1 + 3 + 5 = 9, 1 + 3 + 5 + 7 = 16. So the sum of the first n odd numbers is always n x n." What kind of reasoning is this?

A. deductive: the examples prove it for every n

B. neither: it is not an argument

C. deductive: the conclusion is true

D. inductive: it generalizes from a few examples

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q2
Logic, Set 10: Deduction and induction · MAIN · Q2/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

In an inductive argument, the premises can all be true while the conclusion is false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q3
Logic, Set 10: Deduction and induction · MAIN · Q3/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

n x n + n + 11 is prime for n = 1, 2, 3, ..., 9. A student concludes it is prime for every whole number n. Which value of n shows the conclusion is false?

A. 10

B. 8

C. 4

D. 9

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q4
Logic, Set 10: Deduction and induction · MAIN · Q4/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Which conclusion follows DEDUCTIVELY from "All members of the chess club are in grade 10 or grade 11, and Jun is a member of the chess club"?

A. Jun is good at chess.

B. Jun is in grade 10 or grade 11.

C. Jun is in grade 11.

D. Most grade 10 students are in the chess club.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q5
Logic, Set 10: Deduction and induction · MAIN · Q5/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

"All the cookies are gone, so someone ate them." Which part is the conclusion?

A. nobody ate them
B. all the cookies are gone
C. cookies are tasty
D. someone ate them

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q1
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

A premise is the statement that an argument is trying to prove.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q2
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

"It rained every day this week. So it will rain tomorrow." What is the premise?

A. It rained every day this week.
B. It will rain tomorrow.
C. It never rains.
D. Rain is good for plants.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q3
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

Every argument has at least one premise and one conclusion.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q4
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

"Since 12 is even, 12 is divisible by 2." What is the conclusion?

A. 12 is even
B. 12 is odd
C. 12 is divisible by 2
D. 2 is even

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q5
Logic, Set 10: Deduction and induction · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3, MP8

"Most tickets in the box are red, and I drew a ticket from the box" makes "my ticket is red" certain.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q1
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

"Most tickets in the box are red, and I drew a ticket" makes "my ticket is red" certain.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q2
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Which word signals that a conclusion is only likely?

A. must
B. probably
C. every
D. therefore

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q3
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

If a pattern holds for 100 examples, it must hold for the 101st.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q4
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

A die has been rolled 5 times and showed an even number each time. Which conclusion is certain?

A. The die is unfair.

B. The next roll will be even.

C. Each of those 5 rolls was even.

D. The die only has even numbers.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B2-Q5
Logic, Set 10: Deduction and induction · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

The claim "n x n - n + 17 is prime for every whole number n" holds for n = 1 to 16. Find a whole number n that makes n x n - n + 17 NOT prime.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q1
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Points are placed on a circle and every pair is joined by a straight line. With 1, 2, 3, 4, 5 points (in general position), the circle is cut into 1, 2, 4, 8, 16 regions. A student predicts 32 regions for 6 points. What is the actual number?

A. 31
B. 32
C. 36
D. 30

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q2
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

A deductive argument with true premises can have a false conclusion.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q3
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Which statement is the strongest inductive argument?

A. All bolts in the batch passed, so this bolt passed.
B. My two friends like the song, so everyone does.
C. The first bolt passed, so all bolts pass.
D. All 500 randomly chosen bolts from the batch passed, so nearly all bolts in the batch pass.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q4
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Which reasoning is deductive?

A. Most rectangles in the book are squares, so this one is.

B. All squares are rectangles; this figure is a square; so it is a rectangle.

C. Every rectangle I measured had equal diagonals, so all rectangles do.

D. This figure looks like a rectangle, so it probably is one.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q5
Logic, Set 10: Deduction and induction · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

The expression n x n + n + 41 is prime for n = 0, 1, 2, ..., 39. What is the smallest whole number n for which it is NOT prime?

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q1
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

"For every whole number n, n x n - n is even, because 2 x 2 - 2 = 2, 3 x 3 - 3 = 6 and 4 x 4 - 4 = 12 are even." Which statement about this argument is correct?

A. The claim is true, but three examples do not prove it.

B. The claim is false, because 1 x 1 - 1 = 0.

C. The argument proves the claim, because three cases are checked.

D. The claim is false for large n.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q2
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

If every case of a finite list has been checked, the conclusion about that list is deductive.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q3
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Give a whole number n from 1 to 20 for which "2 to the power n, minus 1, is prime" is FALSE, even though n is prime. (2 to the power n is 2 multiplied by itself n times.)

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q4
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

A student checks that every whole number from 4 to 100 that is even can be written as a sum of two primes. Which conclusion is justified?

A. Every even number greater than 2 is a sum of two primes.

B. Every even number is a sum of two primes.

C. Every even number from 4 to 100 is a sum of two primes.

D. No odd number is a sum of two primes.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q5
Logic, Set 10: Deduction and induction · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.ARGUMENT.02 (local) · MP3, MP8

Set 11: The structure of words

Local outcome LOGIC.WORD.01: Use letter counts, positions and alphabetical order to test word patterns within a stated rule

Puzzle set · genre: word structure · MP7, MP6

MAIN: Use letter counts, positions and order to test word patterns
BACKWARD 1: Strategy: list the letters with their positions
FORWARD 1: ADVANCED APPLICATION: Patterns across several words
FORWARD 2: EXPERT CHALLENGE: Count and construct letter strings under rules

Which word has its letters in alphabetical order (each letter the same as or later in the alphabet than the one before)?

A. CHART
B. BEGAN
C. FLOWER
D. ALMOST

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q1
Logic, Set 11: The structure of words · MAIN · Q1/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many times does the letter S appear in the word ASSESSMENTS?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q2
Logic, Set 11: The structure of words · MAIN · Q2/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Number words are written in capital letters (ONE, TWO, ..., TEN). Which number word has exactly as many letters as the number it names?

A. FOUR
B. THREE
C. FIVE
D. TWO

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q3
Logic, Set 11: The structure of words · MAIN · Q3/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Take the first letter of each word: "Every Apple Tastes Sweet." What word do the first letters spell?

A. TEAS
B. EAST
C. EATS
D. SEAT

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q4
Logic, Set 11: The structure of words · MAIN · Q4/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Write the letters of STEAM in alphabetical order. Which letter is in the middle position?

A. A
B. M
C. S
D. E

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q5
Logic, Set 11: The structure of words · MAIN · Q5/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

The letters of BELLOW are in alphabetical order.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q1
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many times does the letter E appear in REFERENCE?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q2
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

What is the 4th letter of the word PLANET?

A. E
B. A
C. L
D. N

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q3
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Take the first letter of each word: "Owls Watch Lakes." What do the first letters spell?

A. OLW
B. OWL
C. WOL
D. LOW

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q4
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many letters does the word EIGHT have?

Letters: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q5
Logic, Set 11: The structure of words · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

In the list GO, CAT, LAMP, PLANT, each word has one more letter than the word before. Which word could come next?

A. BALLOON
B. TREE
C. GARDEN
D. HOUSE

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q1
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many of the number words ONE, TWO, THREE, ..., TEN have their letters in alphabetical order?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q2
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Which word has its letters in strictly reverse alphabetical order (each letter earlier in the alphabet than the one before it)?

A. SPELL
B. TRACK
C. STONE
D. SPOKE

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q3
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Write the letters of each number word ONE to TEN in alphabetical order; for example, ONE becomes ENO. Which number word becomes EENSV? Give its value as a number.

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q4
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Every word whose letters are in alphabetical order reads differently backward unless all its letters are the same.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q5
Logic, Set 11: The structure of words · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many 3-letter strings can be made from the letters A, B, C, D (letters may repeat) whose letters are in alphabetical order (each letter the same as or later than the one before)?

Strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q1
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many 4-letter strings using only the letters X and Y have no two equal letters next to each other?

Strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q2
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

How many 4-letter strings using the letters A, B, C (letters may repeat) have no letter next to itself?

Strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q3
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

A 6-letter string uses each of the letters A, B, C exactly twice, and no letter is next to itself. Which string follows the rule?

A. ABCABC
B. AABBCC
C. ABBCAC
D. ACBBCA

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q4
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

The five letters of LEVEL (L, E, V, E, L) are rearranged into 5-letter strings. How many different strings have no two equal letters next to each other?

Strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q5
Logic, Set 11: The structure of words · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.WORD.01 (local) · MP7, MP6

Set 12: When is "if ... then" false?

Local outcome LOGIC.CONDITIONAL.01: Find the truth value of a conditional and identify the one case that makes it false

Formal logic set · genre: formal logic · MP3, MP6

MAIN: Find when a conditional is true and the one case that makes it false
BACKWARD 1: Find the "if" part and the "then" part
BACKWARD 2: Truth values of the parts
FORWARD 1: ADVANCED APPLICATION: Conditionals in mathematics
FORWARD 2: EXPERT CHALLENGE: Nested and chained conditionals

A rule: "If a package weighs more than 5 kg, then it has a yellow label." Which package BREAKS the rule?

A. a 7 kg package with a yellow label
B. a 7 kg package with a blue label
C. a 3 kg package with a blue label
D. a 3 kg package with a yellow label

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q1
Logic, Set 12: When is "if ... then" false? · MAIN · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Complete the truth table for "if P then Q".

Fill in T or F in every empty cell.

P	Q	if P then Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q2
Logic, Set 12: When is "if ... then" false? · MAIN · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Find a whole number n that is a counterexample to "If n is odd, then n is prime." (A counterexample makes the "if" part true and the "then" part false.)

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q3
Logic, Set 12: When is "if ... then" false? · MAIN · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If 5 > 8, then 2 + 2 = 5" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q4
Logic, Set 12: When is "if ... then" false? · MAIN · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If it is a weekend day, the library opens at noon." Today the library opened at 9 a.m. What follows?

A. The rule was broken.
B. Today is a weekend day.
C. Today is not a weekend day.
D. Nothing follows.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q5
Logic, Set 12: When is "if ... then" false? · MAIN · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In "If the sun sets, the street lights turn on," what is the "if" part (the hypothesis)?

A. the street lights turn on
B. the lights are bright
C. it is night
D. the sun sets

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q1
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In "Students may enter the lab only if they wear goggles," which is the "then" part when it is written as "If ..., then ..."?

A. they wear goggles
B. students enter the lab
C. they leave the lab
D. the lab is open

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q2
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Rewrite "Every multiple of 10 ends in 0" as a conditional.

A. If a number is not a multiple of 10, then it does not end in 0.
B. If a number ends in 0, then it is a multiple of 10.
C. If a number is a multiple of 10, then it ends in 0.
D. A number ends in 0 and is a multiple of 10.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q3
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In "If x = 3, then x + 1 = 4," the hypothesis is "x + 1 = 4".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q4
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In "The alarm rings whenever the door opens," which part is the hypothesis?

A. the door opens
B. the alarm rings
C. the door closes
D. the alarm is loud

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q5
Logic, Set 12: When is "if ... then" false? · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"6 is a factor of 30" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q1
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

"4 squared is 8" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q2
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

"Every square is a triangle" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q3
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

Which statement is true?

A. -5 is greater than 2.
B. -5 is less than 2.
C. 2 is less than -5.
D. -5 equals 5.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q4
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

"1 is a prime number" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B2-Q5
Logic, Set 12: When is "if ... then" false? · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP6

Find a real number x that makes "If x squared is greater than 9, then x is greater than 3" FALSE.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q1
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If a whole number is divisible by 6, then it is divisible by 3" is true for every whole number.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q2
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

For how many whole numbers n from 1 to 12 is "If n is even, then n is a multiple of 4" FALSE?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q3
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

P is false. Which statement MUST be true?

A. If Q then P
B. P and Q
C. Q
D. If P then Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q4
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If you score above 90, you earn a badge." Lin earned a badge. What follows?

A. The rule was broken.
B. Lin scored 90 or below.
C. Nothing about Lin's score follows.
D. Lin scored above 90.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q5
Logic, Set 12: When is "if ... then" false? · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Complete the truth table for "if P then (if Q then R)".

Fill in T or F in every empty cell.

P	Q	R	if Q then R	if P then (if Q then R)
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q1
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

In how many of the 8 rows of a truth table for P, Q, R is "(if P then Q) and (if Q then R)" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q2
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If P then (if Q then R)" has the same truth value as "if (P and Q) then R" in every row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q3
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

"If P then (if Q then R)" has the same truth value as "if (if P then Q) then R" in every row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q4
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Find whole numbers a and b that make the statement "If a + b is even, then a and b are both even" FALSE.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q5
Logic, Set 12: When is "if ... then" false? · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP6

Set 13: Necessary and sufficient conditions

Local outcome LOGIC.CONDITIONAL.02: Translate "only if", "unless", "necessary" and "sufficient" into conditionals

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Translate "only if", "unless", "necessary" and "sufficient"
BACKWARD 1: Write a statement as "if ..., then ..."
BACKWARD 2: Necessary means required; sufficient means enough
FORWARD 1: ADVANCED APPLICATION: Combine conditions
FORWARD 2: EXPERT CHALLENGE: Chains of conditions

"Being at least 16 years old is NECESSARY for getting a driving permit in this town." Which conditional says the same thing?

A. If a person has a driving permit, then the person is at least 16.
B. If a person is at least 16, then the person has a driving permit.
C. If a person is under 16, then the person may still get a permit.
D. A person is at least 16 and has a permit.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q1
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q1/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

"You may enter the museum only if you have a ticket." Which statement means the same?

A. If you have a ticket, then you may enter the museum.
B. You have a ticket and you enter.
C. If you have no ticket, then you may still enter.
D. If you enter the museum, then you have a ticket.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q2
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q2/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

In logic, "P unless Q" means "if not Q, then P". Which statement means the same as "The picnic is held unless it rains"?

A. If it rains, the picnic is held.
B. If it does not rain, the picnic is held.
C. If the picnic is held, it does not rain.
D. If it does not rain, the picnic is not held.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q3
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q3/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Being a square is a NECESSARY condition for being a rectangle.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q4
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q4/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

n is a whole number. Which condition is NECESSARY for "n is divisible by 6"?

A. n is divisible by 12
B. n is divisible by 9
C. n is even
D. n ends in the digit 6

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q5
Logic, Set 13: Necessary and sufficient conditions · MAIN · Q5/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Rewrite "All birds have feathers" as a conditional.

A. If an animal is a bird, then it has feathers.
B. If an animal is not a bird, then it has no feathers.
C. Birds and feathers.
D. If an animal has feathers, then it is a bird.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q1
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

Rewrite "When the bell rings, class ends" as a conditional.

A. If class ends, then the bell rings.
B. The bell rings and class ends.
C. If the bell does not ring, class does not end.
D. If the bell rings, then class ends.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q2
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

"If x = 2, then x + x = 4" has hypothesis "x + x = 4".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q3
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

What is the conclusion of "If a number ends in 5, it is divisible by 5"?

A. it is 5
B. it is divisible by 5
C. a number ends in 5
D. it is odd

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q4
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

"Every even number is divisible by 2" can be written "If a number is even, then it is divisible by 2".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q5
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

Having fuel is the only thing a gas car needs in order to run.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q1
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Having fuel is sufficient for a gas car to run.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q2
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Which is SUFFICIENT for a whole number to be even?

A. It ends in 1.
B. It is a multiple of 10.
C. It is a multiple of 3.
D. It is greater than 2.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q3
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

If A is sufficient for B, then whenever B happens, A happens.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q4
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Being a mammal is ___ for being a dog.

A. necessary
B. both necessary and sufficient
C. sufficient
D. neither necessary nor sufficient

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B2-Q5
Logic, Set 13: Necessary and sufficient conditions · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

"n is even" is necessary for "n is divisible by 6". Give a whole number n that shows it is NOT sufficient: n is even but not divisible by 6.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q1
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

n is a whole number. Which condition is SUFFICIENT for "n is divisible by 4"?

A. n is greater than 4
B. n is even
C. n is divisible by 2 and by 3
D. n is divisible by 8

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q2
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

"A triangle is equilateral only if it is isosceles (has at least two equal sides)." Which statement is correct?

A. Being equilateral is necessary for being isosceles.
B. Being isosceles is sufficient for being equilateral.
C. Being isosceles is necessary for being equilateral.
D. The two conditions mean the same thing.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q3
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

"You get the discount unless you pay late." Using "P unless Q" = "if not Q, then P", which person is guaranteed the discount?

A. no one
B. everyone
C. someone who paid late
D. someone who did not pay late

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q4
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

A condition can be both necessary and sufficient for another.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q5
Logic, Set 13: Necessary and sufficient conditions · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Facts: A is sufficient for B. B is sufficient for C. Which statement must be true?

A. C is necessary for A.
B. B is sufficient for A.
C. C is sufficient for A.
D. A is necessary for C.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q1
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

For whole numbers n from 1 to 60, condition X: "n is divisible by 4"; condition Y: "n is divisible by 6". How many n satisfy Y but not X (showing X is not necessary for Y)?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q2
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Which condition on a whole number n is BOTH necessary and sufficient for "n is divisible by 10"?

A. n is divisible by 5

B. n is divisible by 20

C. n is divisible by 2 and by 5

D. n is even

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q3
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

If A is necessary for B and B is necessary for A, then A and B are always both true or both false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q4
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Give a whole number n for which "n is a multiple of 3" holds but "n is a multiple of 9" fails, showing that being a multiple of 3 is not sufficient for being a multiple of 9.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q5
Logic, Set 13: Necessary and sufficient conditions · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONDITIONAL.02 (local) · MP6, MP3

Set 14: Logic grid puzzles

Local outcome LOGIC.GRID.01: Solve a matching puzzle with a grid by combining clues until one assignment remains

Puzzle set · genre: logic grids · MP1, MP6

MAIN: Solve a matching puzzle by combining clues
BACKWARD 1: Strategy: mark X for "no" and O for "yes" in a grid
FORWARD 1: ADVANCED APPLICATION: Grids with relative clues
FORWARD 2: EXPERT CHALLENGE: Two-attribute grids and decisive clues

Three students, Ama, Bex and Cho, each play one instrument: drums, flute or violin (all different). Ama does not play the flute. Bex plays neither the drums nor the flute. Which instrument does each student play?

Choose one column for each row. Each column is used once.

	drums	flute	violin
Ama			
Bex			
Cho			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q1
Logic, Set 14: Logic grid puzzles · MAIN · Q1/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four friends, Dara, Eli, Faye and Gus, each ordered a different drink: cocoa, juice, milk or tea. Dara did not order a hot drink (cocoa and tea are hot). Eli ordered tea. Faye ordered neither cocoa nor juice. Who ordered what?

Choose one column for each row. Each column is used once.

	cocoa	**juice**	**milk**	**tea**
Dara				
Eli				
Faye				
Gus				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q2
Logic, Set 14: Logic grid puzzles · MAIN · Q2/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four runners wore bibs 1, 2, 3 and 4 (all different): Hal, Ivy, Jun and Kit. Ivy's number is twice Hal's. Hal's number is even. Jun's number is odd. Kit's number is greater than Jun's. What number did each runner wear?

Choose one column for each row. Each column is used once.

	1	2	3	4
Hal				
Ivy				
Jun				
Kit				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q3
Logic, Set 14: Logic grid puzzles · MAIN · Q3/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three pets, a cat, a dog and a rabbit, belong to Lea, Max and Noa (one each). The cat's owner sits next to Max. Lea owns neither the dog nor the cat. Noa does not own the dog. Who owns which pet?

Choose one column for each row. Each column is used once.

	cat	dog	rabbit
Lea			
Max			
Noa			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q4
Logic, Set 14: Logic grid puzzles · MAIN · Q4/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four lockers, numbered 1 to 4 in a row, belong to Oz, Pia, Quin and Rae (one each). Pia's locker is directly to the right of Oz's (one number higher). Rae's locker is at an end of the row. Quin's locker number is even. Rae's number is lower than Quin's. Oz's locker is not number 2. Who has which locker?

Choose one column for each row. Each column is used once.

	1	2	3	4
Oz				
Pia				
Quin				
Rae				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q5
Logic, Set 14: Logic grid puzzles · MAIN · Q5/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Two friends, Ann and Bo, each pick a different color: red or blue. Ann does not pick red. Who picks which color?

Choose one column for each row. Each column is used once.

	red	blue
Ann		
Bo		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q1
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

In a 3 by 3 grid (3 people, 3 jobs, one each), a person's row has X in two cells. What follows?

A. Nothing follows yet.
B. The person has the job in the third cell.
C. The person has no job.
D. Another person has that job.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q2
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

In a matching puzzle where each option is used once, if Kim has the bike, no one else has the bike.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q3
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three cousins, Cy, Di and Ed, are ages 8, 9 and 10 (all different). Cy is the youngest. Ed is not 10. How old is each?

Choose one column for each row. Each column is used once.

	8	9	10
Cy			
Di			
Ed			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q4
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three people and three snacks, one each. Gil does not have the apple or the pear. Which snack does Gil have, if the snacks are apple, pear and plum?

A. apple
B. it cannot be known
C. pear
D. plum

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q5
Logic, Set 14: Logic grid puzzles · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four friends, Ari, Bea, Cal and Dev, finished a quiz with scores 6, 7, 8 and 9 (all different). Bea scored 1 more than Dev. Ari scored the most. Cal did not score 6. What did each score?

Choose one column for each row. Each column is used once.

	6	7	8	9
Ari				
Bea				
Cal				
Dev				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q1
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Five boxes in a row, positions 1 to 5, hold a hat, a kite, a lamp, a mug and a net (one each). The kite is in an even position. The lamp is directly left of the mug. The hat is at position 5. The net is not next to the kite. Which item is in each position?

Choose one column for each row. Each column is used once.

	hat	kite	lamp	mug	net
1					
2					
3					
4					
5					

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q2
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Rae, Sol and Tom own a cat, a dog and a fish, one each. Clue 1: Rae does not own the cat. Clue 2: Sol does not own the dog. These two clues allow three matchings. Which additional clue leaves exactly one?

A. Rae does not own the fish.

B. Tom owns the cat.

C. Tom does not own the dog.

D. Sol does not own the fish.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q3
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three students, Uma, Vik and Wyn, each chose a different sport (golf, judo or swimming) and a different day (Monday, Tuesday, Wednesday). The judo student trains on Monday. Uma trains on Wednesday. Uma does not play golf. Wyn trains on Tuesday. Which sport did each choose?

Choose one column for each row. Each column is used once.

	golf	judo	swimming
Uma			
Vik			
Wyn			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q4
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

In a matching puzzle with 4 people and 4 options (one each), if three people's options are known, the fourth person's option is also known.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q5
Logic, Set 14: Logic grid puzzles · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four students, Al, Bo, Cy and Di, each have a different favorite number from 1 to 4 and a different color (red, blue, green, gold). The red student's number is 1 more than the blue student's number. Al is gold. Bo's number is 4. Cy's number is twice Di's number. What is each student's number?

Choose one column for each row. Each column is used once.

	1	2	3	4
Al				
Bo				
Cy				
Di				

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q1
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Three people wear a red, a blue and a green hat, one each, so there are 6 possible matchings. Clues have the form "person X does not wear color Y". What is the smallest number of such clues that can leave exactly one matching?

A. 3
B. 4
C. 1
D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q2
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Four people each get a different one of 4 prizes. How many matchings are possible if one person, Ash, cannot get prize 1 or prize 2?

Matchings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q3
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

In a 3-person, 3-option matching puzzle, the clues "A not x" and "B not x" together force C to have x.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q4
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Five players wear the numbers 1 to 5 (all different). Al's number plus Bo's number is 9. Give one possible pair: Al's number a and Bo's number b.

Al a = _____

Bo b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q5
Logic, Set 14: Logic grid puzzles · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.GRID.01 (local) · MP1, MP6

Set 15: Converse, inverse and contrapositive

Local outcome LOGIC.CONDITIONAL.03: Form the converse, inverse and contrapositive and decide which are equivalent to the original

Formal logic set · genre: formal logic · MP3, MP7

MAIN: Form the converse, inverse and contrapositive and judge equivalence
BACKWARD 1: Negate each part
BACKWARD 2: Hypothesis and conclusion
FORWARD 1: ADVANCED APPLICATION: Which forms are equivalent?
FORWARD 2: EXPERT CHALLENGE: Converses that hold and fail

What is the CONTRAPOSITIVE of "If a shape is a triangle, then it has three sides"?

A. A shape is a triangle and does not have three sides.
B. If a shape is not a triangle, then it does not have three sides.
C. If a shape does not have three sides, then it is not a triangle.
D. If a shape has three sides, then it is a triangle.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q1
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q1/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

Which statement is logically EQUIVALENT to "If n is a multiple of 4, then n is even"?

A. If n is even, then n is a multiple of 4.
B. n is a multiple of 4 and n is even.
C. If n is not a multiple of 4, then n is not even.
D. If n is not even, then n is a multiple of 4.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q2
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q2/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

The converse of "If a whole number is a multiple of 6, then it is even" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q3
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q3/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

The statement "If x > 2, then x squared > 4" is true. Find a real number x that shows its CONVERSE is false.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q4
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q4/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

"If P, then Q" is true. Which statement MUST also be true?

A. If not Q, then not P.
B. P and Q.
C. If Q, then P.
D. If not P, then not Q.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q5
Logic, Set 15: Converse, inverse and contrapositive · MAIN · Q5/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

What is the negation of "the shape has three sides"?

A. the shape has no sides
B. the shape has at least three sides
C. the shape does not have three sides
D. the shape has four sides

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q1
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

What is the negation of "n is even" for a whole number n?

A. n is zero
B. n is odd
C. n is prime
D. n is negative

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q2
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

The negation of "x > 3" is "x < 3".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q3
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

What is the negation of "it is raining"?

A. it is not raining
B. it will rain
C. it is sunny
D. it rained yesterday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q4
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

"Not (not P)" says the same as "not P".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q5
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP3, MP7

In "If it snows, school is closed," what is the conclusion?

A. school is open
B. school is closed
C. it snows
D. it is cold

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q1
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

Swap the parts of "If P, then Q". What do you get?

A. If not Q, then not P.

B. P and Q.

C. If not P, then not Q.

D. If Q, then P.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q2
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

In "If x = 5, then x + 1 = 6," the conclusion is "x = 5".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q3
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

Which statement has hypothesis "a figure is a square"?

A. Every rectangle is a square.
B. A figure is a rectangle only if it has four sides.
C. If a figure is a square, then it is a rectangle.
D. If a figure is a rectangle, then it is a square.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q4
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

Swapping and negating both parts of "If P, then Q" gives "If not Q, then not P".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B2-Q5
Logic, Set 15: Converse, inverse and contrapositive · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3, MP7

Complete the truth table for "if P then Q" and its contrapositive "if not Q then not P".

Fill in T or F in every empty cell.

P	Q	if P then Q	if not Q then not P
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q1
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

Which two statements are ALWAYS equivalent to each other?

A. the converse and the inverse
B. the original and the converse
C. the original and the inverse
D. the converse and the contrapositive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q2
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

For how many whole numbers n from 1 to 20 is the INVERSE of "If n is a multiple of 4, then n is even" false?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q3
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

If a conditional is false, its contrapositive is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q4
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

"If a quadrilateral is a square, then its diagonals are equal." Which related statement is FALSE?

A. the contrapositive: if the diagonals are not equal, it is not a square

B. the converse: if the diagonals are equal, the quadrilateral is a square

C. the original statement

D. "a square has equal diagonals"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q5
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

For whole numbers, both "If n is odd, then n squared is odd" and its converse are true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q1
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

The statement "If n is prime and greater than 2, then n is odd" is true. Its converse is "If n is odd, then n is prime and greater than 2". Give a whole number n that shows the converse is false.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q2
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

A statement and its converse are both true. Which statement must be true?

A. P or Q

B. not P

C. P if and only if Q

D. P and Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q3
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

In how many of the 4 rows of a truth table do "if P then Q" and its converse "if Q then P" have DIFFERENT truth values?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q4
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

C is the conditional "if P then Q". Which statement is correct?

A. If C is false, its converse is false.

B. If C is true, its contrapositive may be false.

C. If C is true, its converse is true.

D. If C is true, its inverse may be true or false.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q5
Logic, Set 15: Converse, inverse and contrapositive · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP3, MP7

Set 16: Biconditionals: "if and only if"

Local outcome LOGIC.CONDITIONAL.04: Find truth values of biconditionals and use them as definitions

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Find truth values of "if and only if" statements and use them as definitions
BACKWARD 1: Conditionals first
BACKWARD 2: The converse
FORWARD 1: ADVANCED APPLICATION: Biconditionals as definitions and tests
FORWARD 2: EXPERT CHALLENGE: Chains and opposites of biconditionals

Complete the truth table for "P if and only if Q".

Fill in T or F in every empty cell.

P	Q	P if and only if Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q1
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q1/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

For real numbers x, "x squared = 9 if and only if x = 3" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q2
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q2/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Find a real number x that makes "x > 5 if and only if x > 3" FALSE.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q3
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q3/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Which statement is a correct DEFINITION of an even whole number?

A. A whole number is even only if it is greater than 1.
B. A whole number is even if and only if it is divisible by 2.
C. A whole number is even if and only if it ends in 0.
D. A whole number is even if it is divisible by 4.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q4
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q4/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"P if and only if Q" is true and P is false. What follows?

A. Nothing about Q.
B. Q is true.
C. Q is false.
D. P and Q are both true.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q5
Logic, Set 16: Biconditionals: "if and only if" · MAIN · Q5/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"If 2 > 3, then 4 > 5" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q1
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

"If 3 > 2, then 4 > 5" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q2
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

When is "if P then Q" false?

A. when P is true and Q is false
B. when both are false
C. when Q is true
D. when P is false

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q3
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

"If 4 > 1, then 2 > 6" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q4
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

In how many of the 4 rows is "if P then Q" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q5
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP6, MP3

What is the converse of "If it is a dog, then it barks"?

A. It is a barking dog.
B. If it is not a dog, then it does not bark.
C. If it does not bark, then it is not a dog.
D. If it barks, then it is a dog.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q1
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

The converse of "If n = 2, then n is even" is "If n is not 2, then n is not even".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q2
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

The converse of a true statement is always true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q3
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

"P if and only if Q" means "if P then Q" AND what?

A. if Q then P
B. P or Q
C. if not P then Q
D. not P and not Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q4
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

If a conditional is true but its converse is false, the biconditional is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B2-Q5
Logic, Set 16: Biconditionals: "if and only if" · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.03 (local) · MP6, MP3

For how many whole numbers n from 1 to 20 is "n is even if and only if n is a multiple of 4" TRUE?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q1
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Complete the table to compare "P if and only if Q" with "(if P then Q) and (if Q then P)".

Fill in T or F in every empty cell.

P	Q	P if and only if Q	(if P then Q) and (if Q then P)
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q2
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Which definition of a SQUARE is correct (works in both directions)?

A. A quadrilateral is a square if and only if its diagonals are equal.
B. A quadrilateral is a square if and only if it has four equal sides and four right angles.
C. A quadrilateral is a square if and only if it has four right angles.
D. A quadrilateral is a square if and only if it has four equal sides.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q3
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"A whole number is divisible by 10 if and only if it ends in 0" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q4
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Find a whole number n that makes "n is divisible by 3 if and only if n is odd" FALSE.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q5
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"P exclusive-or Q" has the opposite truth value of "P if and only if Q" in every row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q1
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

"P iff Q" and "Q iff R" are both true. Which must be true?

A. P exclusive-or R
B. P and R
C. P iff R
D. not Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q2
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

In how many of the 8 rows of a truth table for P, Q, R is "(P iff Q) iff R" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q3
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Find whole numbers a and b, each from 1 to 9, that make "a + b is even if and only if a is even" TRUE and a different from b.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q4
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Which statement about whole numbers is TRUE?

A. n is even if and only if n is a multiple of 4.
B. n is a multiple of 3 if and only if n is a multiple of 9.
C. n is odd if and only if n is prime.
D. n is odd if and only if n + 1 is even.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q5
Logic, Set 16: Biconditionals: "if and only if" · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONDITIONAL.04 (local) · MP6, MP3

Set 17: Truth-tellers and liars

Local outcome LOGIC.KNIGHTS.01: Find who tells the truth when each speaker either always tells the truth or always lies

Puzzle set · genre: truth-teller puzzles · MP1, MP3

MAIN: Decide who tells the truth when each speaker always tells the truth or always lies
BACKWARD 1: Strategy: suppose, then test
FORWARD 1: ADVANCED APPLICATION: Statements about counts
FORWARD 2: EXPERT CHALLENGE: Larger systems and impossible statements

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Ana says: "Ben is a liar." Ben says: "Ana and I are both truth-tellers." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Ana		
Ben		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q1
Logic, Set 17: Truth-tellers and liars · MAIN · Q1/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Cal says: "Dee and I are different kinds." Dee says: "Cal is a liar." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Cal		
Dee		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q2
Logic, Set 17: Truth-tellers and liars · MAIN · Q2/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Eve says: "Fin and Gia are both liars." Fin says: "Eve is a truth-teller." Gia says: "Fin is a liar." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Eve		
Fin		
Gia		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q3
Logic, Set 17: Truth-tellers and liars · MAIN · Q3/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Jo says: "Kai is a liar." Kai says: "Lu is a liar." Lu says: "Jo and Kai are both liars." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Jo		
Kai		
Lu		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q4
Logic, Set 17: Truth-tellers and liars · MAIN · Q4/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Mo says: "Exactly one of Ned and Ola is a liar." Ned says: "Mo is a liar." Ola says: "Ned and I are the same kind." Who is a liar?

A. Ned only
B. Mo only
C. Mo and Ned
D. Ola only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q5
Logic, Set 17: Truth-tellers and liars · MAIN · Q5/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A truth-teller can say "I am a liar."

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q1
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A liar can say "I am a truth-teller."

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q2
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Vic says: "2 + 2 = 5." What is Vic?

A. either
B. neither
C. a liar
D. a truth-teller

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q3
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Wes is a truth-teller and says: "Xia is a liar." What is Xia?

A. neither
B. a truth-teller
C. either
D. a liar

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q4
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Yan is a liar and says: "Zed is a liar." What is Zed?

A. a truth-teller
B. a liar
C. either
D. neither

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q5
Logic, Set 17: Truth-tellers and liars · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Ash says: "Exactly one of us three is a truth-teller." Bly says: "Exactly two of us three are truth-tellers." Cru says: "None of us three is a truth-teller." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Ash		
Bly		
Cru		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q1
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Dov says: "At least one of Ela and Fyn is a liar." Ela says: "Dov and Fyn are both truth-tellers." Fyn says: "Dov is a liar." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Dov		
Ela		
Fyn		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q2
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Pat says: "Exactly two of us three are truth-tellers." Quy says: "Pat is a liar." Ren says: "Quy and Pat are the same kind." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Pat		
Quy		
Ren		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q3
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. If two people each say "The other one is a liar," then exactly one of them is a liar.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q4
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Four people each say exactly the same sentence: "Exactly one of us four is a truth-teller." How many of them are truth-tellers?

Truth-tellers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q5
Logic, Set 17: Truth-tellers and liars · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Uri says: "Exactly three of us four are liars." Vee says: "Uri is a truth-teller." Wil says: "Vee is a liar." Xan says: "Wil is a truth-teller and Uri is a liar." Mark each person as a truth-teller or a liar.

Choose one column for each row.

	truth-teller	liar
Uri		
Vee		
Wil		
Xan		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q1
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Which statement can NO person on the island say?

A. "I am a liar."
B. "2 + 2 = 5."
C. "2 + 2 = 4."
D. "I am a truth-teller."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q2
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. Five people stand in a circle. Each says: "The person on my left is a liar." In how many ways can the five be truth-tellers and liars so that every statement fits?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q3
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. A person says: "I am a liar or 2 + 2 = 4." This person must be a truth-teller.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q4
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

On this island every person is either a truth-teller, who always tells the truth, or a liar, who always lies. You meet Gil and Hal. Gil says: "At least one of us two is a truth-teller." Hal says nothing. Which statement MUST be true?

A. Hal is a liar.
B. Gil is a truth-teller.
C. Both are liars.
D. If Hal is a truth-teller, then Gil is a truth-teller.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q5
Logic, Set 17: Truth-tellers and liars · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.KNIGHTS.01 (local) · MP1, MP3

Set 18: Truth tables

Local outcome LOGIC.TRUTH.01: Complete truth tables for compound statements

Formal logic set · genre: formal logic · MP6, MP8

- MAIN: Complete truth tables for compound statements
- BACKWARD 1: The basic connectives
- BACKWARD 2: Work from the inside out
- FORWARD 1: ADVANCED APPLICATION: Three-variable tables
- FORWARD 2: EXPERT CHALLENGE: Read and build statements from tables

Complete the truth table for "P and (not Q)".

Fill in T or F in every empty cell.

P	Q	not Q	P and (not Q)
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q1
Logic, Set 18: Truth tables · MAIN · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(not P) or Q".

Fill in T or F in every empty cell.

P	Q	not P	(not P) or Q
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q2
Logic, Set 18: Truth tables · MAIN · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(P or Q) and not (P and Q)".

Fill in T or F in every empty cell.

P	Q	P or Q	P and Q	(P or Q) and not (P and Q)
T	T			
T	F			
F	T			
F	F			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q3
Logic, Set 18: Truth tables · MAIN · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

A truth table for three statements P, Q and R lists every combination of truth values. How many rows does it have?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q4
Logic, Set 18: Truth tables · MAIN · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(P and Q) or R".

Fill in T or F in every empty cell.

P	Q	R	P and Q	(P and Q) or R
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q5
Logic, Set 18: Truth tables · MAIN · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "P and Q".

Fill in T or F in every empty cell.

P	Q	P and Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q1
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

Complete the truth table for "P or Q" (inclusive).

Fill in T or F in every empty cell.

P	Q	P or Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q2
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

Complete the truth table for "not P".

Fill in T or F in every empty cell.

P	not P
T	
F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q3
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

"P and Q" is true when P is true and Q is false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q4
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

"P or Q" is true when P is false and Q is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q5
Logic, Set 18: Truth tables · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP6, MP8

In "not (P and Q)", which part is computed first?

A. not P
B. not Q
C. P and Q
D. P or Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q1
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

"not (P and Q)" and "(not P) and Q" always have the same truth value.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q2
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

P is true and Q is false. Is "not (P and Q)" true or false? Enter 1 for true or 0 for false.

Value (1 or 0): _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q3
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

P is false, Q is true. What is the value of "(P or Q) and P"?

A. true
B. false
C. both
D. it cannot be found

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q4
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "not (P or Q)".

Fill in T or F in every empty cell.

P	Q	P or Q	not (P or Q)
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B2-Q5
Logic, Set 18: Truth tables · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "P and (Q or R)".

Fill in T or F in every empty cell.

P	Q	R	Q or R	P and (Q or R)
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q1
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(P or Q) and (not R)".

Fill in T or F in every empty cell.

P	Q	R	P or Q	(P or Q) and (not R)
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q2
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

In how many of the 8 rows is "P or (Q and R)" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q3
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

A statement in P and Q is true only in the row P true, Q false. Which statement is it?

A. P and (not Q)
B. not (P and Q)
C. P or (not Q)
D. (not P) and Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q4
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

A truth table for 4 statements has 16 rows.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q5
Logic, Set 18: Truth tables · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

A statement in P, Q, R is true in exactly the rows where an odd number of P, Q, R are true. Which statement is it?

A. P or Q or R
B. P and Q and R
C. P exclusive-or (Q exclusive-or R)
D. (P and Q) or R

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q1
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

How many different possible output columns can a truth table with two statement letters P and Q have (4 rows, each output T or F)?

Columns: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q2
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Complete the truth table for "(P and not Q) or (not P and Q)".

Fill in T or F in every empty cell.

P	Q	P and not Q	not P and Q	(P and not Q) or (not P and Q)
T	T			
T	F			
F	T			
F	F			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q3
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

In how many of the 16 rows of a table for P, Q, R, S is "(P and Q) or (R and S)" true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q4
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Every truth-table column for P and Q can be produced by some statement using only "not", "and", "or".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q5
Logic, Set 18: Truth tables · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP6, MP8

Set 19: Tautology, contradiction and equivalence

Local outcome LOGIC.TRUTH.02: Classify compound statements as tautologies, contradictions or contingencies and test equivalence with a table

Formal logic set · genre: formal logic · MP7, MP3

MAIN: Classify statements as tautologies, contradictions or contingencies, and test equivalence
BACKWARD 1: Fill in truth tables
BACKWARD 2: Always, never, sometimes
FORWARD 1: ADVANCED APPLICATION: Equivalences by truth table
FORWARD 2: EXPERT CHALLENGE: Laws of logic

Which statement is a TAUTOLOGY (true in every row of its truth table)?

A. P or Q

B. (P and Q) or not P or not Q

C. P and Q and not Q

D. if P then Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q1
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q1/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Which statement is a CONTRADICTION (false in every row)?

A. P or not P

B. P or Q

C. if P then Q

D. P and Q and not Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q2
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q2/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Complete the truth table for "(if P then Q) or (if Q then P)".

Fill in T or F in every empty cell.

P	Q	if P then Q	if Q then P	(if P then Q) or (if Q then P)
T	T			
T	F			
F	T			
F	F			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q3
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q3/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"not (P and Q)" is logically equivalent to "(not P) or (not Q)".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q4
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q4/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

How is "if P then Q" classified?

A. a contingency: true in some rows and false in one

B. a contradiction

C. neither true nor false

D. a tautology

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q5
Logic, Set 19: Tautology, contradiction and equivalence · MAIN · Q5/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Complete the truth table for "P or not P".

Fill in T or F in every empty cell.

P	not P	P or not P
T		
F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q1
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

Complete the truth table for "P and not P".

Fill in T or F in every empty cell.

P	not P	P and not P
T		
F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q2
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

Complete the truth table for "not P or Q".

Fill in T or F in every empty cell.

P	Q	not P or Q
T	T	
T	F	
F	T	
F	F	

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q3
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

How many rows does a truth table for P and Q have?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q4
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

A truth table for three statements P, Q and R has 6 rows.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q5
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.TRUTH.01 (local) · MP7, MP3

A statement is true in every row of its table. What is it called?

A. a contingency
B. a tautology
C. a premise
D. a contradiction

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q1
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

A statement is false in every row. What is it called?

A. a tautology
B. a contingency
C. a contradiction
D. a conclusion

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q2
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

A contingency is true in every row of its truth table.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q3
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

A column reads T, T, F, T. How is the statement classified?

A. it cannot be told
B. a contradiction
C. a tautology
D. a contingency

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q4
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"7 is odd and 7 is not odd" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B2-Q5
Logic, Set 19: Tautology, contradiction and equivalence · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Complete the table to compare "not (P or Q)" with "(not P) and (not Q)".

Fill in T or F in every empty cell.

P	Q	not (P or Q)	(not P) and (not Q)
T	T		
T	F		
F	T		
F	F		

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q1
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"if P then Q" is equivalent to "if Q then P".

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q2
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Which statement is equivalent to "not (if P then Q)"?

A. not P or Q

B. P and not Q

C. not P and not Q

D. if not P then not Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q3
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

How many of these are tautologies? "P or not P"; "P and not P"; "if P then Q"; "(if P then Q) or (if Q then P)"; "P iff not P".

Tautologies: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q4
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

The negation of a tautology is also a tautology.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q5
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Which statement is equivalent to "P or (P and Q)"?

A. P
B. P and Q
C. P or Q
D. Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q1
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"((if P then Q) and P) implies Q", written "if ((if P then Q) and P) then Q", is a tautology.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q2
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

"if ((if P then Q) and Q) then P" is a tautology.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q3
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Of the 16 possible truth-table columns for statements in P and Q, how many are contingencies?

Contingencies: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q4
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Which pair are equivalent?

A. "not (P and Q)" and "not P and not Q"
B. "P and (Q or R)" and "(P and Q) or R"
C. "if (P or Q) then R" and "(if P then R) or (if Q then R)"
D. "if P then (Q and R)" and "(if P then Q) and (if P then R)"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q5
Logic, Set 19: Tautology, contradiction and equivalence · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.TRUTH.02 (local) · MP7, MP3

Set 20: What is on top? Stacking and occlusion

Local outcome LOGIC.SPATIAL.03: Infer stacking order from what is visible and predict views of a stack

Puzzle set · genre: occlusion and stacking · MP7, MP1

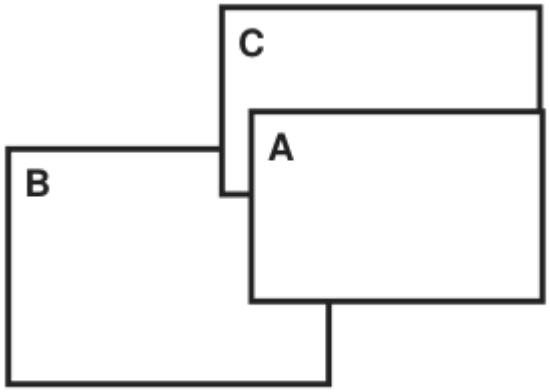
MAIN: Infer stacking order from what is visible

BACKWARD 1: Strategy: the cut-off sheet is underneath

FORWARD 1: ADVANCED APPLICATION: Larger stacks

FORWARD 2: EXPERT CHALLENGE: How much does a picture tell?

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 3 sheets: A, B, C. Put the sheets in order from the BOTTOM sheet to the TOP sheet.

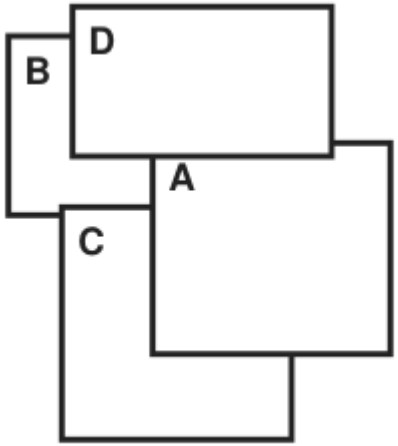


Put these in order:

A sheet A
B sheet B
C sheet C

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q1
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 4 sheets: A, B, C, D. List the sheets starting with the one at the BOTTOM and ending with the one on TOP.



Put these in order:

A sheet A
B sheet B
C sheet C
D sheet D

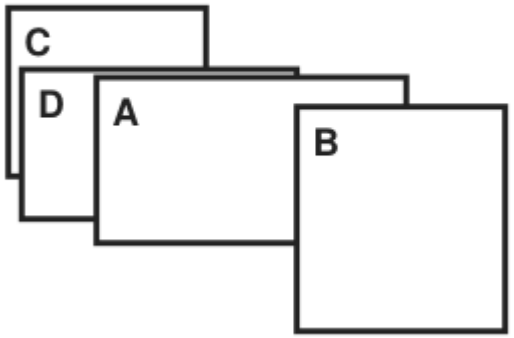
REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q2
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Four round discs are stacked with their centers lined up, from bottom to top: W (radius 5 cm), X (radius 3 cm), Y (radius 4 cm), Z (radius 1 cm). Looking straight down from above, which discs can be seen?

A. W and Z
B. Z only
C. W, Y and Z
D. W, X, Y and Z

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q3
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 4 sheets: A, B, C, D. Put the sheets in order from the BOTTOM sheet to the TOP sheet.



Put these in order:

A sheet A
B sheet B
C sheet C
D sheet D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q4
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Square cards are placed one at a time on an 8 by 8 grid, each covering a block of cells (rows and columns numbered 1 to 8; "rows 1-4, columns 1-4" means a 4 by 4 block). In order: A covers rows 1-4, columns 1-4; B covers rows 3-6, columns 3-6; C covers rows 2-5, columns 2-5; D covers rows 1-3, columns 4-7; E covers rows 5-7, columns 1-3. Seen from above, how many of the five cards can be seen at least partly?

Cards seen: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q5
Logic, Set 20: What is on top? Stacking and occlusion · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Where sheets X and Y overlap, the outline of X runs straight through and the outline of Y stops at X's edge. What follows?

A. X is under Y.

B. It cannot be told.

C. Y is under X.

D. They are side by side.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q1
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Pairs found: P under Q, Q under R, P under R. Order the sheets from bottom to top.

Put these in order:

P sheet P

Q sheet Q

R sheet R

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q2
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

A sheet whose whole outline can be seen must be on top of every sheet it overlaps.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q3
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Discs are stacked with centers lined up, from bottom to top: radius 2 cm, then 6 cm. Which can be seen from above?

A. only the 2 cm disc
B. both discs
C. neither disc
D. only the 6 cm disc

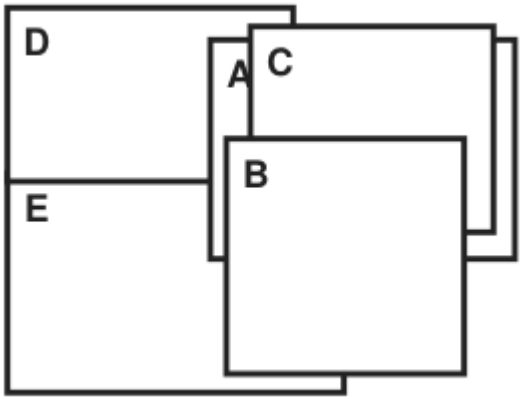
REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q4
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Five sheets are stacked. Exactly one of them is on top of all four others. How many sheets are under it?

Sheets under it: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q5
Logic, Set 20: What is on top? Stacking and occlusion · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 5 sheets: A, B, C, D, E. Put the five sheets in order from the BOTTOM sheet to the TOP sheet.

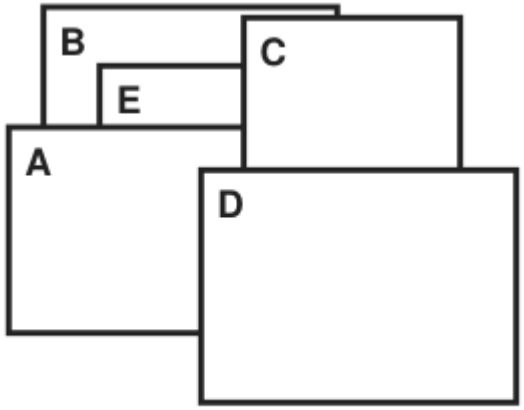


Put these in order:

A sheet A
B sheet B
C sheet C
D sheet D
E sheet E

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q1
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 5 sheets: A, B, C, D, E. List the five sheets starting with the BOTTOM sheet and ending with the TOP sheet.



Put these in order:

A sheet A
B sheet B
C sheet C
D sheet D
E sheet E

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q2
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Cards are placed one at a time on an 8 by 8 grid: A covers rows 2-3, columns 2-3; B covers rows 1-4, columns 1-4; C covers rows 4-6, columns 4-6; D covers rows 1-2, columns 5-6; E covers rows 3-6, columns 1-4. Seen from above, how many of the five cards can be seen at least partly?

Cards seen: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q3
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Four discs with centers lined up are stacked, bottom to top: radius 6, 2, 5, 3 (cm). How many can be seen from above?

A. 3
B. 1
C. 2
D. 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q4
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

If two sheets do not overlap at all, the picture from above cannot show which one is higher.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q5
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Three sheets: A and B do not overlap each other, and sheet C overlaps both and is on top of both. How many different bottom-to-top orders are consistent with the picture from above?

Orders: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q1
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Discs of radius 1, 2, 3 and 4 cm are stacked with centers lined up. In which stacking order (bottom to top) can ALL four discs be seen from above?

A. 3, 4, 2, 1
B. 4, 3, 2, 1
C. 1, 2, 3, 4
D. 4, 2, 3, 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q2
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Discs of radius 1, 2, 3 and 4 cm are stacked in some order with centers lined up. Over all 24 stacking orders, in how many is EXACTLY one disc visible from above?

Orders: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q3
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

From a picture of a stack of sheets, the TOP sheet can always be identified if every sheet overlaps at least one other sheet.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q4
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Cards on a grid: P covers rows 1-3, columns 1-3; Q covers rows 2-4, columns 2-4. P was placed first. Which cell shows card P?

A. row 1, column 2

B. row 3, column 3

C. row 2, column 2

D. row 4, column 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q5
Logic, Set 20: What is on top? Stacking and occlusion · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.03 (local) · MP7, MP1

Set 21: Valid inference forms

Local outcome LOGIC.INFERENCE.01: Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Formal logic set · genre: formal logic · MP3

MAIN: Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism
BACKWARD 1: When is a conditional true?
BACKWARD 2: Using "or" and "not"
FORWARD 1: ADVANCED APPLICATION: Longer chains of valid steps
FORWARD 2: EXPERT CHALLENGE: Valid forms with several variables

Premises: "If the reservoir is full, the dam gates open." "The reservoir is full." What conclusion follows?

A. The reservoir is not full.

B. The dam gates do not open.

C. Nothing follows.

D. The dam gates open.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q1
Logic, Set 21: Valid inference forms · MAIN · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Premises: "If a number is divisible by 9, it is divisible by 3." "The number 52 is not divisible by 3." What follows?

A. 52 is divisible by 3.
B. 52 is divisible by 9.
C. 52 is not divisible by 9.
D. Nothing follows.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q2
Logic, Set 21: Valid inference forms · MAIN · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Arrange the steps of the argument so that each line follows from earlier ones and the conclusion comes last.

Put these in order:

A If the trail is icy, the park closes.
B If the park closes, the race is cancelled.
C The trail is icy.
D So the park closes.
E So the race is cancelled.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q3
Logic, Set 21: Valid inference forms · MAIN · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Premises: "The key is in the drawer or in the coat." "The key is not in the drawer." What follows?

A. The key is in neither place.
B. The key is in the coat.
C. The key is in the drawer.
D. Nothing follows.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q4
Logic, Set 21: Valid inference forms · MAIN · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

This argument is valid: "If P then Q. If Q then R. So if P then R."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q5
Logic, Set 21: Valid inference forms · MAIN · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

"If P then Q" is false when P is false and Q is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q1
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"If P then Q" is false when P is false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q2
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"If P then Q" is true and P is true. What is Q?

A. true
B. either
C. neither
D. false

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q3
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"If P then Q" is true and Q is false. What is P?

A. neither
B. either
C. false
D. true

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q4
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"If P then Q" is true and Q is true. P must be true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q5
Logic, Set 21: Valid inference forms · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.CONDITIONAL.01 (local) · MP3

"P or Q" is true and P is true. Then Q must be false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q1
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

"P or Q" is true and P is true. Then Q is false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q2
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

"The meeting is on Monday or Tuesday." It is not on Monday. When is it?

A. Monday
B. Tuesday
C. neither day
D. it cannot be told

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q3
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

Which is the negation of "the light is on"?

A. the light is dim
B. the lamp is broken
C. the light is bright
D. the light is not on

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q4
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

"P or Q" is true and both P and Q could be true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B2-Q5
Logic, Set 21: Valid inference forms · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP3

Premises: "If it is a holiday, the bank is closed." "If the bank is closed, Mara cannot deposit." "Mara deposited." What follows?

A. It is a holiday.
B. It is not a holiday.
C. The bank is closed.
D. Nothing follows.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q1
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Valid: "P or Q. If P then R. If Q then R. So R."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q2
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Premises: "Ana or Ben will host." "If Ben hosts, the party is on Friday." "The party is not on Friday." Who hosts?

A. both
B. Ben
C. Ana
D. it cannot be told

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q3
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Arrange into a valid argument ending with the conclusion "The fountain is off."

Put these in order:

A If the fountain is on, the pump runs.

B The pump does not run.

C So the fountain is off.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q4
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Valid: "If P then Q. Not Q. So not P."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q5
Logic, Set 21: Valid inference forms · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Valid: "If P then (Q and R). Not R. So not P."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q1
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Valid: "If P then (Q or R). Not Q. So R."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q2
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Premises: "If A then B." "If C then D." "A or C." Which conclusion follows?

A. B or D
B. B and D
C. not B
D. A and C

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q3
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

An argument has premises "P or Q" and "if P then R" and conclusion "Q or R". In how many of the 8 rows of its truth table are both premises true?

Rows: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q4
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

The argument "P or Q. If P then R. So Q or R." is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q5
Logic, Set 21: Valid inference forms · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

Set 22: Tempting invalid inferences

Local outcome LOGIC.INFERENCE.02: Recognize affirming the consequent and denying the antecedent, and show why they fail

Formal logic set · genre: formal logic · MP3, MP6

- MAIN: Recognize affirming the consequent and denying the antecedent
- BACKWARD 1: Valid forms first
- BACKWARD 2: A counterexample shows invalidity
- FORWARD 1: ADVANCED APPLICATION: Spot invalid forms in disguise
- FORWARD 2: EXPERT CHALLENGE: Invalid or just unconvincing?

"If a phone is waterproof, it survives a fall into a pool. Leo's phone survived a fall into a pool. So Leo's phone is waterproof." What is wrong with this argument?

A. A premise is false.
B. It denies the antecedent.
C. Nothing: it is modus ponens.
D. It affirms the consequent: the phone may have survived for another reason.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q1
Logic, Set 22: Tempting invalid inferences · MAIN · Q1/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

"If n is a multiple of 4, then n is even. 6 is not a multiple of 4. So 6 is not even." What is wrong?

A. It denies the antecedent: not P does not give not Q.
B. Nothing: it is modus tollens.
C. It affirms the consequent.
D. The conditional is false.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q2
Logic, Set 22: Tempting invalid inferences · MAIN · Q2/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

This form is valid: "If P then Q. Q. So P."

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q3
Logic, Set 22: Tempting invalid inferences · MAIN · Q3/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Use whole numbers to show that "If n is a multiple of 6, then n is even. n is even. So n is a multiple of 6." is invalid: give a value of n that makes both premises true and the conclusion false.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q4
Logic, Set 22: Tempting invalid inferences · MAIN · Q4/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Which argument is VALID?

A. "If it rains, the game is off. The game is off. So it is raining."
B. "If it rains, the game is off. It is not raining. So the game is not off."
C. "If it rains, the game is off. So if the game is off, it rains."
D. "If it rains, the game is off. The game is not off. So it is not raining."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q5
Logic, Set 22: Tempting invalid inferences · MAIN · Q5/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

"If P then Q. P. So Q." What is this form called?

A. modus tollens
B. affirming the consequent
C. modus ponens
D. denying the antecedent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q1
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

"If P then Q. Not Q. So not P." What is this form called?

A. denying the antecedent
B. modus tollens
C. modus ponens
D. affirming the consequent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q2
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

Modus ponens is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q3
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

Modus tollens is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q4
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

In modus ponens, which part of the conditional is given as a fact?

A. the "if" part
B. neither part
C. the "then" part
D. the negation of the "if" part

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q5
Logic, Set 22: Tempting invalid inferences · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3, MP6

An argument form is invalid if there is one case with all premises true and the conclusion false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q1
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

"If x > 5, then x > 2. x > 2. So x > 5." Which value of x shows it is invalid?

A. 1
B. 6
C. 10
D. 3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q2
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

A valid argument can have true premises and a false conclusion.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q3
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

"If an animal is a robin, it has wings. A bat has wings. So a bat is a robin." Which fact shows the form is invalid?

A. Some birds cannot fly.

B. Bats fly at night.

C. A bat has wings but is not a robin.

D. Robins have wings.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q4
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

If one case with true premises has a true conclusion, the argument is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B2-Q5
Logic, Set 22: Tempting invalid inferences · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3, MP6

"The alarm sounds only if there is smoke. There is smoke. So the alarm sounds." Is this valid?

A. Yes: it is modus ponens.
B. No: "A only if S" means "if A then S"; from S, A does not follow.
C. No: the conclusion is false.
D. Yes: smoke always sets off alarms.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q1
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

"Every square is a rectangle. This shape is not a square. So it is not a rectangle." This argument is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q2
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Show that "If x > 10, then x > 3. x is not greater than 10. So x is not greater than 3." is invalid: give a number x that makes the premises true and the conclusion false.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q3
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Which argument has the SAME form as "If it is a cat, it is a mammal. It is a mammal. So it is a cat."?

A. "If n is 2, n is even. n is even. So n is 2."
B. "n is 2 or n is 3. n is not 3. So n is 2."
C. "If n is 2, n is even. n is 2. So n is even."
D. "If n is 2, n is even. n is odd. So n is not 2."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q4
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Denying the antecedent and affirming the consequent are both invalid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q5
Logic, Set 22: Tempting invalid inferences · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

"If a number is divisible by 4, it is even. 12 is even. So 12 is divisible by 4." The conclusion is true, so the argument is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q1
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

For whole numbers n from 1 to 30, the argument "If n is a multiple of 6, then n is even. n is even. So n is a multiple of 6." has true premises for every even n. For how many of these n is the conclusion false?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q2
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Which argument is INVALID, even though every premise and the conclusion are true?

A. "If 9 is even, then 9 is a whole number. 9 is even. So 9 is a whole number."
B. "9 is odd or 9 is even. 9 is not even. So 9 is odd."
C. "If 9 is odd, then 9 is a whole number. 9 is a whole number. So 9 is odd."
D. "If 9 is odd, then 9 is a whole number. 9 is odd. So 9 is a whole number."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q3
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Valid: "If P then Q. If not P then Q. So Q."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q4
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

Valid: "If P then Q. If Q then P. Not P. So not Q."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q5
Logic, Set 22: Tempting invalid inferences · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INFERENCE.02 (local) · MP3, MP6

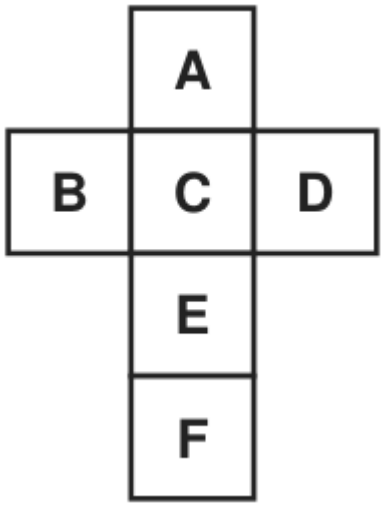
Set 23: Cube nets

Local outcome LOGIC.SPATIAL.04: Match a cube to its net: which faces are opposite and which are adjacent

Puzzle set · genre: cube nets · MP7, MP1

- MAIN: Match a cube to its net: opposite and adjacent faces
- BACKWARD 1: Strategy: skip one square to find the opposite face
- FORWARD 1: ADVANCED APPLICATION: Harder nets
- FORWARD 2: EXPERT CHALLENGE: Counting and designing nets

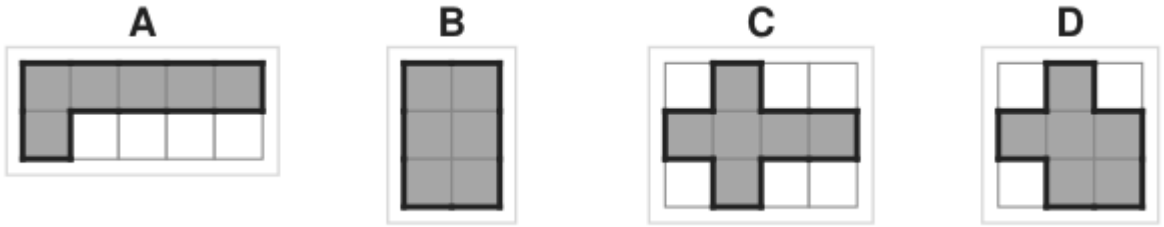
The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which letter is on the face OPPOSITE the face A?



A. C
B. E
C. B
D. F

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q1
Logic, Set 23: Cube nets · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

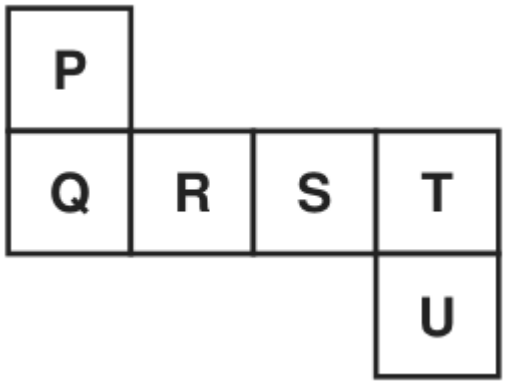
Each figure is made of six squares. Which figure can be folded along the lines between squares into a closed cube (with no overlapping squares)?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q2
Logic, Set 23: Cube nets · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. For each face P, Q and R of this net, which letter is on the opposite face?

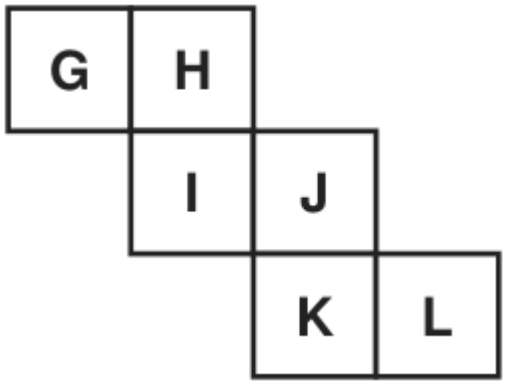


Choose one column for each row. Each column is used once.

	S	T	U
P			
Q			
R			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q3
Logic, Set 23: Cube nets · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

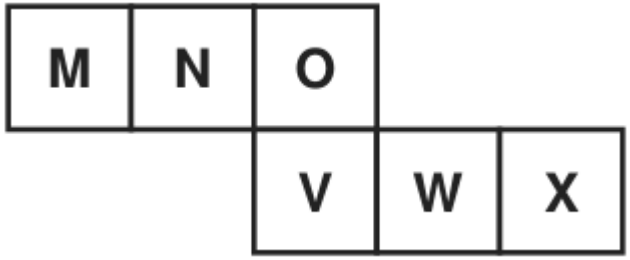
The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which two letters can NEVER be seen at the same time when you look at the cube?



A. G and H
B. H and J
C. H and I
D. G and J

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q4
Logic, Set 23: Cube nets · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. On the cube made from this net, faces M and W share an edge.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q5
Logic, Set 23: Cube nets · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

A straight row of four squares, numbered 1 to 4, is part of a cube net. Which square is opposite square 1?

A. 2
B. 3
C. 4
D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q1
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

Two squares that share an edge in a net always share an edge on the cube.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q2
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

How many pairs of opposite faces does a cube have?

Pairs: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q3
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

On a cube, how many faces share an edge with any one face?

A. 4
B. 6
C. 3
D. 5

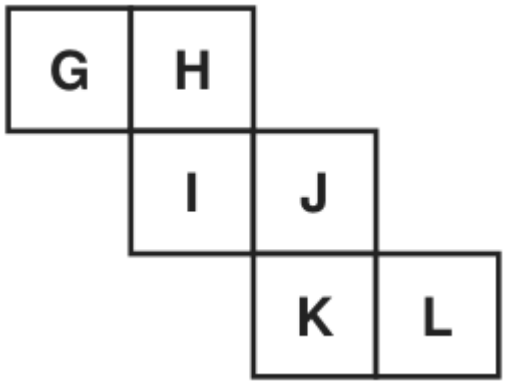
REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q4
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

Two faces that are opposite on the cube can be seen at the same time.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q5
Logic, Set 23: Cube nets · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

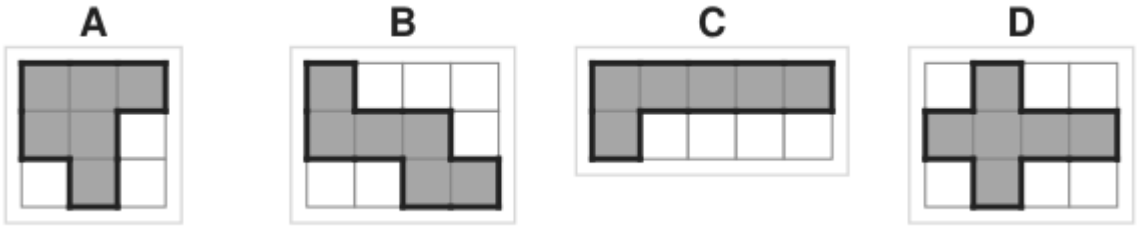
The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which letter is opposite H?



A. G
B. J
C. I
D. K

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q1
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

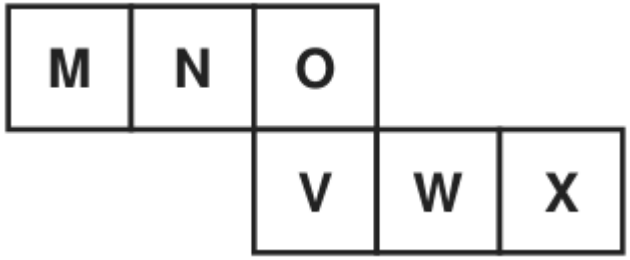
Each of the four figures is made of six squares. How many of them can be folded into a closed cube?



Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q2
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. For each face M, N and V of this net, which letter is on the opposite face?



Choose one column for each row. Each column is used once.

	O	W	X
M			
N			
V			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q3
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

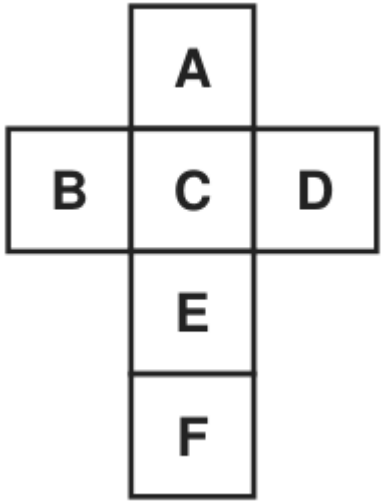
Every net of a cube has exactly 6 squares.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q4
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. In the cross-shaped net (A on top; B, C, D across; E and F below C), which two faces each share an edge with all of B, C and D?



A. B and D
B. E and F
C. A and E
D. A and F

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q5
Logic, Set 23: Cube nets · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

Count the different nets of a cube: shapes of six squares joined edge to edge that fold into a cube, where shapes that are turns or flips of each other count as the same. How many are there?

Nets: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q1
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The faces of a cube are colored so that faces sharing an edge always have different colors. What is the least number of colors needed?

Colors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q2
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

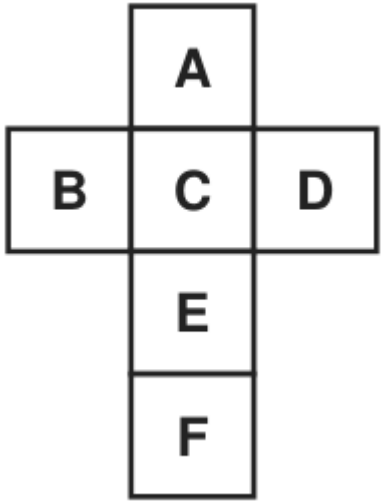
On a standard die, opposite faces add to 7. In a cross-shaped net with 1, 2, 3, 4 in the long column and 5, 6 on the two sides, the numbers can follow this rule.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q3
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. The numbers 1 to 6 are written on the cross-shaped net with A = 1, B = 2, C = 3, D = 5, E = 6 and F = 4. Which statement is true?



A. Every pair of opposite faces adds to 7.

B. Faces 3 and 4 touch.

C. Faces 2 and 6 are opposite.

D. Faces 1 and 2 are opposite.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q4
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

A cube net can contain four squares in a 2 by 2 block.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q5
Logic, Set 23: Cube nets · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.04 (local) · MP7, MP1

Set 24: Valid versus sound

Local outcome LOGIC.INFERENCE.03: Tell whether an argument is valid, and whether it is also sound

Formal logic set · genre: formal logic · MP3

MAIN: Tell whether an argument is valid, and whether it is also sound
BACKWARD 1: Is the form valid?
BACKWARD 2: Are the premises true?
FORWARD 1: ADVANCED APPLICATION: Classify harder arguments
FORWARD 2: EXPERT CHALLENGE: Validity, soundness and truth together

"All mammals breathe air. Whales are mammals. So whales breathe air." How is this argument best described?

A. sound but not valid
B. valid but not sound
C. invalid
D. valid and sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q1
Logic, Set 24: Valid versus sound · MAIN · Q1/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"All birds can swim. A sparrow is a bird. So a sparrow can swim." How is this argument best described?

A. sound but not valid
B. valid but not sound
C. valid and sound
D. invalid

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q2
Logic, Set 24: Valid versus sound · MAIN · Q2/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

An argument can be sound even if its conclusion is false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q3
Logic, Set 24: Valid versus sound · MAIN · Q3/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"If a number is divisible by 10, it is even. 14 is even. So 14 is divisible by 10." How is this described?

A. sound but not valid
B. valid but not sound
C. valid and sound
D. invalid, so it cannot be sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q4
Logic, Set 24: Valid versus sound · MAIN · Q4/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Which argument is SOUND?

A. "Every odd number is prime. 9 is odd. So 9 is prime."
B. "Every prime greater than 2 is odd. 13 is a prime greater than 2. So 13 is odd."
C. "If 13 is odd, 13 is prime. 13 is prime. So 13 is odd."
D. "Every prime is odd. 2 is prime. So 2 is odd."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q5
Logic, Set 24: Valid versus sound · MAIN · Q5/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"If P then Q. P. So Q." is a valid form.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q1
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

"If P then Q. Q. So P." is a valid form.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q2
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

A valid argument guarantees what?

A. The argument is convincing.
B. The premises are true.
C. If the premises are true, the conclusion is true.
D. The conclusion is true.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q3
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

"P or Q. P. So not Q." is a valid form.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q4
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

An argument with a true conclusion is always valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q5
Logic, Set 24: Valid versus sound · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.INFERENCE.01 (local) · MP3

"Every rectangle has four equal sides" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q1
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every rectangle is a square" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q2
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every multiple of 2 is a multiple of 4" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q3
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every prime number is odd" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q4
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Water freezes at 10 degrees Celsius at normal air pressure" is true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B2-Q5
Logic, Set 24: Valid versus sound · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.ARGUMENT.01 (local) · MP3

"Every whole number that is divisible by 4 is divisible by 8. 12 is divisible by 4. So 12 is divisible by 8." How is it described?

A. valid but not sound
B. invalid
C. sound but not valid
D. valid and sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q1
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

A valid argument with a false conclusion can have all of its premises true.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q2
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"No reptile has fur. A snake is a reptile. So a snake has no fur." How is it described?

A. sound but not valid
B. valid but not sound
C. invalid
D. valid and sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q3
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

If an argument is invalid, its conclusion must be false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q4
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Which statement is TRUE?

A. Every sound argument is valid.
B. Every argument with true premises is sound.
C. Every valid argument is sound.
D. Every argument with a true conclusion is sound.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q5
Logic, Set 24: Valid versus sound · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Which combination is IMPOSSIBLE for an argument?

A. invalid, all premises true, conclusion true
B. invalid, all premises true, conclusion false
C. valid, all premises true, conclusion false
D. valid, a false premise, conclusion true

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q1
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

An argument whose premises can never all be true at the same time (they contradict each other) is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q2
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Of the 8 combinations (valid or invalid; premises all true or not; conclusion true or false), how many are impossible for an argument?

Impossible combinations: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q3
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

"If x = 3, then x squared = 9. x squared = 9. So x = 3." For real x, this argument is:

A. sound: both premises can be true

B. invalid: x = -3 makes the premises true and the conclusion false

C. valid: squaring is unique

D. valid but not sound

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q4
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Adding a false premise to a sound argument keeps it valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q5
Logic, Set 24: Valid versus sound · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3

Set 25: Moves and rearrangements

Local outcome LOGIC.SEQUENCE.01: Carry out stated moves on a row of tokens and find moves that reach a target

Puzzle set · genre: token moves · MP1, MP6

MAIN: Carry out stated moves on a row of tokens and find moves that reach a target
BACKWARD 1: Strategy: number the positions
FORWARD 1: ADVANCED APPLICATION: Fewest moves under different rules
FORWARD 2: EXPERT CHALLENGE: Moves that can never work

Tiles in a row each show a different letter. The row reads T, E, A, M. Move the leftmost tile to the far right end. Then swap the first two tiles. What is the row now, from left to right?

Put these in order:

T T
E E
A A
M M

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q1
Logic, Set 25: Moves and rearrangements · MAIN · Q1/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps two tiles that are next to each other. What is the least number of moves that turn D, C, B, A into A, B, C, D?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q2
Logic, Set 25: Moves and rearrangements · MAIN · Q2/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Which single move turns R, O, P, E into O, P, E, R?

A. Swap the two middle tiles.

B. Move E from the right end to the left end.

C. Swap R and O.

D. Move R from the left end to the right end.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q3
Logic, Set 25: Moves and rearrangements · MAIN · Q3/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps two tiles that are next to each other. What is the least number of moves that turn C, A, B, E, D into A, B, C, D, E?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q4
Logic, Set 25: Moves and rearrangements · MAIN · Q4/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Reversing the whole row twice puts every tile back where it started.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-M-Q5
Logic, Set 25: Moves and rearrangements · MAIN · Q5/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Row: K, I, T. Swap the first two tiles. What is the row?

Put these in order:

K K
I I
T T

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q1
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Row: P, A, N. Move the leftmost tile to the far right. What is the row?

Put these in order:

P P

A A

N N

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q2
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. How many pairs of tiles are in the wrong alphabetical order in C, B, A?

Pairs: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q3
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Swapping two neighboring tiles twice returns the row to how it was.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q4
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps neighboring tiles. Least number of moves from B, A, C to A, B, C?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-B1-Q5
Logic, Set 25: Moves and rearrangements · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move takes ANY tile and puts it at the far right end. What is the least number of moves that turn B, C, D, A into A, B, C, D?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q1
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move takes ANY tile and puts it at the far right end. What is the least number of moves that turn A, C, D, B into A, B, C, D?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q2
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. Row: S, T, O, P. Move the tile in position 3 to the far left end, then swap the last two tiles. What is the row?

Put these in order:

S S
T T
O O
P P

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q3
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps neighboring tiles. What is the least number of moves that turn E, D, C, B, A into A, B, C, D, E?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q4
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. If one move swaps neighboring tiles, any arrangement of 4 tiles can be reached from any other.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F1-Q5
Logic, Set 25: Moves and rearrangements · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move reverses any three tiles that are next to each other (for example A, B, C becomes C, B, A). Starting from A, B, C, D, E, can you reach B, A, C, D, E?

A. No: each move keeps every tile in an odd or an even position, and A would need to move from position 1 to position 2.

B. Yes: any arrangement can be reached with enough moves.

C. Yes: reverse A, B, C, then reverse B, A, ... in a few moves.

D. No: because each move changes three tiles at once.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q1
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move reverses any three neighboring tiles. Starting from A, B, C, D, E, how many different arrangements can be reached (counting the start)?

Arrangements: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q2
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps neighboring tiles. The row C, A, D, B, E must become A, B, C, D, E. Give the least number of moves m.

m = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q3
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps ANY two tiles (not only neighbors). D, C, B, A can be turned into A, B, C, D in 2 moves.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q4
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Tiles in a row each show a different letter. One move swaps ANY two tiles. What is the least number of moves that turn B, C, D, E, A into A, B, C, D, E?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S25-F2-Q5
Logic, Set 25: Moves and rearrangements · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SEQUENCE.01 (local) · MP1, MP6

Set 26: Quantifiers over a stated domain

Local outcome LOGIC.QUANT.01: Decide the truth of "all", "some" and "no" statements over a stated finite domain

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Decide the truth of "all", "some" and "no" statements over a stated domain

BACKWARD 1: Read "all", "some" and "no"

BACKWARD 2: Stay inside the stated domain

FORWARD 1: ADVANCED APPLICATION: Quantifiers with two conditions

FORWARD 2: EXPERT CHALLENGE: Nested quantifiers

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "All chess members have a bus pass."

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q1
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Some member aged 16 has a bus pass."

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q2
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). Which statement is TRUE?

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. All robotics members have a bus pass.

B. Every member is at least 16.

C. No art member has a bus pass.

D. Some chess member is 16.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q3
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). How many members make "the member is 15 or has no bus pass" true?

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q4
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 10. Which statement is FALSE?

A. Some number in the domain is prime.
B. Every number in the domain is less than 10.
C. Some number in the domain is a multiple of 7.
D. No number in the domain is negative.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-M-Q5
Logic, Set 26: Quantifiers over a stated domain · MAIN · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Some students walk to school" is true when ...

A. most students walk
B. at least one student walks to school
C. exactly one student walks
D. every student walks to school

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q1
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"No student walks to school" is true when ...

A. most students ride

B. some students do not walk

C. not even one student walks

D. exactly one student walks

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q2
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"All the lights are on" is true if exactly one light is off.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q3
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Some numbers in {2, 4, 6} are odd" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q4
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Every number in {3, 5, 7} is odd" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B1-Q5
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is {1, 2, 3}. "Every number in the domain is less than 4" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q1
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is {10, 20, 30}. "Some number in the domain is odd" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q2
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the days Monday to Friday. Which is true?

A. No day in the domain is a weekend day.
B. Every day in the domain starts with M.
C. Some day in the domain is Sunday.
D. Some day in the domain is Saturday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q3
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"All prime numbers are odd" is true when the domain is {3, 5, 7, 11}.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q4
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"All prime numbers are odd" is true when the domain is all whole numbers.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-B2-Q5
Logic, Set 26: Quantifiers over a stated domain · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Every member without a bus pass is 16."

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q1
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Some member with a bus pass is older than every robotics member."

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q2
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 20. How many numbers make "the number is even and not a multiple of 3" true?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q3
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). Which statement is FALSE?

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

A. Every chess member is 15 or 17.

B. All members aged 17 play chess.

C. Some member has no bus pass.

D. No member aged 15 is in robotics.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q4
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is empty (it has no members). "Every member of the domain is green" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F1-Q5
Logic, Set 26: Quantifiers over a stated domain · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 5. "For every number x, there is a number y with y > x" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q1
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 5. "There is a number y such that for every number x, $y \geq x$" is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q2
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is all whole numbers. Which statement is TRUE?

A. For every x and every y, $y > x$.

B. There is an x such that for every y, $y > x$.

C. For every x, there is a y with $y > x$.

D. There is a y such that for every x, $y > x$.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q3
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). For how many members x is the statement "some other member is the same age as x" true?

Name	Age	Club	Bus pass
Ava	15	chess	yes
Ben	16	robotics	no
Cy	15	robotics	yes
Dee	17	chess	yes
Eli	16	art	no

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q4
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

The domain is the whole numbers 1 to 9. Give a number x that makes "for every y in the domain, x + y > 5" true.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S26-F2-Q5
Logic, Set 26: Quantifiers over a stated domain · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

Set 27: Negating quantified statements

Local outcome LOGIC.QUANT.02: Negate statements with "all", "some", "no", and nested quantifiers

Formal logic set · genre: formal logic · MP6, MP3

MAIN: Negate statements with "all", "some", "no", and nested quantifiers
BACKWARD 1: The meaning of each quantifier
BACKWARD 2: Negating simple statements
FORWARD 1: ADVANCED APPLICATION: Negate quantified statements with conditions
FORWARD 2: EXPERT CHALLENGE: Nested quantifiers and their negations

What is the negation of "All the tickets were sold"?

A. At least one ticket was not sold.
B. All the tickets were unsold.
C. No ticket was sold.
D. Some tickets were sold.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q1
Logic, Set 27: Negating quantified statements · MAIN · Q1/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

What is the negation of "Some student brought a calculator"?

A. Some student did not bring a calculator.
B. Exactly one student brought a calculator.
C. Every student brought a calculator.
D. No student brought a calculator.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q2
Logic, Set 27: Negating quantified statements · MAIN · Q2/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

What is the negation of "No bridge in the town is closed"?

A. No bridge in the town is open.
B. Some bridge in the town is closed.
C. Every bridge in the town is closed.
D. Some bridge in the town is open.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q3
Logic, Set 27: Negating quantified statements · MAIN · Q3/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The domain is the whole numbers. The statement "Every whole number greater than 1 has an even square" is false. Give a number n that makes its negation true: a whole number greater than 1 whose square is odd.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q4
Logic, Set 27: Negating quantified statements · MAIN · Q4/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The negation of "Every square is a rectangle" is "No square is a rectangle".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-M-Q5
Logic, Set 27: Negating quantified statements · MAIN · Q5/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

"Some cards are red" is false if exactly one card is red.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q1
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"All cards are red" is false if one card is blue.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q2
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"No card is red" is true if one card is red.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q3
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Not all cards are red" means ...

A. no card is red
B. every card is blue
C. exactly one card is red
D. at least one card is not red

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q4
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

"Some cards are red" and "some cards are not red" can both be true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B1-Q5
Logic, Set 27: Negating quantified statements · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP6, MP3

What is the negation of "the bus is late"?

A. the bus is not late
B. the bus is early
C. the train is late
D. the bus is very late

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q1
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

The negation of "$x \geq 4$" is "$x \leq 4$".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q2
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

What is the negation of "the number is even and positive"?

A. the number is zero
B. the number is odd and negative
C. the number is odd or not positive
D. the number is even or positive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q3
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

"Not (the door is locked)" means the door is not locked.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q4
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

What is the negation of "it rained on Monday or Tuesday"?

A. it did not rain on Monday
B. it rained on neither Monday nor Tuesday
C. it rained on both days
D. it rained on Wednesday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-B2-Q5
Logic, Set 27: Negating quantified statements · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.01 (local) · MP6, MP3

What is the negation of "Every student who studied passed"?

A. Some student who passed did not study.
B. Some student who studied did not pass.
C. No student who studied passed.
D. Every student who did not study failed.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q1
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The domain is the real numbers. What is the negation of "For every x, x squared ≥ 0"?

A. For every x, x squared ≤ 0.

B. There is an x with x squared ≥ 0.

C. There is an x with x squared < 0.

D. For every x, x squared < 0.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q2
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The negation of "Some prime number is even" is "Every prime number is odd" (for whole numbers greater than 1).

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q3
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The statement "Every whole number from 1 to 30 that ends in 9 is prime" is false. Give a number that shows it (a witness for its negation).

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q4
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The negation of "No dog is a cat" is "Every dog is a cat".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F1-Q5
Logic, Set 27: Negating quantified statements · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

What is the negation of "For every student, there is a club the student likes"?

A. There is a student who likes no club.

B. There is a club that no student likes.

C. No student likes any club.

D. For every student, there is a club the student does not like.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q1
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The domain is the whole numbers. What is the negation of "There is a number y such that for every x, y > x"?

A. There is no number y.
B. There is a y with $y \leq x$ for some x.
C. For every y and every x, $y \leq x$.
D. For every y, there is an x with $y \leq x$.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q2
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The negation of "For every x there is a y with $x + y = 0$" is "There is an x such that for every y, $x + y \neq 0$".

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q3
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

The domain is the whole numbers 1 to 10. The statement "For every x there is a y with y = 2x" is false in this domain. Give an x that shows it.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q4
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

For any statement about a domain, either the statement or its negation is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S27-F2-Q5
Logic, Set 27: Negating quantified statements · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.QUANT.02 (local) · MP6, MP3

Set 28: Routes and paths

Local outcome LOGIC.GRAPH.01: Trace routes in a drawn network and decide which routes exist under stated rules

Puzzle set · genre: routes and graphs · MP1, MP7

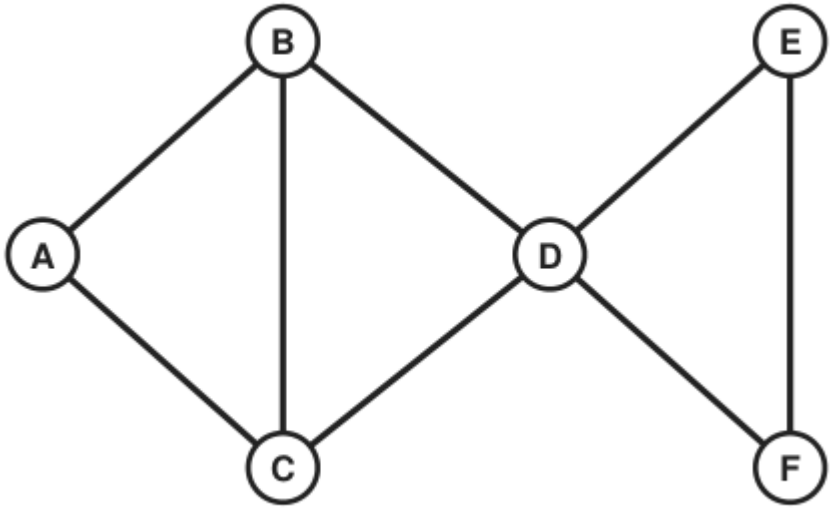
MAIN: Trace routes in a network and decide which routes exist

BACKWARD 1: Strategy: mark the route as you trace it

FORWARD 1: ADVANCED APPLICATION: Routes with conditions

FORWARD 2: EXPERT CHALLENGE: Every road, every town

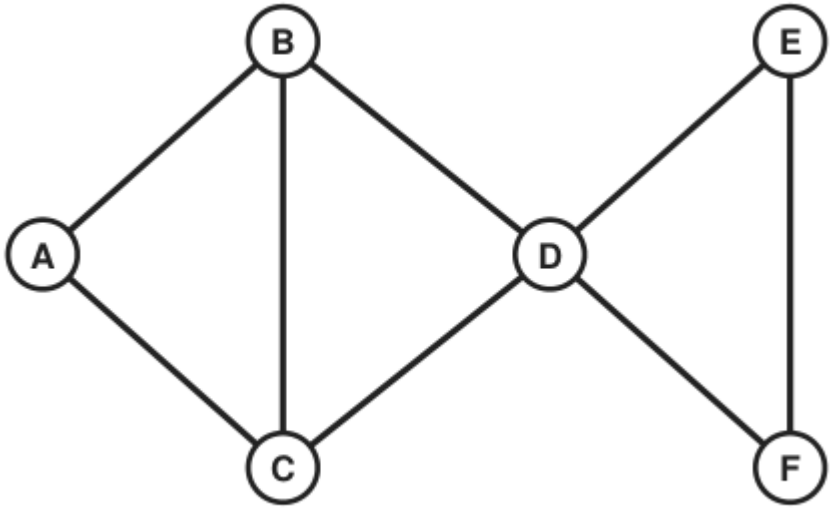
Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads on a route from A to F?



Roads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q1
Logic, Set 28: Routes and paths · MAIN · Q1/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

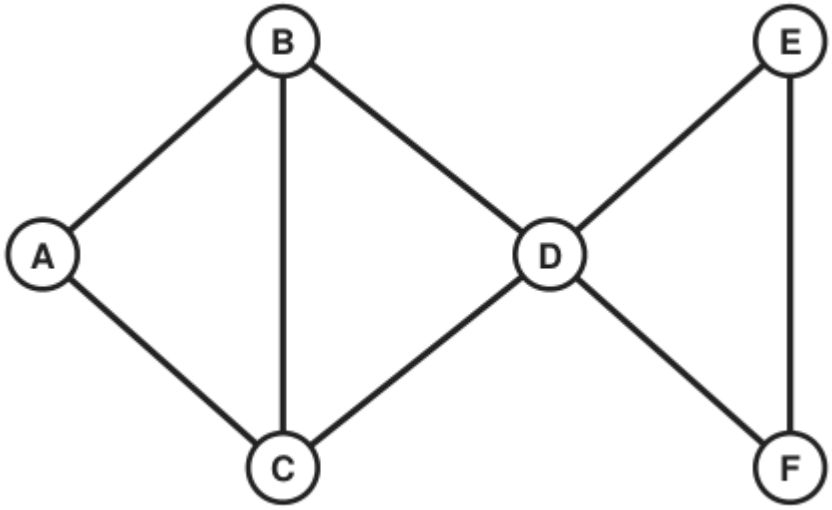
Roads are the straight lines; towns are the circles. Roads meet only at towns. How many different routes go from A to F without visiting any town twice?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q2
Logic, Set 28: Routes and paths · MAIN · Q2/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

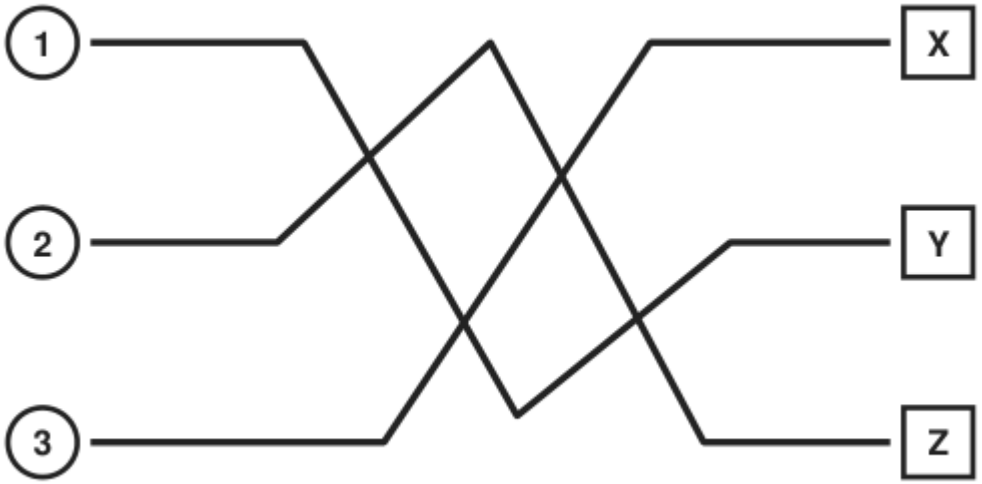
Roads are the straight lines; towns are the circles. Roads meet only at towns. Which town, if it were closed (with all its roads), would cut the network into two parts that cannot reach each other?



A. B
B. E
C. C
D. D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q3
Logic, Set 28: Routes and paths · MAIN · Q3/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

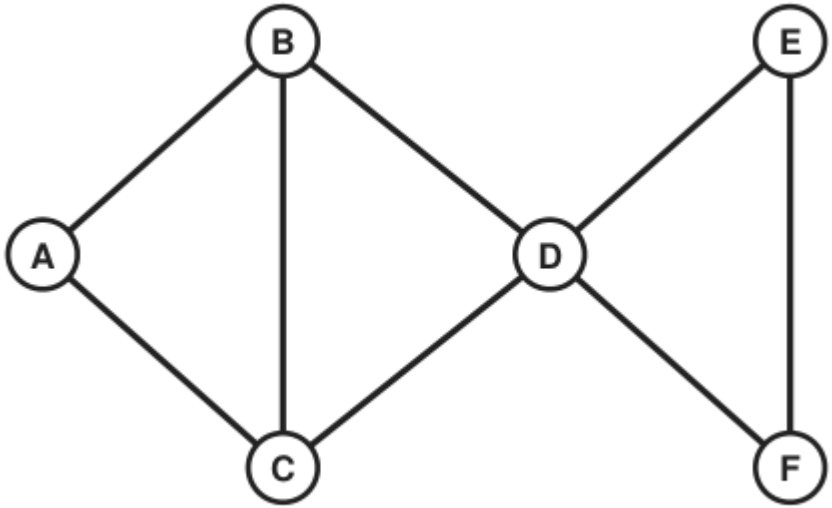
Walkers 1, 2 and 3 each follow their own line from the left to a gate on the right. Where lines cross, there is no turning: each walker keeps going straight along their own line. Which gate does walker 2 reach?



A. Z
B. Y
C. X
D. none of the gates

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q4
Logic, Set 28: Routes and paths · MAIN · Q4/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. A walk can use every road exactly once (towns may be visited more than once).

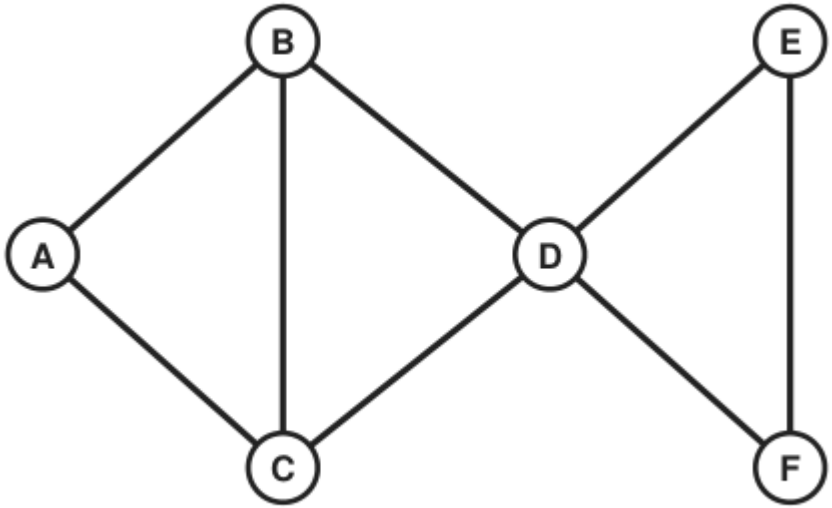


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-M-Q5
Logic, Set 28: Routes and paths · MAIN · Q5/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

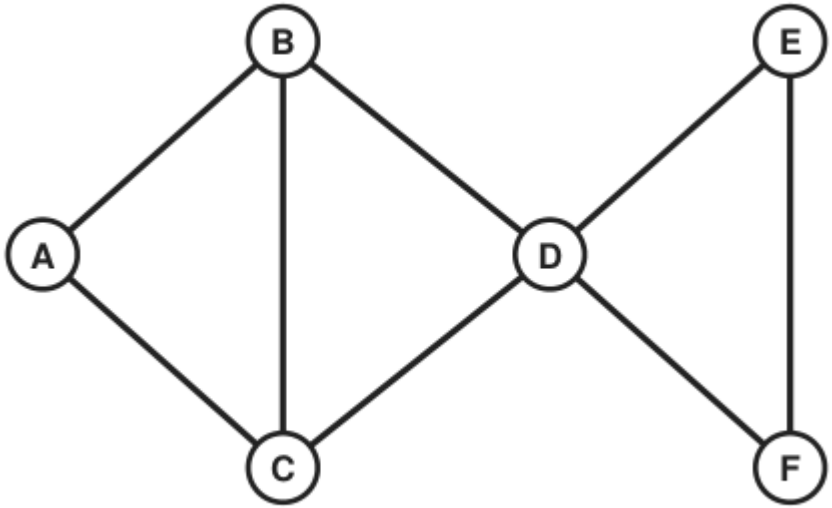
Roads are the straight lines; towns are the circles. Roads meet only at towns. How many roads meet at town D?



Roads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q1
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

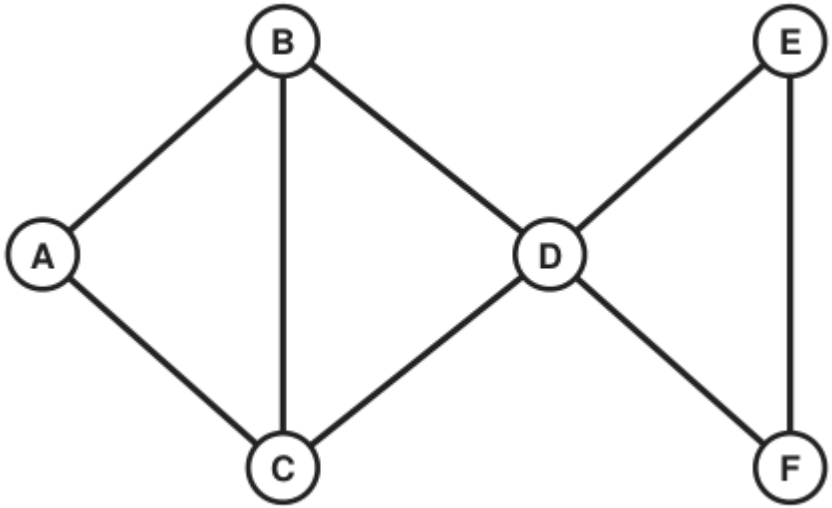
Roads are the straight lines; towns are the circles. Roads meet only at towns. Which towns can be reached from A using exactly one road?



A. only B
B. D
C. B and C
D. B, C and D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q2
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. There is a road directly from A to D.

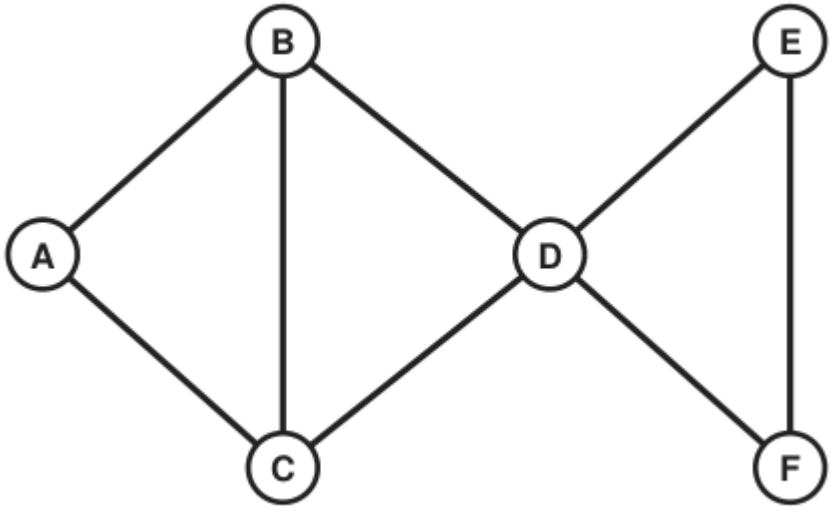


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q3
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes go from E to F without visiting a town twice?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q4
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

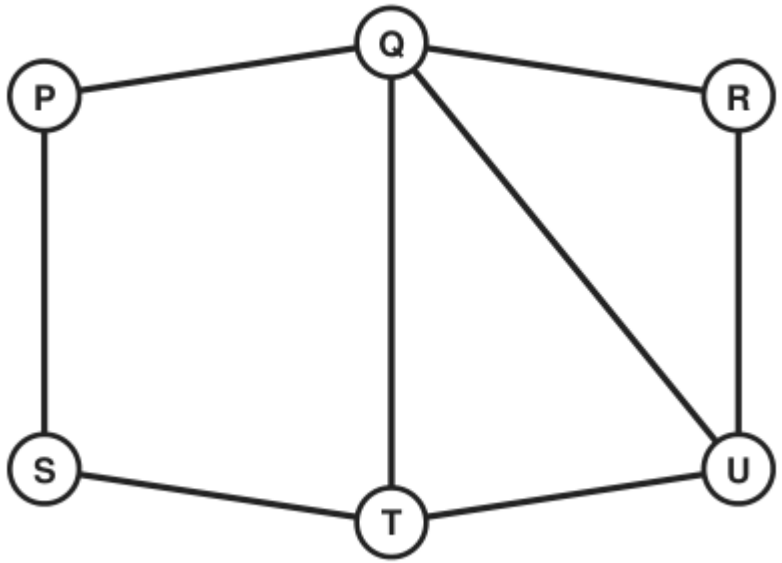
In a network, a town with 3 roads must be the start or end of a walk that uses every road once.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-B1-Q5
Logic, Set 28: Routes and paths · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

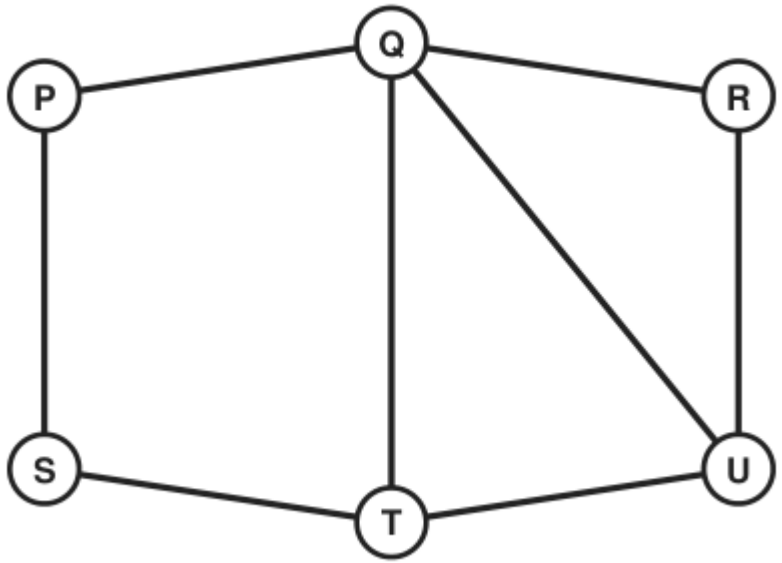
Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes go from P to U without visiting a town twice?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q1
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

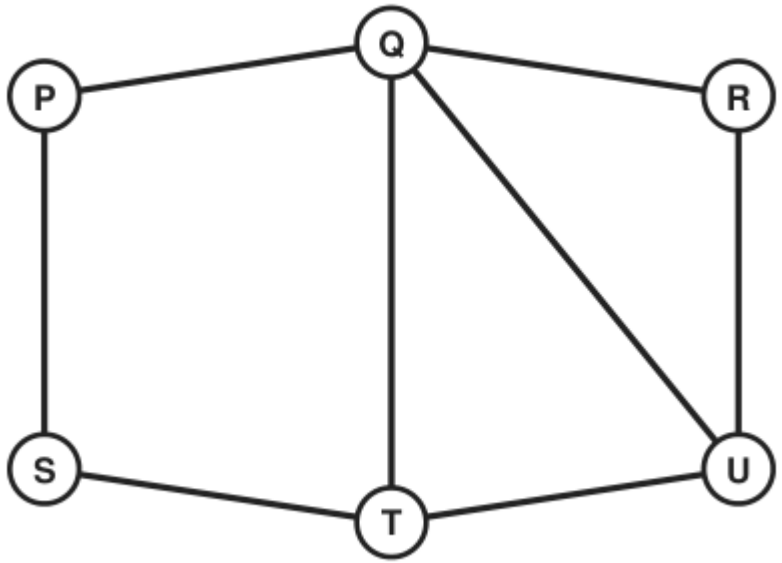
Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes from P to U (no town twice) pass through T?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q2
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. A walk can use every road exactly once.

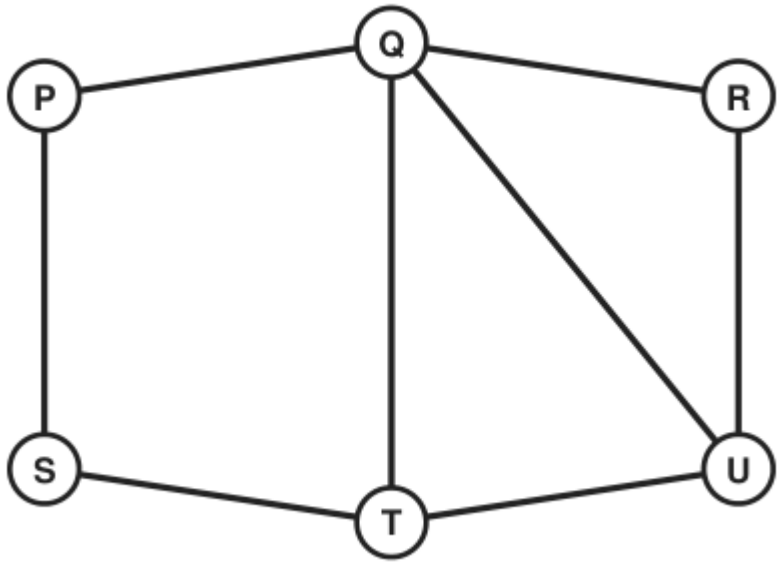


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q3
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

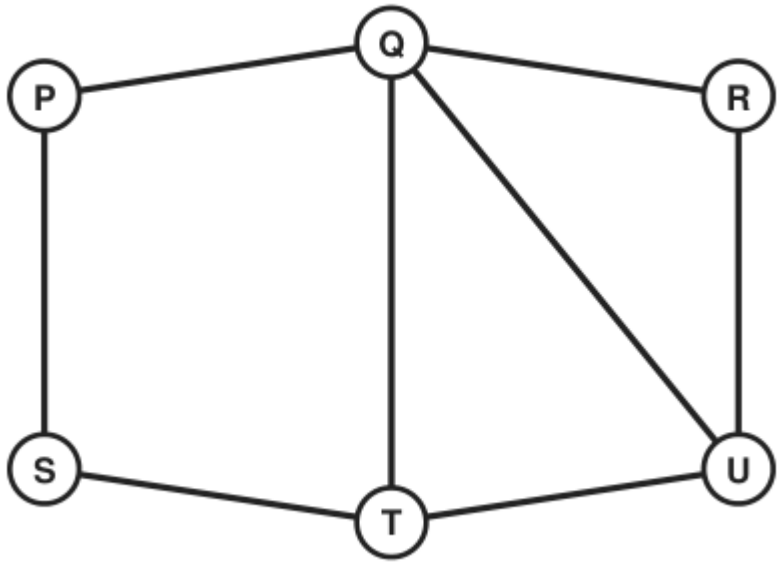
Roads are the straight lines; towns are the circles. Roads meet only at towns. Which single road, if closed, makes T unreachable from P by any route?



A. P-S
B. none of these roads
C. Q-T
D. S-T

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q4
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads on a route from S to R?



Roads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F1-Q5
Logic, Set 28: Routes and paths · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

In a network of 6 towns, the numbers of roads at the towns are 1, 2, 2, 3, 3 and 3. Every road has two ends, each at a town. How many roads are there?

Roads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q1
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

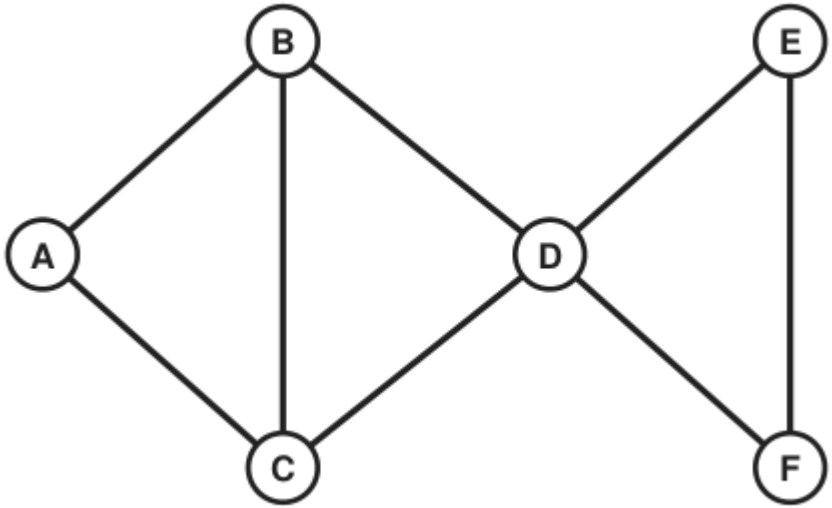
Some network has exactly three towns with an odd number of roads.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q2
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

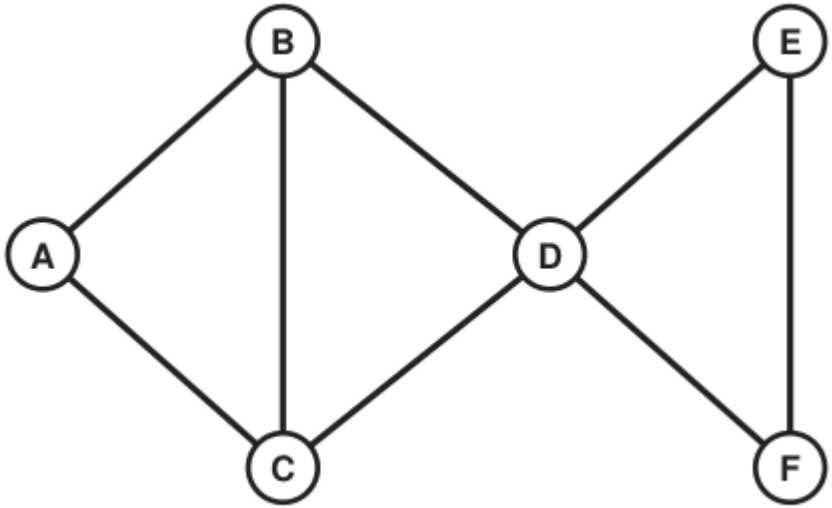
Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads that must be ADDED so that a walk can use every road exactly once and return to its starting town?



A. 2
B. 1
C. 3
D. 0

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q3
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes from A to F visit EVERY town exactly once?



Routes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q4
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

A network has towns with these numbers of roads: 3, 3, 2, 2 and k. For a walk that uses every road exactly once to be possible in a connected network, at most two towns may have an odd number of roads. Give a whole number k from 1 to 6 that keeps this possible.

k = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S28-F2-Q5
Logic, Set 28: Routes and paths · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.GRAPH.01 (local) · MP1, MP7

Set 29: Witnesses and counterexamples

Local outcome LOGIC.QUANT.03: Give a witness for a "some" statement and a counterexample to an "all" statement

Formal logic set · genre: formal logic · MP3

- MAIN: Give a witness for a "some" statement and a counterexample to an "all" statement
- BACKWARD 1: Read "every" and "some" statements
- BACKWARD 2: Test a number against a property
- FORWARD 1: ADVANCED APPLICATION: Counterexamples in algebra
- FORWARD 2: EXPERT CHALLENGE: Witnesses that are hard to find

Give a witness for the true statement "Some whole number between 20 and 30 is prime."

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q1
Logic, Set 29: Witnesses and counterexamples · MAIN · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give a counterexample to "Every whole number that is divisible by 3 is odd."

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q2
Logic, Set 29: Witnesses and counterexamples · MAIN · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Which number is a counterexample to "Every number that ends in 1 is prime"?

A. 21
B. 31
C. 12
D. 11

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q3
Logic, Set 29: Witnesses and counterexamples · MAIN · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give real numbers a and b that are a counterexample to "For all real numbers a and b, (a + b) squared = a squared + b squared."

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q4
Logic, Set 29: Witnesses and counterexamples · MAIN · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3

One example where a statement is true proves "Every number has this property."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-M-Q5
Logic, Set 29: Witnesses and counterexamples · MAIN · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3

What kind of example disproves "Every bird can fly"?

A. a flying insect
B. a bird that can fly
C. a bird that cannot fly
D. an animal that is not a bird

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q1
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.01 (local) · MP3

What kind of example proves "Some square numbers are even"?

A. an odd square number

B. an even number that is not a square

C. every square number

D. an even square number, such as 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q2
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.01 (local) · MP3

"All cats are black" is shown true by one black cat.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q3
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.01 (local) · MP3

"All cats are black" is shown false by one white cat.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q4
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.01 (local) · MP3

"No number is negative" is shown false by the number 1.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B1-Q5
Logic, Set 29: Witnesses and counterexamples · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.01 (local) · MP3

15 is prime.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q1
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3

27 is divisible by 3 and odd.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q2
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3

What is 7 squared minus 7?

Value: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q3
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3

2 is both prime and even.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q4
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Which number is a multiple of 4 but not of 8?

A. 12
B. 10
C. 16
D. 8

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-B2-Q5
Logic, Set 29: Witnesses and counterexamples · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give a real number x that is a counterexample to "For every real number x, x squared ≥ x."

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q1
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give positive whole numbers a and b that are a counterexample to "If a divides b x c, then a divides b or a divides c", using c = 2.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q2
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Which value of n is a counterexample to "n squared + n + 1 is prime for every whole number n ≥ 1"?

A. 1

B. 4

C. 3

D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q3
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3

The statement "For every real x, if x > 1 then x squared > x" has no counterexample.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q4
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give whole numbers a and b, each from 1 to 9, that show "a - b = b - a" can be false.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F1-Q5
Logic, Set 29: Witnesses and counterexamples · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give a witness for "Some whole number from 1 to 100 leaves remainder 2 when divided by 3, remainder 3 when divided by 4, and remainder 4 when divided by 5."

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q1
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Is there a whole number n from 1 to 100 with n squared ending in the digit 2?

A. No: but only because 2 is even.

B. Yes: n = 12.

C. Yes: n = 22.

D. No: squares end only in 0, 1, 4, 5, 6 or 9.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q2
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Give whole numbers a, b and c (each from 1 to 20) with a squared + b squared = c squared and a < b.

a = _____

b = _____

c = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q3
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.QUANT.03 (local) · MP3

How many counterexamples to "Every odd number from 1 to 30 is prime" are there in that range?

Counterexamples: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q4
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.QUANT.03 (local) · MP3

To disprove "There is a whole number n with n + 5 = n", one example is enough.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S29-F2-Q5
Logic, Set 29: Witnesses and counterexamples · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.QUANT.03 (local) · MP3

Set 30: Sets and Venn diagrams

Local outcome LOGIC.SETS.01: Mark subset, intersection and complement relations on Venn diagrams

Formal logic set · genre: formal logic · MP7, MP6

MAIN: Mark subset, intersection and complement relations on Venn diagrams
BACKWARD 1: Regions of a two-circle diagram
BACKWARD 2: Set words: and, or, not
FORWARD 1: ADVANCED APPLICATION: Three-circle diagrams and counts
FORWARD 2: EXPERT CHALLENGE: Inclusion and exclusion

Circle A is the set of students in soccer and circle B the set of students in band. Mark the region for students in soccer but NOT in band.

Mark the regions on the Venn diagram of soccer (A), band (B) (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q1
Logic, Set 30: Sets and Venn diagrams · MAIN · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Circle A is the set of even numbers and circle B the set of multiples of 3 (whole numbers 1 to 30 make up the outside box). Mark the region that contains 12.

Mark the regions on the Venn diagram of even (A), multiples of 3 (B) (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q2
Logic, Set 30: Sets and Venn diagrams · MAIN · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Three circles: A (likes math), B (likes art), C (likes music). Mark every region for people who like EXACTLY two of the three subjects.

Mark the regions on the Venn diagram of math (A), art (B), music (C) (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q3
Logic, Set 30: Sets and Venn diagrams · MAIN · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

In a class of 30 students, 18 play soccer, 12 play in the band, and 5 do both. How many students do neither?

Students: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q4
Logic, Set 30: Sets and Venn diagrams · MAIN · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Every member of set A is also in set B (A is a subset of B). On a two-circle diagram, mark the region that must be EMPTY.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-M-Q5
Logic, Set 30: Sets and Venn diagrams · MAIN · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Mark the region for things in BOTH A and B.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q1
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Mark the region for things in NEITHER A nor B.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q2
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Mark every region for things in A (whether or not they are in B).

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q3
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

How many regions does a two-circle Venn diagram have, counting the outside?

Regions: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q4
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Mark every region for things in B but not in A.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B1-Q5
Logic, Set 30: Sets and Venn diagrams · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

The intersection of A and B contains things that are ...

A. in neither
B. in A or in B
C. in A and in B
D. in A but not B

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q1
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

The union of A and B contains things that are ...

A. in A or in B (or both)
B. in exactly one
C. in neither
D. in A and in B

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q2
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

The complement of A contains everything in the box that is not in A.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q3
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

A = {1, 2, 3, 4}, B = {3, 4, 5}. How many numbers are in the union of A and B?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q4
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

A = {1, 2, 3, 4}, B = {3, 4, 5}. How many numbers are in the intersection?

Count: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-B2-Q5
Logic, Set 30: Sets and Venn diagrams · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.CONNECTIVE.02 (local) · MP7, MP6

Three circles A, B, C. Mark every region for things in A or B but NOT in C.

Mark the regions on the Venn diagram of A, B, C (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q1
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Three circles A, B, C. Mark every region for things in at least two of the sets.

Mark the regions on the Venn diagram of A, B, C (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q2
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Of 40 students, 22 take French, 17 take Spanish and 9 take both. How many take exactly one of the two?

Students: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q3
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Every member of A is in B, and no member of B is in C. On a three-circle diagram (A, B, C), mark every region that must be EMPTY.

Mark the regions on the Venn diagram of A, B, C (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q4
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

If A is a subset of B, then the intersection of A and B is A.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F1-Q5
Logic, Set 30: Sets and Venn diagrams · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

In a group of 50 people, 25 like tea, 30 like coffee and 15 like both. How many like neither?

People: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q1
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Sets A, B, C have 10, 12 and 8 members. A and B share 4, A and C share 3, B and C share 2, and all three share 1. How many members are in the union?

Members: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q2
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

In a club of 20, s members swim and r members run; every member does at least one. Give possible numbers s and r if exactly 5 do both.

swim s = _____

run r = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q3
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Two sets can have 8 and 6 members and a union of 15 members.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q4
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

Sets A and B are inside a box of 30 things. A has 12 things, B has 9, and A and B share 4. How many things are in the box but in neither A nor B?

Things: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S30-F2-Q5
Logic, Set 30: Sets and Venn diagrams · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP7, MP6

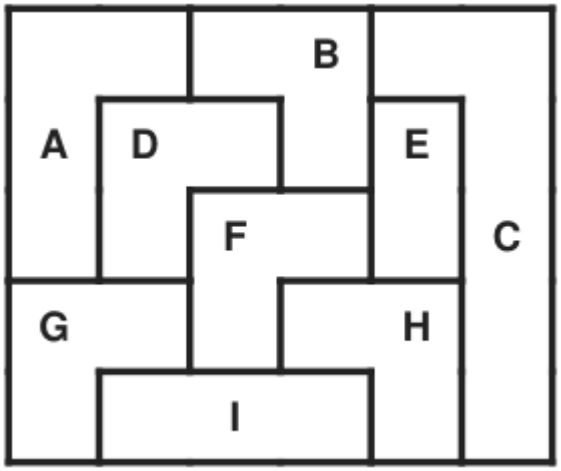
Set 31: Maps and neighbors

Local outcome LOGIC.GRAPH.02: Turn a map into a neighbor (adjacency) graph and reason about neighbor counts and colorings

Puzzle set · genre: region adjacency · MP7, MP1

MAIN: Turn a map into a neighbor graph and reason about neighbors and colorings
BACKWARD 1: Strategy: draw the neighbor graph
FORWARD 1: ADVANCED APPLICATION: Neighbor counts as clues
FORWARD 2: EXPERT CHALLENGE: Colorings: count and prove

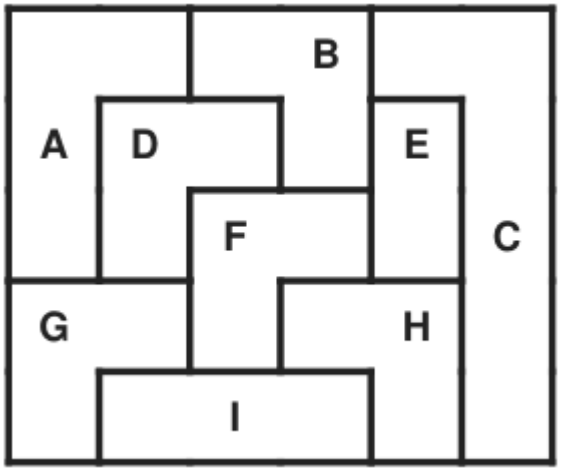
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region F have?



Neighbors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q1
Logic, Set 31: Maps and neighbors · MAIN · Q1/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

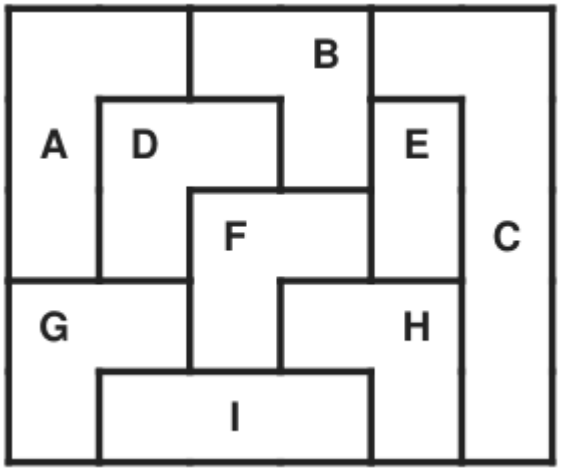
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Which region has the most neighbors?



A. C
B. F
C. D
D. B

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q2
Logic, Set 31: Maps and neighbors · MAIN · Q2/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Regions A and I are neighbors.

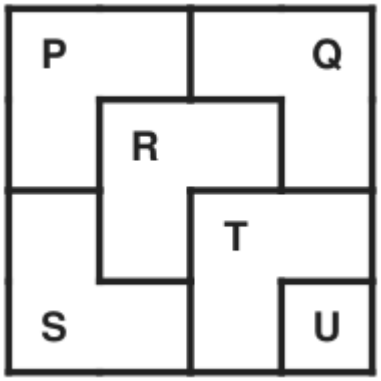


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q3
Logic, Set 31: Maps and neighbors · MAIN · Q3/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

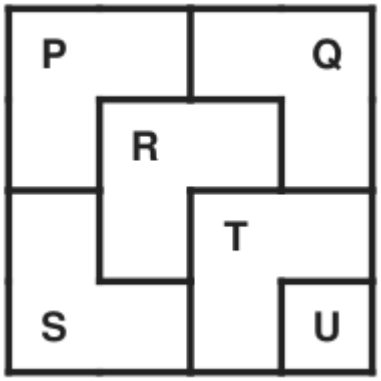
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Each region gets one color and neighbors must get different colors. What is the least number of colors needed for this map?



Colors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q4
Logic, Set 31: Maps and neighbors · MAIN · Q4/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Color each region red, blue or green so that neighbors differ. P is red and Q is blue. Give every region's color.

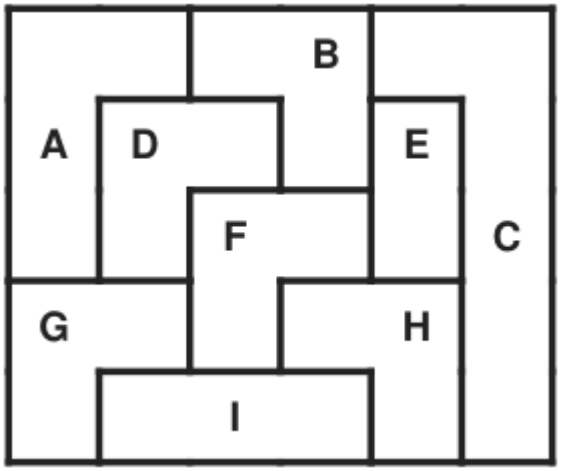


Choose one column for each row.

	red	blue	green
P			
Q			
R			
S			
T			
U			

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-M-Q5
Logic, Set 31: Maps and neighbors · MAIN · Q5/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

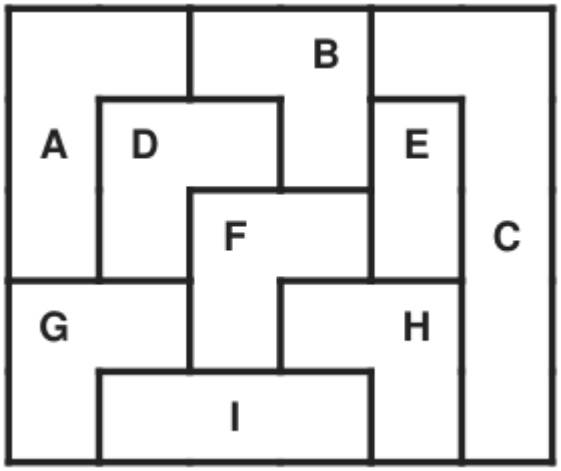
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region C have?



Neighbors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q1
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Regions B and D are neighbors.



A. True

B. False

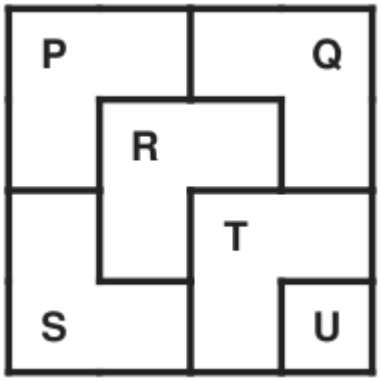
REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q2
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Two regions touch only at a single corner point. Are they neighbors under this rule?

A. Yes
B. It depends on their size
C. No
D. Only if they are the same color

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q3
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region U have?



Neighbors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q4
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

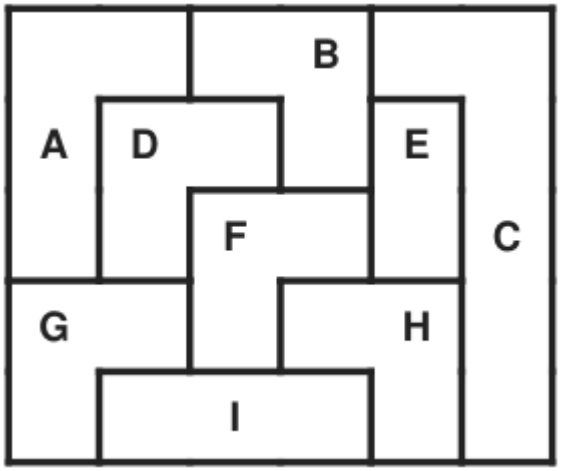
If three regions all border each other, a coloring with different colors for neighbors needs at least 3 colors.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-B1-Q5
Logic, Set 31: Maps and neighbors · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

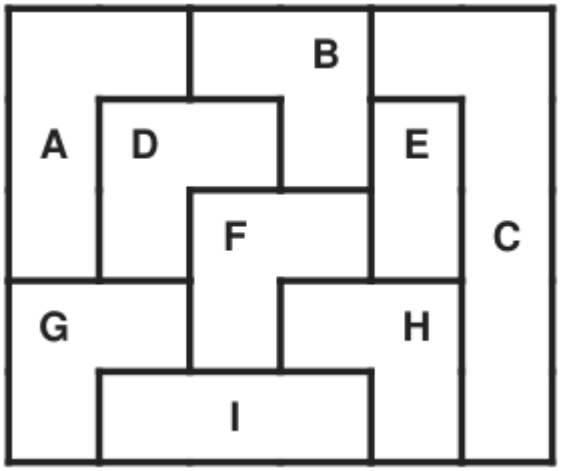
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. What is the total of the neighbor counts of all nine regions?



Total: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q1
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

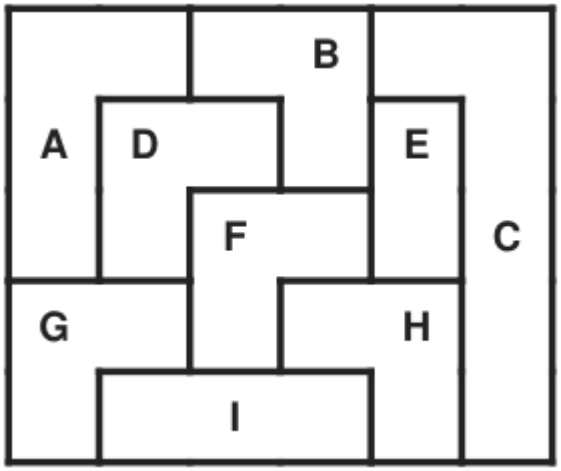
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many pairs of regions share a border?



Pairs: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q2
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

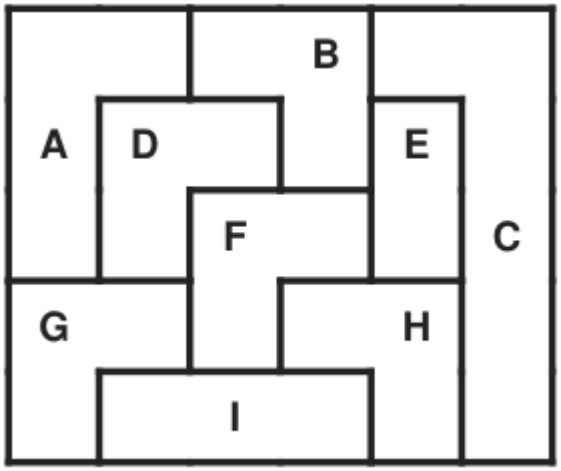
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. What is the least number of colors needed so that neighbors differ?



Colors: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q3
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. In any coloring of this map with 3 colors where neighbors differ, regions A and F have the same color.

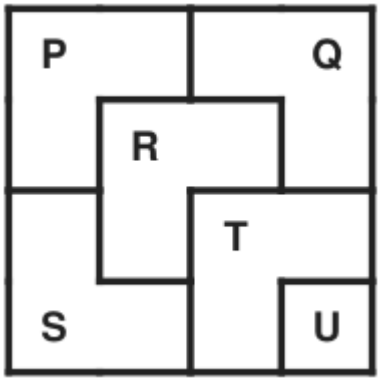


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q4
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

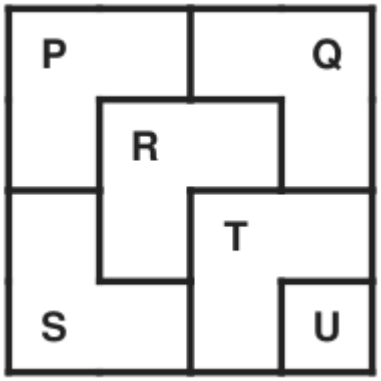
Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Which ONE region could be removed so that only 2 colors are needed for the rest?



A. S
B. U
C. T
D. R

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F1-Q5
Logic, Set 31: Maps and neighbors · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many ways can the map be colored with red, blue and green so that neighbors differ?



Colorings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q1
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Some map drawn on a page needs at least 4 colors so that neighbors differ.

A. True

B. False

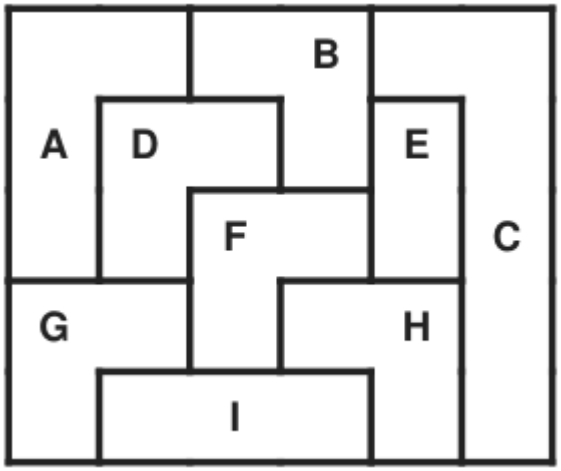
REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q2
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

A map has regions in a ring: region 1 borders 2, 2 borders 3, and so on, and the last region borders region 1 (and there are no other borders). Give a number of regions n, from 3 to 10, for which 2 colors are NOT enough.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q3
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many ways can this map be colored with red, blue and green so that neighbors differ, if A is red?



Colorings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q4
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

If a map can be colored with 4 colors so that neighbors differ, it can also be colored with 3 colors.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S31-F2-Q5
Logic, Set 31: Maps and neighbors · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.GRAPH.02 (local) · MP7, MP1

Set 32: Categorical arguments

Local outcome LOGIC.SETS.02: Test categorical arguments with Venn diagrams, including the existence assumption

Formal logic set · genre: formal logic · MP3, MP7

- MAIN: Test categorical arguments with Venn diagrams, including the existence assumption
- BACKWARD 1: Mark statements on a Venn diagram
- BACKWARD 2: Valid means no counterexample
- FORWARD 1: ADVANCED APPLICATION: Harder categorical forms
- FORWARD 2: EXPERT CHALLENGE: Existence and hidden assumptions

Premise: "No reptile is warm-blooded." Circle A is reptiles and circle B is warm-blooded animals. Mark the region this premise says is EMPTY.

Mark the regions on the Venn diagram of reptiles (A), warm-blooded (B) (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q1
Logic, Set 32: Categorical arguments · MAIN · Q1/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All violinists are musicians. All musicians practice. So all violinists practice." Is this valid?

A. No: some violinists may not practice.

B. Yes, but only if every musician is a violinist.

C. Yes: every violinist is inside the musicians, who are all inside those who practice.

D. No: the conclusion reverses a premise.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q2
Logic, Set 32: Categorical arguments · MAIN · Q2/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All squares are rectangles. Some rectangles are not rhombuses. So some squares are not rhombuses." Is this valid?

A. No: the rectangles that are not rhombuses may all lie outside the squares.

B. Yes: the conclusion is true.

C. Yes: squares are rectangles, so they share every property.

D. No: no rectangle is a square.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q3
Logic, Set 32: Categorical arguments · MAIN · Q3/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B. Some C are A. So some C are B." This form is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q4
Logic, Set 32: Categorical arguments · MAIN · Q4/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All dragons breathe fire. So some dragons breathe fire." When does this follow?

A. Always: "all" includes "some".
B. Only if dragons are real animals.
C. Never: the premise is about all dragons.
D. Only if at least one dragon exists (the existence assumption).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-M-Q5
Logic, Set 32: Categorical arguments · MAIN · Q5/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B." Mark the region that must be EMPTY.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q1
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

"No A is B." Mark the region that must be EMPTY.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q2
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

"Some A is B." Mark the region where something must be.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q3
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

"Some A is not B." Mark the region where something must be.

Mark the regions on the Venn diagram of A, B (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q4
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

"All A are B" and "No A is B" can both be true if A has no members.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B1-Q5
Logic, Set 32: Categorical arguments · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.SETS.01 (local) · MP3, MP7

An argument is valid if a situation exists with true premises and a false conclusion.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q1
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q1/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

"All cats are mammals. All dogs are mammals. So all cats are dogs." Which situation shows it is invalid?

A. a world where all mammals are dogs

B. the real world: cats and dogs are both mammals but cats are not dogs

C. a world with no mammals

D. a world with no cats

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q2
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q2/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

"All A are B. All B are C. So all A are C." is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q3
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q3/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

"Some A are B. Some B are C. So some A are C." is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q4
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q4/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

Which premise pair makes "So no A is C" follow?

A. "All A are B." and "Some B are not C."

B. "Some A are B." and "No B is C."

C. "All A are B." and "No B is C."

D. "No A is B." and "No B is C."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-B2-Q5
Logic, Set 32: Categorical arguments · BACKWARD 2 (prerequisite) · Q5/5 · Outcome LOGIC.INFERENCE.03 (local) · MP3, MP7

"No A is B. All C are B. So no C is A." This form is valid.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q1
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B. No C is A. So no C is B." This form is valid.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q2
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"Some athletes are vegetarians. All vegetarians eat beans. So ..." Which conclusion follows?

A. all athletes eat beans
B. no athlete eats meat
C. all bean eaters are athletes
D. some athletes eat beans

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q3
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

Premises: "All A are B" and "No B is C". On a three-circle diagram (A, B, C), mark every region that the premises make EMPTY.

Mark the regions on the Venn diagram of A, B, C (on screen: click each region).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q4
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B. Some A are C. So some B are C." This form is valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F1-Q5
Logic, Set 32: Categorical arguments · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All A are B. All A are C. So some B are C." This form is valid without any existence assumption.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q1
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

Which added premise makes "All A are B. All A are C. So some B are C." valid?

A. At least one A exists.
B. All B are A.
C. At least one B exists.
D. At least one C exists.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q2
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

How many of these forms are valid (with no existence assumption)? (1) All A are B; all B are C; so all A are C. (2) No A is B; all C are A; so no C is B. (3) All A are B; some C are not B; so some C are not A. (4) Some A are B; all B are C; so some A are C.

Valid forms: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q3
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"All unicorns have one horn" is true if no unicorns exist.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q4
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

"No square is a circle. Some circles are shaded. So some shaded things are not squares." Is it valid?

A. No: shaded things may all be squares.
B. Yes: a shaded circle is not a square.
C. No: it needs squares to exist.
D. Yes, but only if all circles are shaded.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S32-F2-Q5
Logic, Set 32: Categorical arguments · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SETS.02 (local) · MP3, MP7

Set 33: Word ladders

Local outcome LOGIC.WORD.02: Find shortest one-letter-change paths through a supplied word list

Puzzle set · genre: word ladders · MP1, MP7

MAIN: Find shortest one-letter-change paths through a supplied word list
BACKWARD 1: Strategy: change the letters that must change, one at a time
FORWARD 1: ADVANCED APPLICATION: Longer ladders and detours
FORWARD 2: EXPERT CHALLENGE: Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Put these four words in ladder order, starting with SAND, so that each word differs from the next by one letter.

Put these in order:

A SAND
B SEND
C SENT
D RENT

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q1
Logic, Set 33: Word ladders · MAIN · Q1/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps in a ladder from BAND to BOAT?

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q2
Logic, Set 33: Word ladders · MAIN · Q2/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word fills the gap in the ladder SENT, ?, RANT?

A. BENT
B. RENT
C. PANT
D. SEND

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q3
Logic, Set 33: Word ladders · MAIN · Q3/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Give a word from the list that is one step from both BEND and BEAT.

word = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q4
Logic, Set 33: Word ladders · MAIN · Q4/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. A ladder from LAND to BELT can be done in 3 steps.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-M-Q5
Logic, Set 33: Word ladders · MAIN · Q5/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word is one step from BEST?

A. REST
B. BOAT
C. POST
D. RENT

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q1
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

In how many positions do the words MOST and PART differ?

Positions: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q2
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

A ladder from MAST to LAST can be done in one step.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q3
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from PORT to MOST?

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q4
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

If two words differ in 3 positions, every ladder between them has at least 3 steps.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-B1-Q5
Logic, Set 33: Word ladders · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from BEND to PORT?

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q1
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. How many different shortest ladders go from BAND to BOAT?

Ladders: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q2
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word needs the MOST steps to reach from SAND (using shortest ladders)?

A. BOLT
B. PORT
C. MOST
D. LAST

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q3
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Give a word from the list that is exactly one step from LAND and also one step from SEND.

word = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q4
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. BOAT is one step from BOLT.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F1-Q5
Logic, Set 33: Word ladders · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from SAND to MOST?

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q1
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Suppose PANT is removed from the list. What is the least number of steps from RANT to PART now? Enter 0 if no ladder is possible.

Steps: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q2
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Every ladder from BAND to BOAT passes through BEND.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q3
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. How many words on the list are exactly one step from BENT?

Words: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q4
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

A ladder must go from a 4-letter word to another 4-letter word that differs in all 4 positions. What is the least possible number of steps?

A. 3
B. 8
C. 4
D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S33-F2-Q5
Logic, Set 33: Word ladders · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.WORD.02 (local) · MP1, MP7

Set 34: Translate clues into conditions

Local outcome LOGIC.CONSTRAINT.02: Translate puzzle clues into exact conditions and test candidate arrangements against them

Puzzle set · genre: clue translation · MP2, MP6

MAIN: Translate puzzle clues into exact conditions and test candidate arrangements against them
BACKWARD 1: Strategy: write each clue as a condition on numbers
FORWARD 1: ADVANCED APPLICATION: Harder clue words
FORWARD 2: EXPERT CHALLENGE: Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition says exactly the same thing as the clue "Exactly one locker lies between Cal and Dee"?

A. C - D = 2
B. |C - D| = 1
C. |C - D| >= 2
D. |C - D| = 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q1
Logic, Set 34: Translate clues into conditions · MAIN · Q1/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. Put the friends in locker order, from locker 1 to locker 5.

Put these in order:

A Ava
B Ben
C Cal
D Dee
E Eli

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q2
Logic, Set 34: Translate clues into conditions · MAIN · Q2/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. The arrangement Ava, Cal, Eli, Dee, Ben (lockers 1 to 5) breaks exactly one of the five clues.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q3
Logic, Set 34: Translate clues into conditions · MAIN · Q3/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. The arrangement Ava, Dee, Eli, Ben, Cal (lockers 1 to 5) breaks one clue. Which one?

A. Clue 2
B. Clue 5
C. Clue 1
D. Clue 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q4
Logic, Set 34: Translate clues into conditions · MAIN · Q4/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

A row has nine lockers numbered 1 to 9. Eli's locker must meet every condition: it is odd-numbered; it is not at either end (not 1 and not 9); and it is at least 2 lockers away from locker 5 (|E - 5| >= 2). Give one locker number Eli could have.

E = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-M-Q5
Logic, Set 34: Translate clues into conditions · MAIN · Q5/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ava is somewhere left of Ben"?

A. A = B - 1

B. A > B

C. A < B

D. A + 1 < B

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q1
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ben and Cal are next to each other"?

A. B - C = 1
B. |B - C| = 1
C. |B - C| = 2
D. |B - C| <= 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q2
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. With five lockers, the clue "Eli is not at either end" means E is 2, 3 or 4.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q3
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. The clue "Dee is directly left of Ava" means the same as D < A.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q4
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Dee is in locker 2, and exactly one locker lies between Cal and Dee. Which locker is Cal in?

Locker: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-B1-Q5
Logic, Set 34: Translate clues into conditions · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "At least two lockers lie between Ava and Eli"?

A. |A - E| >= 3

B. A - E >= 3

C. |A - E| = 3

D. |A - E| >= 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q1
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. In these puzzles, "X unless Y" means "if Y is false, then X is true." Which condition matches the clue "Ben is in locker 3 unless Cal is"?

A. C = 3

B. B = 3 or C = 3

C. B is not 3 and C is not 3

D. B = 3 and C is not 3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q2
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Six friends, Ava, Ben, Cal, Dee, Eli and Fay, each use one of six lockers numbered 1 to 6 from left to right. "Left of" means a lower number. Clues: (1) Ava is somewhere left of Ben. (2) Exactly two lockers lie between Cal and Dee. (3) Fay is somewhere right of Cal. (4) Dee is in an odd-numbered locker. (5) Cal is somewhere left of Ava. (6) Ben is not in locker 6. Put the friends in locker order, from locker 1 to locker 6.

Put these in order:

A Ava
B Ben
C Cal
D Dee
E Eli
F Fay

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q3
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. The two conditions A < B and B < A + 2 together say that Ben is directly right of Ava.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q4
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Seven lockers are numbered 1 to 7 from left to right. Give lockers for Ava and Ben that meet every clue: Ava is somewhere left of Ben; at least two lockers lie between them; Ben's locker is even-numbered.

A = _____

B = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F1-Q5
Logic, Set 34: Translate clues into conditions · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. If clue 3 is removed, how many arrangements fit the other four clues?

Arrangements: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q1
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. How many arrangements fit both clues "Eli is not at either end" and "Ben and Cal are next to each other" (and no other clues)?

Arrangements: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q2
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is in an even-numbered locker. If clue 5 is replaced by "Dee is in locker 2", the puzzle still has exactly one answer, and it is the same answer.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q3
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ava and Ben are not both at the ends"?

A. Exactly one of Ava and Ben is in locker 1 or 5

B. Ava and Ben are both in lockers 2, 3 or 4

C. Neither Ava nor Ben is in locker 1 or 5

D. At most one of Ava and Ben is in locker 1 or 5

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q4
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. In these puzzles, "X unless Y" means "if Y is false, then X is true." Under the clue "Ben is in locker 3 unless Cal is", an arrangement with Ava in locker 3 is allowed.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S34-F2-Q5
Logic, Set 34: Translate clues into conditions · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONSTRAINT.02 (local) · MP2, MP6

Set 35: Must, could, cannot

Local outcome LOGIC.CONSTRAINT.03: Decide what must be true, what could be true and what cannot be true in a constrained arrangement

Puzzle set · genre: scheduling puzzles · MP1, MP3

- MAIN: Decide what must be true, what could be true and what cannot be true in a constrained arrangement
- BACKWARD 1: Strategy: list every case, then read must, could, cannot
- FORWARD 1: ADVANCED APPLICATION: Must, could, cannot under an added condition
- FORWARD 2: EXPERT CHALLENGE: Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which statement MUST be true?

A. Kites is on Friday.

B. Fossils is on Monday.

C. Gears is on Monday or Tuesday.

D. Honey is on Friday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q1
Logic, Set 35: Must, could, cannot · MAIN · Q1/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which statement CANNOT be true?

A. Jazz is on Tuesday.

B. Kites is on Wednesday.

C. Fossils is on Friday.

D. Honey is on Wednesday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q2
Logic, Set 35: Must, could, cannot · MAIN · Q2/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Fossils could be on Friday.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q3
Logic, Set 35: Must, could, cannot · MAIN · Q3/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. How many complete schedules meet all four conditions?

Schedules: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q4
Logic, Set 35: Must, could, cannot · MAIN · Q4/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose Honey is on Wednesday. Using Monday = 1, Tuesday = 2, Wednesday = 3, Thursday = 4 and Friday = 5, give a day number Kites could be on.

Kites day = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-M-Q5
Logic, Set 35: Must, could, cannot · MAIN · Q5/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. How many schedules meet both conditions?

Schedules: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q1
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Then Q must be on day 1.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q2
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Then R must be on day 3.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q3
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

In a scheduling puzzle, what does it mean to say a statement CANNOT be true?

A. It is not mentioned in any of the conditions.
B. It is false in every schedule that meets the conditions.
C. It is false in at least one schedule that meets the conditions.
D. It is true in exactly one schedule that meets the conditions.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q4
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Which statement COULD be true?

A. P is on day 1.
B. R is on day 1.
C. P is on day 2.
D. Q is on day 3.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-B1-Q5
Logic, Set 35: Must, could, cannot · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Kites is on Monday, which statement MUST be true?

A. Fossils is on Wednesday.

B. Jazz is on Thursday.

C. Honey is on Friday.

D. Gears is on Tuesday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q1
Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Kites is on Friday, how many schedules are possible?

Schedules: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q2
Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Jazz is on Wednesday, then Honey must be on Friday.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q3
Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which ONE added condition would leave exactly one possible schedule?

A. Fossils is on Wednesday.

B. Kites is on Thursday.

C. Fossils is on Friday.

D. Honey is on Wednesday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q4
Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose Honey is on Friday. Using Monday = 1 to Friday = 5, give the day numbers of Gears and Jazz in one schedule that meets every condition.

Gears day = _____

Jazz day = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F1-Q5

Logic, Set 35: Must, could, cannot · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. How many of the five talks could be on Friday?

Talks: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q1
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose condition 3 is replaced by "Fossils is on Monday." Which statement must be true?

A. Kites is on Friday.

B. Honey is on Friday.

C. Jazz is on Wednesday or Thursday.

D. Gears is on Tuesday.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q2
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Removing condition 4 adds exactly two new schedules.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q3
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Add a fifth condition: no two of Gears, Honey and Jazz are on consecutive days. How many schedules remain?

Schedules: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q4
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Gears is on Monday, then Fossils and Kites could be on Tuesday and Thursday.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S35-F2-Q5
Logic, Set 35: Must, could, cannot · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.CONSTRAINT.03 (local) · MP1, MP3

Set 36: Search every case

Local outcome LOGIC.SEARCH.01: Organize a complete case analysis and count or find every solution

Puzzle set · genre: complete search · MP1, MP8

MAIN: Organize a complete case analysis and count or find every solution
BACKWARD 1: Strategy: make an organized list
FORWARD 1: ADVANCED APPLICATION: Counting with extra conditions
FORWARD 2: EXPERT CHALLENGE: Searches with many cases

How many three-digit whole numbers (100 to 999) have digits that add up to 5?

Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q1
Logic, Set 36: Search every case · MAIN · Q1/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

A 12-hour digital clock shows times from 1:00 to 12:59: the hour (1 to 12, no leading zero) and then two minute digits (00 to 59). At how many times do the shown digits add up to exactly 22?

Times: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q2
Logic, Set 36: Search every case · MAIN · Q2/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Find the two-digit whole number that equals 3 times the sum of its digits.

Number = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q3
Logic, Set 36: Search every case · MAIN · Q3/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

In how many ways can you make 30 cents using nickels (5 cents), dimes (10 cents) and quarters (25 cents)? You may use any number of each coin, including none; the order of the coins does not matter.

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q4
Logic, Set 36: Search every case · MAIN · Q4/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Exactly 6 of the 24 orderings of the letters A, B, C and D have A before B and also C before D.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-M-Q5
Logic, Set 36: Search every case · MAIN · Q5/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

How many two-digit numbers have digits that add up to 6?

Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q1
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

In how many ways can you make 20 cents using only nickels and dimes? The order of the coins does not matter.

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q2
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

There are exactly 4 two-digit numbers whose digits multiply to 12.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q3
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Ana lists the ways to make 15 cents from nickels and dimes: "3 nickels; 1 dime and 1 nickel." Is her list complete? The order of the coins does not matter.

A. No: a nickel then a dime is a third way

B. No: 2 dimes and no nickels is missing

C. No: 1 dime alone is missing

D. Yes: she used 0 dimes and 1 dime, and 2 dimes is too much

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q4
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

There are 5 two-digit numbers whose digits add up to 3.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-B1-Q5
Logic, Set 36: Search every case · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

How many three-digit numbers have three different digits that add up to 6?

Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q1
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

In how many ways can 10 be written as a sum of 1s and 2s if the order matters? (For example, 3 can be written 3 ways: 1 + 1 + 1, 1 + 2 and 2 + 1.)

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q2
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Find a three-digit number whose three digits are all odd, all different, and add up to 15.

Number = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q3
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

There are more ways to make 50 cents from nickels and dimes than from dimes and quarters. The order of the coins does not matter.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q4
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

How many rectangles of any size (including squares) can be traced along the lines of a grid that is 2 squares tall and 3 squares wide?

A. 15
B. 18
C. 17
D. 6

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F1-Q5
Logic, Set 36: Search every case · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

A 12-hour digital clock shows times from 1:00 to 12:59: the hour (1 to 12, no leading zero) and then two minute digits (00 to 59). At how many times do the shown digits add up to exactly 6?

Times: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q1
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

In how many ways can you make 1 dollar (100 cents) using nickels, dimes and quarters? The order of the coins does not matter.

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q2
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

How many positive whole numbers less than 1000 have digits that add up to 4?

Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q3
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Every whole number of cents from 8 to 30 can be paid exactly using only 3-cent and 5-cent stamps.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q4
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Give a whole number of cents from 1 to 10 that CANNOT be paid exactly using only 3-cent and 5-cent stamps.

Amount = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S36-F2-Q5
Logic, Set 36: Search every case · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SEARCH.01 (local) · MP1, MP8

Set 37: Direct proof with numbers

Local outcome LOGIC.PROOF.01: Build a direct argument about even, odd and multiples from definitions

Formal logic set · genre: formal logic · MP3, MP7

- MAIN: Build a direct argument about even, odd and multiples from definitions
- BACKWARD 1: Use the definitions of even and odd
- FORWARD 1: ADVANCED APPLICATION: Direct proofs about multiples
- FORWARD 2: EXPERT CHALLENGE: Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that the sum of two odd integers is even in order.

Put these in order:

A Let m = 2a + 1 and n = 2b + 1, where a and b are integers.

B Then m + n = 2a + 2b + 2.

C So m + n = 2(a + b + 1).

D Since a + b + 1 is an integer, m + n is even.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q1
Logic, Set 37: Direct proof with numbers · MAIN · Q1/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which first line correctly begins a direct proof that the product of any two odd integers is odd?

A. Let m = 2a + 1 and n = 2a + 1, where a is an integer.

B. Let m = 3 and n = 5.

C. Let m = 2a + 1 and n = 2b + 1, where a and b are integers.

D. Assume that m times n is odd.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q2
Logic, Set 37: Direct proof with numbers · MAIN · Q2/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. A proof that an even integer plus an odd integer is odd reaches the line m + n = 2a + 2b + 1. Which line correctly finishes it?

A. Since 4 + 3 = 7 is odd, m + n is odd.

B. m + n is odd because the expression 2a + 2b + 1 contains a 1.

C. m + n = 2(a + b + 1), so m + n is even.

D. m + n = 2(a + b) + 1, and a + b is an integer, so m + n is odd.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q3
Logic, Set 37: Direct proof with numbers · MAIN · Q3/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Checking that 3 + 5, 7 + 9 and 11 + 13 are all even proves that the sum of any two odd integers is even.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q4
Logic, Set 37: Direct proof with numbers · MAIN · Q4/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. To show that 6n + 4 is even for every integer n, write 6n + 4 = 2(pn + q). Give the numbers p and q.

p = _____

q = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q5
Logic, Set 37: Direct proof with numbers · MAIN · Q5/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which expression is odd for every integer k?

A. 2k + 7

B. k + 1

C. 2k + 4

D. 4k

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-B1-Q1
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. 2k + 6 is even for every integer k.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-B1-Q2
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Write 19 in the form 2k + 1. What is k?

k = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-B1-Q3
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. If n = 2k + 1, then n + 1 is odd.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-B1-Q4
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Which statement is the definition of an even integer?

A. n ends in the digit 0

B. n = 2k + 1 for some integer k

C. n leaves remainder 1 when divided by 2

D. n = 2k for some integer k

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-B1-Q5
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that if n is a multiple of 3, then n squared is a multiple of 9 in a correct order. (Two orders are correct.)

Put these in order:

A Let n = 3k, where k is an integer.

B Then n squared = 9k squared.

C k squared is an integer, because k is an integer.

D So n squared is 9 times an integer, a multiple of 9.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q1
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which is a correct direct proof that the sum of three consecutive integers is a multiple of 3?

A. n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1), and n + 1 is an integer.

B. 1 + 2 + 3 = 6, and 6 is a multiple of 3.

C. n + (n + 1) + (n + 2) = 3n + 3, which is odd.

D. n + n + n = 3n, which is a multiple of 3.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q2
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. If n is odd, then n squared minus 1 is a multiple of 8.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q3
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. (2a + 1)(2b + 1) = 2(2ab + a + b) + 1. For a = 3 and b = 4, what number is 2ab + a + b?

Number = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q4
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. A student proves that every multiple of 4 is even by writing n = 4k = 2(2k). What justifies the last sentence, "so n is even"?

A. n can be any number at all.

B. 2k is an integer, so n is 2 times an integer.

C. 4 is an even number.

D. k must be an even number.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q5
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that the square of an odd integer leaves remainder 1 when divided by 4 in order.

Put these in order:

A Let n = 2k + 1, where k is an integer.

B Then n squared = 4k squared + 4k + 1.

C So n squared = 4(k squared + k) + 1.

D Since k squared + k is an integer, n squared leaves remainder 1 when divided by 4.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F2-Q1
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Claim: the sum of two even integers is a multiple of 4. "Proof: Let m = 2k and n = 2k. Then m + n = 4k." What is wrong?

A. Nothing is wrong; the proof is correct.

B. It should have started with m = 4k.

C. It uses the same k for both, so it only covers the case m = n.

D. It should check an example before starting.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F2-Q2
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. The statement "n squared + n is even for every integer n" can be proved by two cases: n even and n odd.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F2-Q3
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. For odd n = 2k + 1, n squared + 4n + 3 = 2(2k squared + 6k + 4). For k = 5, what number is 2k squared + 6k + 4?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F2-Q4
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Give an integer n from 1 to 20 for which n squared + n + 1 is even. If no such n exists, enter 0.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F2-Q5
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Set 38: Tiling and covering

Local outcome LOGIC.SPATIAL.05: Decide which pieces cover a region exactly, and prove when no covering exists

Puzzle set · genre: tiling · MP1, MP7

MAIN: Decide which pieces cover a region exactly, and prove when no covering exists

BACKWARD 1: Strategy: color the board like a checkerboard

FORWARD 1: ADVANCED APPLICATION: Count tilings and use other pieces

FORWARD 2: EXPERT CHALLENGE: Prove impossibility and count harder tilings

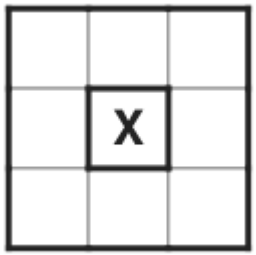
A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many different ways can a rectangle 2 squares tall and 4 squares wide be tiled with dominoes?



Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-M-Q1
Logic, Set 38: Tiling and covering · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. The 3 by 3 board with its center square removed (shown) can be tiled by 4 dominoes.

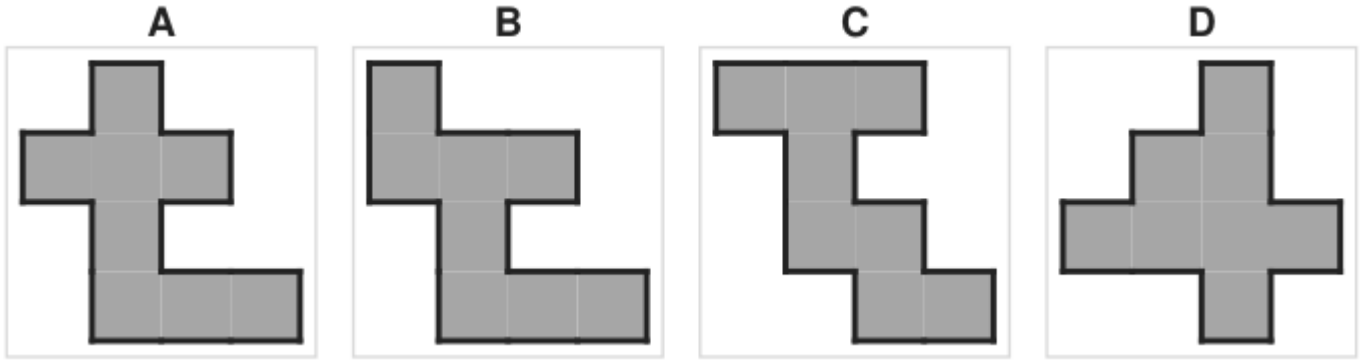


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-M-Q2
Logic, Set 38: Tiling and covering · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. Which region can be tiled by 4 dominoes?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-M-Q3
Logic, Set 38: Tiling and covering · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a 5 by 5 board, and the other 24 squares are to be tiled by 12 dominoes. Rows and columns are numbered 1 to 5. Give the row r and column c of a square whose removal makes a tiling possible.

Row r = _____

Column c = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-M-Q4
Logic, Set 38: Tiling and covering · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

An L-tromino is three squares in an L shape (a 2 by 2 square with one square missing). It may be turned any way. In how many ways can a rectangle 2 squares tall and 3 wide be tiled by two L-trominoes?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-M-Q5
Logic, Set 38: Tiling and covering · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A 4 by 4 board is colored like a checkerboard. How many of its squares are dark?

Dark: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-B1-Q1
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

Every domino placed on a checkerboard covers one dark square and one light square.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-B1-Q2
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region has 7 dark and 5 light squares on a checkerboard coloring. According to the color count, what is the greatest number of dominoes that could be placed on it without overlapping?

Dominoes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-B1-Q3
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region has 6 dark and 6 light squares. What does the color count tell you?

A. It can be tiled in exactly 6 ways.

B. It can certainly be tiled by 6 dominoes.

C. It does not rule out a tiling; you still have to find one or show none exists.

D. It cannot be tiled by dominoes.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-B1-Q4
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region with an even number of squares can always be tiled by dominoes.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-B1-Q5
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many ways can a strip 2 squares tall and 6 squares wide be tiled by dominoes?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F1-Q1
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A straight tromino covers three squares in a row. A 3 by 3 square can be tiled by three straight trominoes in exactly 2 ways.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F1-Q2
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A straight tromino covers three squares in a row, flat or upright. One corner square is removed from a 4 by 4 board. Can the other 15 squares be tiled by 5 straight trominoes?

A. Yes, and there is more than one way.

B. No: the corner gap blocks every tiling.

C. No: 15 squares cannot be split into trominoes.

D. Yes, but in exactly one way.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F1-Q3
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A straight tromino covers three squares in a row, flat or upright. In how many ways can a rectangle 3 squares tall and 4 wide be tiled by straight trominoes?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F1-Q4
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A strip 2 squares tall and n squares wide can be tiled by dominoes in exactly 8 ways. What is n?

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F1-Q5
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many ways can a rectangle 3 squares tall and 4 wide be tiled by dominoes?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F2-Q1
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A board 4 rows by 5 columns has two squares removed: row 1, column 1 and row 3, column 3. The other 18 squares can be tiled by 9 dominoes.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F2-Q2
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

An L-tromino is three squares in an L shape, turned any way. In how many ways can a strip 2 squares tall and 9 wide be tiled by L-trominoes?

A. 6
B. 8
C. 9
D. 3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F2-Q3
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a 3 by 3 board. Rows and columns are numbered 1 to 3. Give a square (row r, column c) whose removal leaves 8 squares that can be tiled by dominoes.

Row r = _____

Column c = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F2-Q4
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a board 3 rows by 5 columns. For how many of the 15 squares does removing it leave 14 squares that can be tiled by dominoes?

Squares: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F2-Q5
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

Set 39: Justify each step of solving an equation

Local outcome LOGIC.PROOF.02: Explain each step in solving an equation and justify the solution method

Formal logic set · genre: formal logic · MP3, MP6

MAIN: Explain each step in solving an equation and justify the solution method
BACKWARD 1: Properties of equality
FORWARD 1: ADVANCED APPLICATION: Steps that need care
FORWARD 2: EXPERT CHALLENGE: Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the steps for solving $4x + 7 = 2x + 19$ in order.

Put these in order:

A $2x + 7 = 19$ (subtract $2x$ from both sides)

B $2x = 12$ (subtract 7 from both sides)

C $x = 6$ (divide both sides by 2)

D Check: $4(6) + 7 = 31$ and $2(6) + 19 = 31$

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q1
Logic, Set 39: Justify each step of solving an equation · MAIN · Q1/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Which reason justifies going from 5x - 4 = 21 to 5x = 25?

A. Use the distributive property on 5x - 4.

B. Multiply both sides by 4.

C. Rewrite 5x - 4 as 4 - 5x.

D. Add 4 to both sides; equal numbers plus 4 are still equal.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q2
Logic, Set 39: Justify each step of solving an equation · MAIN · Q2/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. A student writes: Line 1: 2(x + 3) = 14. Line 2: 2x + 3 = 14. Line 3: 2x = 11. Line 4: x = 5.5. Which line is the first one that does NOT follow from the line before it?

A. Line 2

B. None; every line follows

C. Line 4

D. Line 3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q3
Logic, Set 39: Justify each step of solving an equation · MAIN · Q3/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. From 6 - x = 10, the next line "x = 10 - 6" follows correctly.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q4
Logic, Set 39: Justify each step of solving an equation · MAIN · Q4/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 7 - 2(x - 1) = x - 3. Give x.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q5
Logic, Set 39: Justify each step of solving an equation · MAIN · Q5/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

If a = b, which statement must also be true?

A. a / 5 = b / 3

B. 5a = b

C. a - 5 = b - 5

D. a - 5 = b + 5

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q1
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

If $3x = 12$, then $x = 4$ follows by dividing both sides by 3.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q2
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Which step uses the distributive property?

A. $4x = 20$ becomes $x = 5$

B. $5(x - 2) = 13$ becomes $5x - 10 = 13$

C. $x + 3 = 8$ becomes $x = 5$

D. $x + 2 = 9$ becomes $2 + x = 9$

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q3
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Multiplying both sides of an equation by 0 always gives an equation with the same solutions.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q4
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

To solve x / 4 = 9 in one step, by what number do you multiply both sides?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q5
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. A student solves x squared = 5x by dividing both sides by x and gets x = 5. What is wrong?

A. Nothing; x = 5 is the only solution.

B. The answer should be x = 0 only.

C. x squared = 5x has no solutions.

D. Dividing by x assumes x is not 0, and x = 0 is also a solution.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q1
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

The equation $3(x + 2) = 3x + 6$ is true for every number x.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q2
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 2(x - 4) + 3 = 5x + 7. What is x?

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q3
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solving $4x + 1 = 4x - 3$ leads to the line $1 = -3$. What does this show?

A. The solution is $x = -3$.

B. Every number is a solution.

C. The equation has no solution.

D. The solution is $x = 0$.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q4
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the lines of this solution of (x + 5) / 3 = 4 in order.

Put these in order:

A Assume x is a number with (x + 5) / 3 = 4.

B x + 5 = 12 (multiply both sides by 3)

C x = 7 (subtract 5 from both sides)

D Check: (7 + 5) / 3 = 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q5
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. To solve the equation (square root of (x + 2)) = x, a student squares both sides: x + 2 = x squared, so x = 2 or x = -1. Which answer is correct?

A. Only x = 2; x = -1 gives square root of 1 = 1, not -1.

B. Neither one.

C. Only x = -1.

D. Both x = 2 and x = -1.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q1
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. If each line of a solution follows from the line before it, then every number found at the end must solve the original equation.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q2
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it.
Solve 3(2x - 1) = 4(x + 2) - 1. Give x.

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q3
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Which step can give an equation with MORE solutions than the one before it?

A. Dividing both sides by 2
B. Adding 7 to both sides
C. Multiplying both sides by -3
D. Squaring both sides

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q4
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the lines of this solution of 5 - 3(x - 1) = 2x - 2 in order.

Put these in order:

A 5 - 3x + 3 = 2x - 2 (distributive property)

B 8 - 3x = 2x - 2 (combine 5 and 3)

C 10 = 5x (add 3x and 2 to both sides)

D x = 2 (divide both sides by 5)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q5
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Set 40: Proof by contradiction

Local outcome LOGIC.PROOF.03: Set up and complete an argument by contradiction

Formal logic set · genre: formal logic · MP3

MAIN: Set up and complete an argument by contradiction
BACKWARD 1: Strategy: assume the opposite and look for the impossible
FORWARD 1: ADVANCED APPLICATION: Contradiction with divisibility
FORWARD 2: EXPERT CHALLENGE: Contradiction in larger arguments

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "there is no largest even integer" by contradiction, how should the proof begin?

A. Show that 100 is an even integer.
B. Assume there is a largest even integer; call it N.
C. Assume there is no largest even integer.
D. Assume every even integer is less than 100.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q1
Logic, Set 40: Proof by contradiction · MAIN · Q1/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of a proof that if n squared is even, then n is even, in order.

Put these in order:

A Assume n squared is even, but n is odd.

B Then n = 2k + 1 for some integer k.

C So n squared = 2(2k squared + 2k) + 1, which is odd.

D This contradicts n squared being even, so n must be even.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q2
Logic, Set 40: Proof by contradiction · MAIN · Q2/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Reaching a statement that contradicts a known fact shows that the assumption was false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q3
Logic, Set 40: Proof by contradiction · MAIN · Q3/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Claim: no integer is both even and odd. Suppose n = 2a and n = 2b + 1 for integers a and b. Which line gives the contradiction?

A. 2a = 2b + 1, so a = b.

B. n = 4 is even and not odd.

C. 2a = 2b + 1, so 2(a - b) = 1, but 1 is not 2 times an integer.

D. 2a + 2b + 1 is an odd number.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q4
Logic, Set 40: Proof by contradiction · MAIN · Q4/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To show that 15 is not the sum of two even integers, assume 15 = 2a + 2b for integers a and b. Then a + b must equal what number?

a + b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q5
Logic, Set 40: Proof by contradiction · MAIN · Q5/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "n is not divisible by 4" by contradiction, what do you assume?

A. n is divisible by 4.

B. n is not divisible by 4.

C. n is divisible by 2.

D. n is an odd number.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q1
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "there is no integer x with 2x = 7" by contradiction, what do you assume?

A. Every integer x has 2x = 7.
B. There is an integer x with 2x not equal to 7.
C. There is no integer x with 2x = 7.
D. There is an integer x with 2x = 7.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q2
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. A proof by contradiction begins by assuming that the statement to be proved is true.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q3
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. If assuming P leads to 1 = 0, then P is false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q4
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Assume there is a largest odd integer M. Which number gives the contradiction?

A. 2M

B. M + 2

C. M - 2

D. M + 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q5
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove that no integers a and b satisfy 6a + 9b = 10, which fact gives the contradiction?

A. 6a + 9b = 3(2a + 3b) is a multiple of 3, but 10 is not.

B. a and b would have to be fractions.

C. 10 is even, and 6a + 9b is always odd.

D. 6 and 9 are both odd numbers.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q1
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of the proof that the square root of 2 is not a fraction in order.

Put these in order:

A Assume the square root of 2 = p / q, with p and q integers in lowest terms.
B Then p squared = 2q squared, so p squared is even, so p is even: p = 2m.
C Then 4m squared = 2q squared, so q squared = 2m squared, and q is even.
D Both p and q are even, which contradicts lowest terms.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q2
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Among any 5 integers, two of them leave the same remainder when divided by 4.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q3
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.03 (local) · MP3

A perfect square is divided by 4. What is the largest remainder (0, 1, 2 or 3) that can NEVER occur?

Remainder: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q4
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Give an integer n from 1 to 50 for which n squared + 1 is a multiple of 3. If no such n exists, enter 0.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q5
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Assume there are only finitely many primes, p1, p2, ..., pk, and let N = p1 x p2 x ... x pk + 1. Why is this a contradiction?

A. N is always an even number.

B. N is always a prime number.

C. N is larger than every prime, so N is not prime.

D. N leaves remainder 1 when divided by each listed prime, so a prime factor of N is not on the list.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q1
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.03 (local) · MP3

2 x 3 x 5 x 7 x 11 x 13 + 1 is a prime number.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q2
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of a proof that if $a + b$ is at least 20, then a is at least 10 or b is at least 10, in order.

Put these in order:

A Assume $a + b$ is at least 20, but $a < 10$ and $b < 10$.

B Adding the two inequalities gives $a + b < 20$.

C That contradicts $a + b$ being at least 20.

D So a is at least 10 or b is at least 10.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q3
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Claim: every integer is even. "Proof: Assume some integer is odd, say 3. Then 3 = 2k for no integer k. Contradiction." What is wrong?

A. Nothing; the proof is correct.

B. Nothing impossible was reached; 3 not being 2k agrees with 3 being odd.

C. It should assume that every integer is odd.

D. It should use 5 instead of 3.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q4
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.03 (local) · MP3

At least how many integers must you choose to be sure that two of them leave the same remainder when divided by 6?

Integers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q5
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Set 41: Parity and alternation

Local outcome LOGIC.INVARIANT.01: Use odd/even and alternation to predict outcomes and rule out possibilities

Puzzle set · genre: parity · MP7, MP8

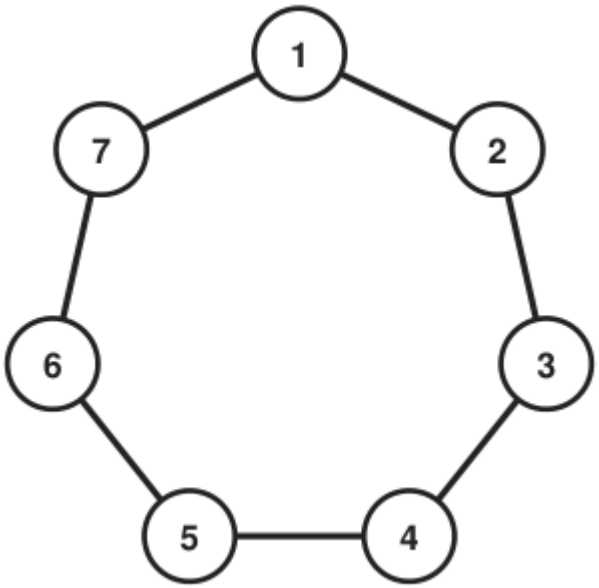
MAIN: Use odd/even and alternation to predict outcomes and rule out possibilities

BACKWARD 1: Strategy: keep track of odd and even only

FORWARD 1: ADVANCED APPLICATION: Parity on boards and rings

FORWARD 2: EXPERT CHALLENGE: Prove impossibility with parity

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Seven gears form a closed ring, each touching the gears on either side (figure). Gear 1 is turned clockwise. What happens?



A. Only gears 1, 3, 5 and 7 turn.

B. Every gear turns, and the directions alternate around the ring.

C. The gears jam: going around the ring, gear 1 would have to turn both ways.

D. Every gear turns clockwise.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-M-Q1
Logic, Set 41: Parity and alternation · MAIN · Q1/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Twelve gears stand in a row (not a ring), each touching the next. Gear 1 turns clockwise. How many of the 12 gears turn counterclockwise?

Gears: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-M-Q2
Logic, Set 41: Parity and alternation · MAIN · Q2/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A frog starts on 0 on a number line and makes exactly 9 jumps, each 1 unit left or right. It can end on 4.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-M-Q3
Logic, Set 41: Parity and alternation · MAIN · Q3/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A lamp starts OFF. Each press of its switch changes it (OFF to ON, or ON to OFF). Give a number of presses from 1 to 20 that leaves the lamp ON.

Presses = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-M-Q4
Logic, Set 41: Parity and alternation · MAIN · Q4/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A token moves along a row of squares numbered 1 to 15. Each move jumps exactly 2 squares left or right (staying on the row). It starts on square 4. How many of the 15 squares can it ever reach, counting square 4?

Squares: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-M-Q5
Logic, Set 41: Parity and alternation · MAIN · Q5/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

The sum of 5 odd numbers is odd.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-B1-Q1
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A coin starts heads up. It is turned over 13 times. Which side is up at the end?

A. Either side, depending on how it is turned
B. Tails
C. Heads
D. It cannot be known without trying

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-B1-Q2
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Nine gears stand in a row, each touching the next. Gear 1 turns clockwise. What is the number of the last gear in the row that turns clockwise?

Gear: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-B1-Q3
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

An even number plus an odd number is even.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-B1-Q4
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Which total CANNOT be the sum of three odd numbers?

A. 15
B. 9
C. 10
D. 3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-B1-Q5
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A token starts on a corner square of a 5 by 5 board. Each move goes to a square sharing a side (going back is allowed). After exactly 7 moves, how many different squares could the token be on?

Squares: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F1-Q1
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Ten gears form a closed ring, each touching the gears on either side. They can all turn.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F1-Q2
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Seven cups stand upside down. A move turns over exactly 4 cups (any 4). Can all seven cups end right side up?

A. No: each move changes the number of upside-down cups by an even amount, so it stays odd.

B. Yes, after exactly 7 moves.

C. Yes: turn over 4 cups, then 3 cups.

D. No: 7 is not a multiple of 4.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F1-Q3
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Put a + or - sign in front of each of the numbers 1, 2, 3, ..., 10 and add. Give a total from 0 to 10 that can be made this way.

Total = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F1-Q4
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Nine gears are arranged in a 3 by 3 square, and each gear touches the gears directly beside, above and below it. A corner gear turns clockwise. How many gears turn clockwise, including the corner gear?

Gears: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F1-Q5
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A token must visit every square of a 5 by 5 board exactly once, each move going to a square that shares a side. It starts on the square next to a corner (row 1, column 2). Is this possible?

A. Yes, but only if it ends on a corner.

B. Yes: go along row 1, then snake back and forth.

C. No: 25 is not a multiple of 5.

D. No: the path must start and end on the corner color, and this square has the other color.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F2-Q1
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Put a + or - sign in front of each of the numbers 1, 2, 3, ..., 10 and add. How many different totals can be made?

Totals: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F2-Q2
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

On a square grid, a path that moves one square up, down, left or right at each step, and ends where it started, always has an even number of steps.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F2-Q3
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Give a number of gears n from 3 to 12 for which a closed ring of n gears, each touching its two neighbors, would jam.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F2-Q4
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A bug walks along the edges of a cube from corner to corner. It starts at one corner and makes exactly 5 moves (going back is allowed). How many of the 8 corners could it end on?

Corners: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F2-Q5
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Set 42: Disprove a claim with a counterexample

Local outcome LOGIC.PROOF.04: Find a counterexample that disproves a general mathematical claim

Formal logic set · genre: formal logic · MP3

MAIN: Find a counterexample that disproves a general mathematical claim
BACKWARD 1: What a counterexample is
FORWARD 1: ADVANCED APPLICATION: Counterexamples to algebra rules
FORWARD 2: EXPERT CHALLENGE: Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: for every whole number n, n squared + n + 11 is prime. Which n is a counterexample?

A. 1
B. 10
C. 3
D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-M-Q1
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q1/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: for all numbers a and b, if a > b, then a squared > b squared. Give a counterexample.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-M-Q2
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q2/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. The claim "every odd number greater than 1 is prime" is disproved by the number 9.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-M-Q3
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q3/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: 2 to the power n, plus 1, is prime for every whole number n from 1 to 6. What is the smallest n that disproves it?

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-M-Q4
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q4/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: every multiple of 4 is a multiple of 8. Which number is NOT a counterexample?

A. 16
B. 4
C. 12
D. 20

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-M-Q5
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q5/5 · Outcome LOGIC.PROOF.04 (local) · MP3

What is a counterexample to the claim "every number with property P has property Q"?

A. A number with property Q that does not have property P

B. A number with neither property

C. A number with both properties

D. A number with property P that does not have property Q

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-B1-Q1
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.PROOF.04 (local) · MP3

One counterexample is enough to show that a claim about every case is false.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-B1-Q2
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.PROOF.04 (local) · MP3

If a claim works for five examples, it is proved true for every case.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-B1-Q3
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Which number is a counterexample to "all prime numbers are odd"?

A. 3
B. 9
C. 2
D. 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-B1-Q4
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: every whole number divisible by 3 is divisible by 6. Give a counterexample from 1 to 30.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-B1-Q5
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: (a + b) squared = a squared + b squared for all numbers a and b. Give whole numbers a and b that disprove it.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F1-Q1
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: the square root of (a + b) equals the square root of a plus the square root of b, for all numbers a and b that are 0 or more. Which pair is a counterexample?

A. a = 0, b = 4

B. a = 9, b = 16

C. a = 0, b = 0

D. a = 1, b = 0

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F1-Q2
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: n factorial plus 1 is prime for every positive whole number n. (n factorial is 1 x 2 x ... x n.) What is the smallest n that disproves it?

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F1-Q3
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. The claim "if n squared is a multiple of 4, then n is a multiple of 4" is disproved by n = 2.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F1-Q4
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: every rectangle with perimeter 20 has area 25. Which rectangle is a counterexample?

A. 2 by 10
B. 3 by 8
C. 5 by 5
D. 4 by 6

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F1-Q5
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: for whole numbers a, b and c, if a divides b x c, then a divides b or a divides c. Give a counterexample with a, b and c each from 2 to 12.

a = _____

b = _____

c = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F2-Q1
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: 2 to the power n, minus 1, is prime whenever n is prime. The smallest counterexample is n = 11.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F2-Q2
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.04 (local) · MP3

Which claim is FALSE?

A. The sum of two even numbers is even.

B. If n squared is odd, then n is odd.

C. If n is prime, then n + 2 is prime.

D. If n is odd, then n squared is odd.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F2-Q3
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: every whole number from 1 to 30 is a sum of two square numbers (0, 1, 4, 9, 16, 25, ...; a square may be used twice). What is the smallest counterexample?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F2-Q4
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: for every positive whole number n, 2 to the power n is greater than n. Why is n = -1 NOT a valid counterexample?

A. The claim is only about positive whole numbers, and -1 is not one.

B. Because a counterexample must be a large number.

C. Because counterexamples cannot be negative.

D. Because 2 to the power -1 is greater than -1.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F2-Q5
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.04 (local) · MP3

Set 43: Proofs about lines and angles

Local outcome LOGIC.PROOF.05: Order and justify the steps of proofs about angles formed by intersecting and parallel lines

Formal logic set · genre: formal logic · MP3

MAIN: Order and justify the steps of proofs about angles formed by intersecting and parallel lines
BACKWARD 1: Name the angle pairs
FORWARD 1: ADVANCED APPLICATION: Perpendicular bisectors and converses
FORWARD 2: EXPERT CHALLENGE: Complete and check angle proofs

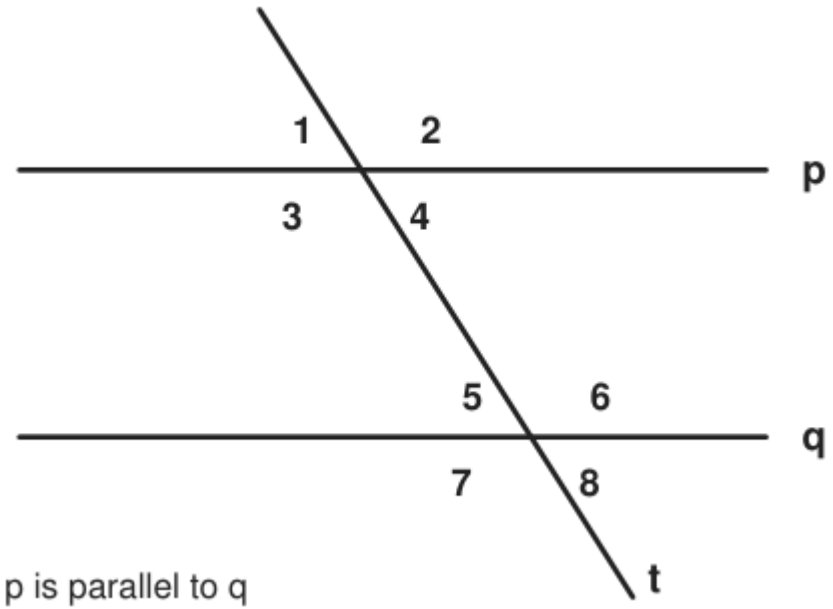
Lines meet at a point, forming angle 1 and angle 2 as vertical angles, and angle 3 next to both (angle 3 forms a linear pair with angle 1 and with angle 2). Put the steps of the proof that angle 1 = angle 2 in a correct order. (Two orders are correct.)

Put these in order:

A angle 1 + angle 3 = 180 degrees (linear pair)
B angle 2 + angle 3 = 180 degrees (linear pair)
C angle 1 + angle 3 = angle 2 + angle 3 (both sums equal 180 degrees)
D angle 1 = angle 2 (subtract angle 3 from both sides)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q1
Logic, Set 43: Proofs about lines and angles · MAIN · Q1/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Which pair of angles are alternate interior angles?

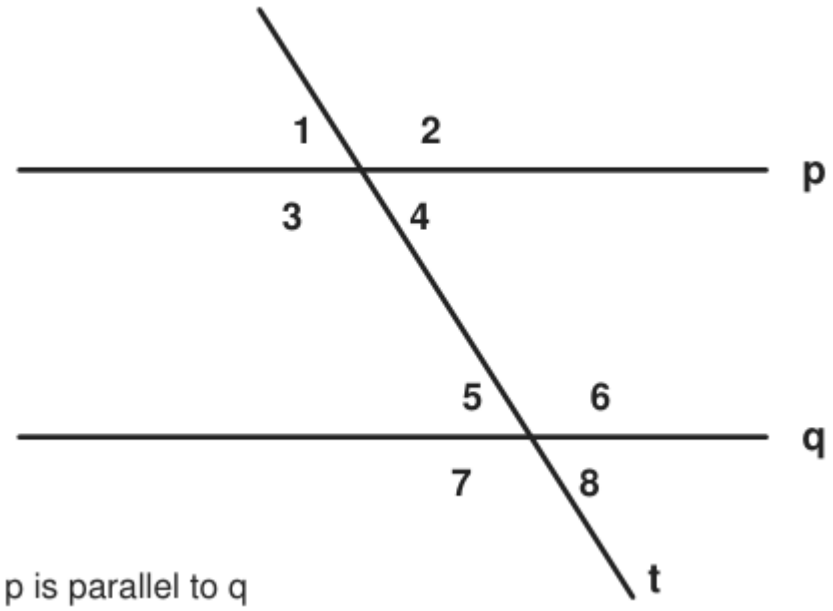


p is parallel to q

A. Angles 2 and 3
B. Angles 4 and 5
C. Angles 4 and 8
D. Angles 1 and 5

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q2
Logic, Set 43: Proofs about lines and angles · MAIN · Q2/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. If angle 2 = 70 degrees, then angle 7 = 70 degrees.



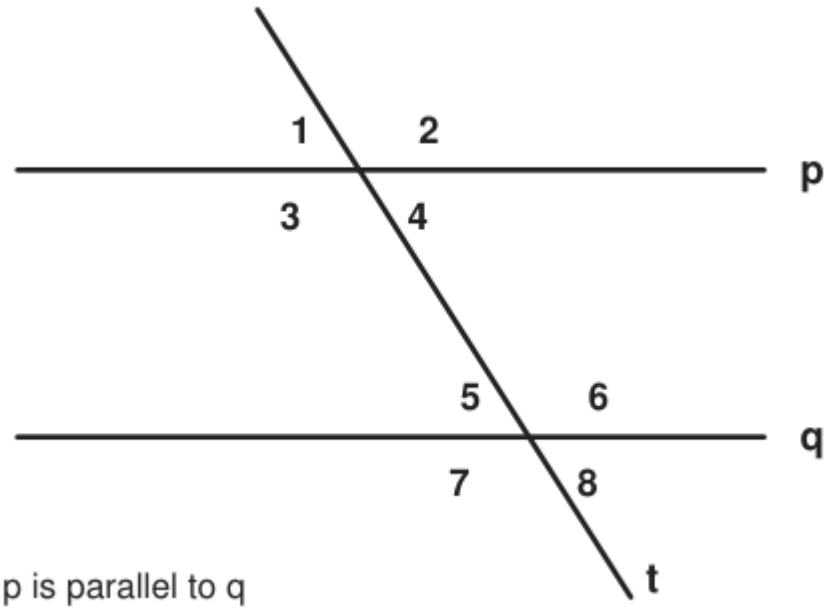
p is parallel to q

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q3
Logic, Set 43: Proofs about lines and angles · MAIN · Q3/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel, and angle 1 = 115 degrees. Give the measure of angle 6 in degrees.

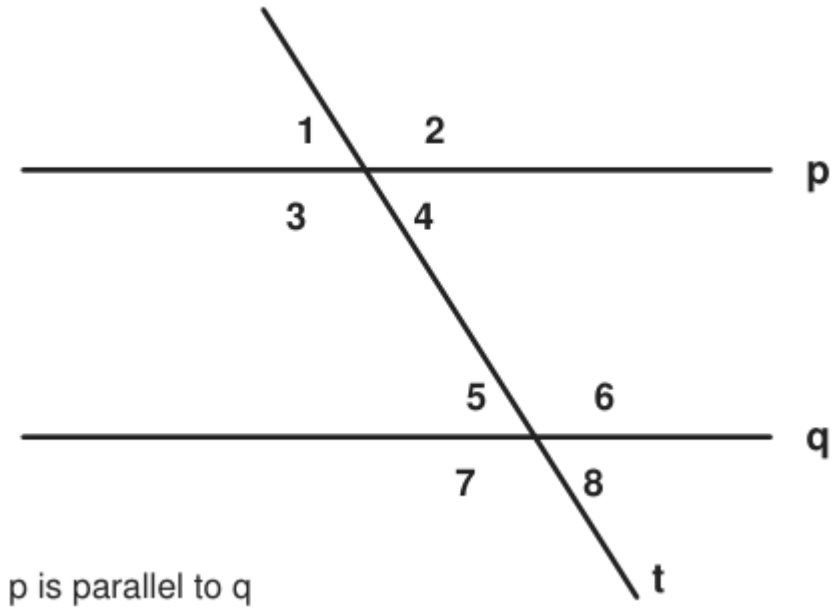


p is parallel to q

angle 6 = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q4
Logic, Set 43: Proofs about lines and angles · MAIN · Q4/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Put the steps of the proof that angle 3 = angle 6 when p is parallel to q in a correct order. (Two orders are correct.)



p is parallel to q

Put these in order:

A p is parallel to q, cut by transversal t (given)
B angle 2 = angle 6 (corresponding angles, p parallel to q)
C angle 2 = angle 3 (vertical angles)
D angle 3 = angle 6 (both equal angle 2)

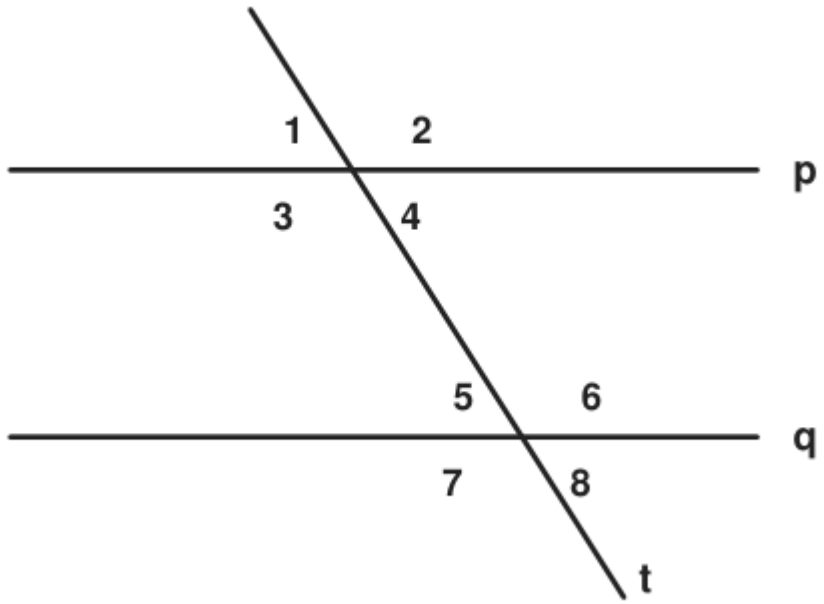
REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q5
Logic, Set 43: Proofs about lines and angles · MAIN · Q5/5 · Outcome LOGIC.PROOF.05 (local) · MP3

Two angles form a linear pair (together they make a straight line). One measures 48 degrees. What does the other measure, in degrees?

Degrees: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q1
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angles 1 and 4 are vertical angles.

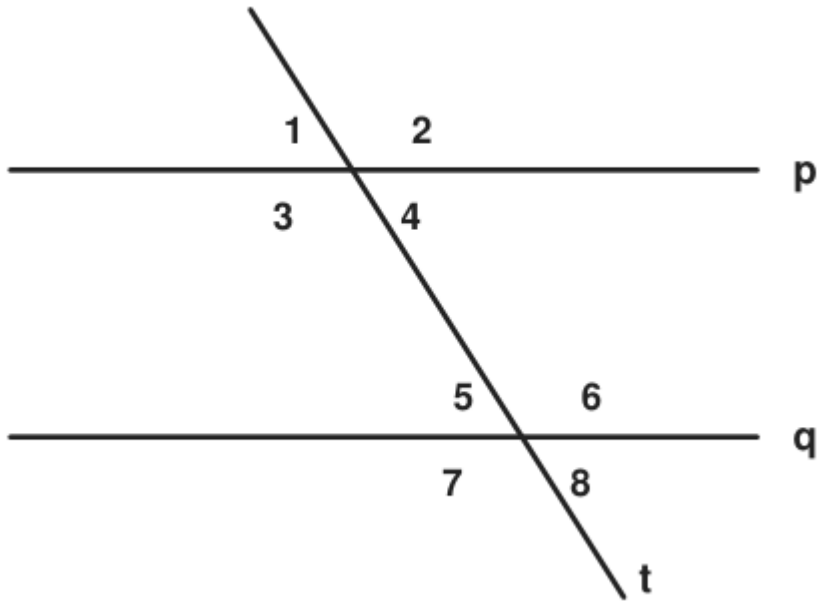


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q2
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.PROOF.05 (local) · MP3

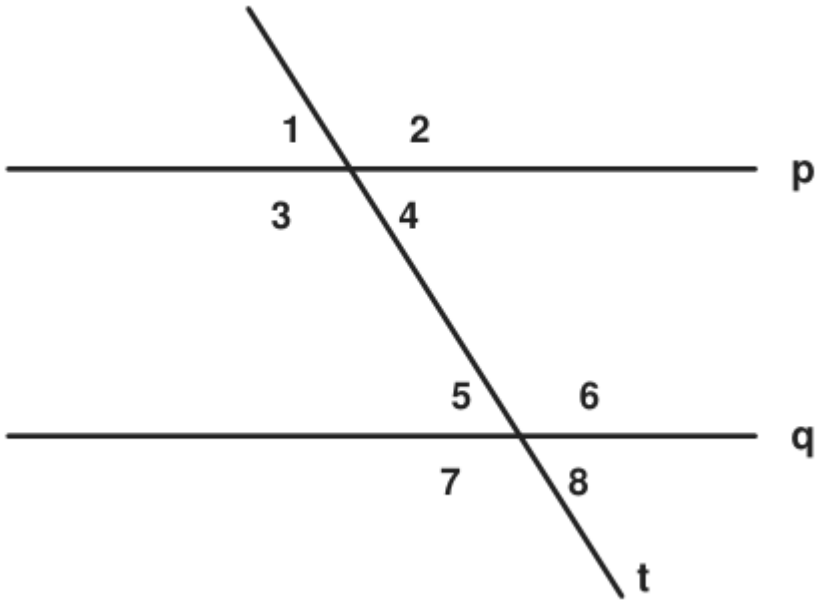
In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). What kind of angle pair are angles 3 and 7?



A. Vertical angles
B. A linear pair
C. Corresponding angles
D. Alternate interior angles

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q3
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angles 3 and 5 are alternate interior angles.

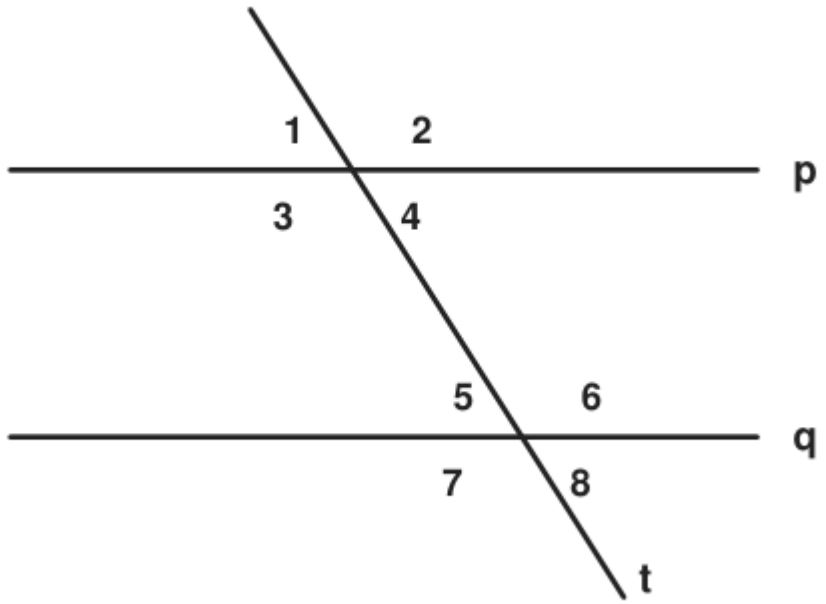


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q4
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angle 3 = 2x + 10 degrees and angle 2 = 3x - 15 degrees. They are vertical angles. Find x.



x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q5
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.PROOF.05 (local) · MP3

M is the midpoint of segment AB, and line PM is perpendicular to AB, so P is on the perpendicular bisector of AB. Put the steps of the proof that PA = PB in a correct order. (Two orders are correct.)

Put these in order:

A AM = BM, and angles PMA and PMB are right angles (midpoint; perpendicular)
B PM = PM (a segment equals itself)
C Triangle PMA is congruent to triangle PMB (SAS)
D PA = PB (corresponding parts of congruent triangles)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q1
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.05 (local) · MP3

Theorem: a point is on the perpendicular bisector of AB exactly when it is the same distance from A and B. Point P has PA = PB = 7. What follows?

A. Nothing follows without more information.

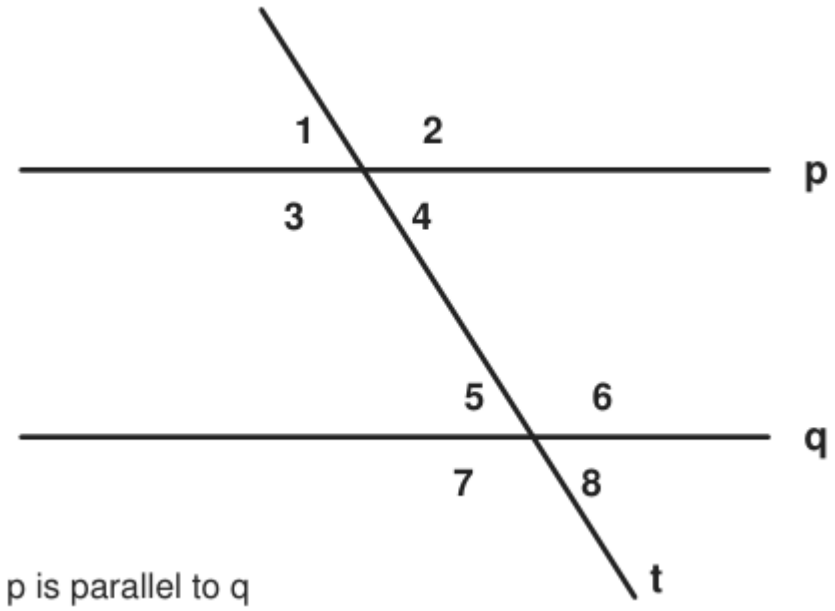
B. P is the midpoint of AB.

C. AB = 7.

D. P is on the perpendicular bisector of AB.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q2
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Then angle 4 + angle 6 = 180 degrees.

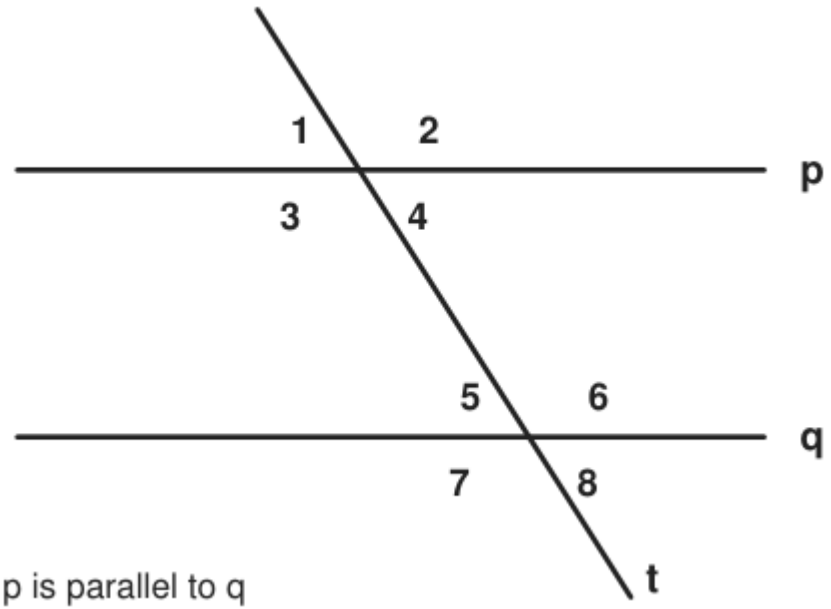


A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q3
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Angle 4 = 3x + 5 degrees and angle 5 = 5x - 35 degrees. Find x.

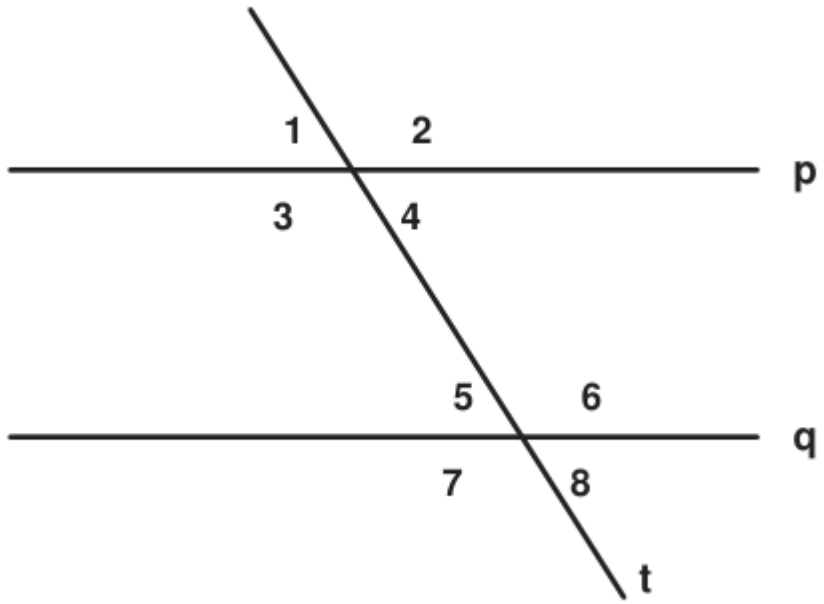


p is parallel to q

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q4
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Here p and q are NOT known to be parallel. Which fact would prove that they are parallel?



A. angle 3 = angle 6

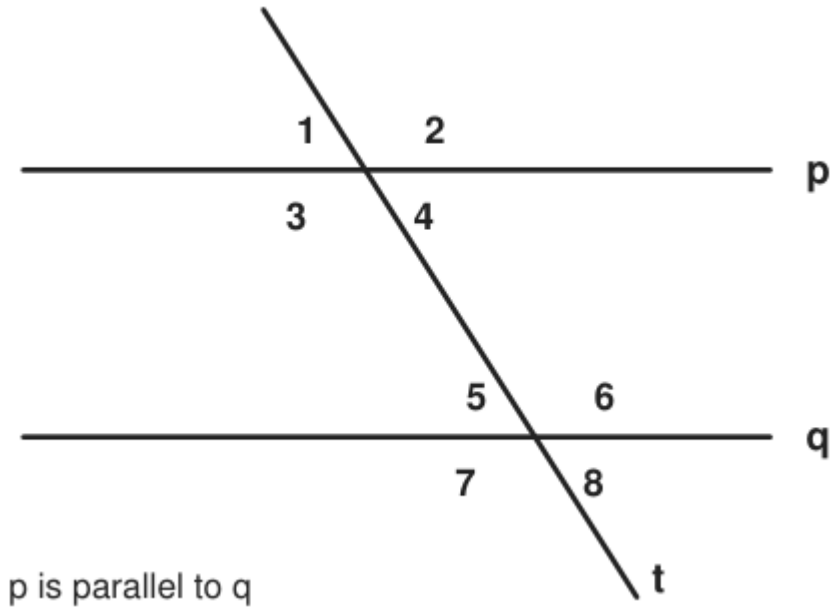
B. angle 1 + angle 2 = 180 degrees

C. angle 1 = angle 4

D. angle 5 = angle 8

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q5
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Put the steps of the proof that angle 4 + angle 6 = 180 degrees when p is parallel to q in a correct order. (Two orders are correct.)



p is parallel to q

Put these in order:

A angle 4 = angle 5 (alternate interior angles, p parallel to q)

B angle 5 + angle 6 = 180 degrees (linear pair)

C angle 4 + angle 6 = 180 degrees (substitute angle 4 for angle 5)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q1
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.05 (local) · MP3

A student writes: "Angle 1 = angle 4 because they look the same size in the diagram." What is wrong with this step?

A. A diagram is not a reason; the step needs the vertical angles theorem.

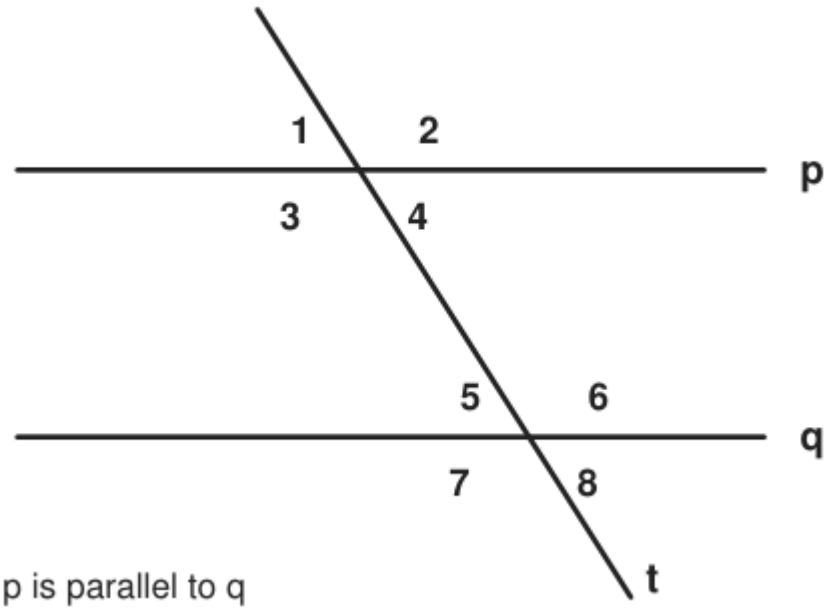
B. The step should say angle 1 + angle 4 = 180 degrees.

C. Nothing; looking at the diagram is enough.

D. Angles 1 and 4 are never equal.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q2
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Angle 1 = 2x + 30 degrees and angle 6 = x + 15 degrees. Find x.

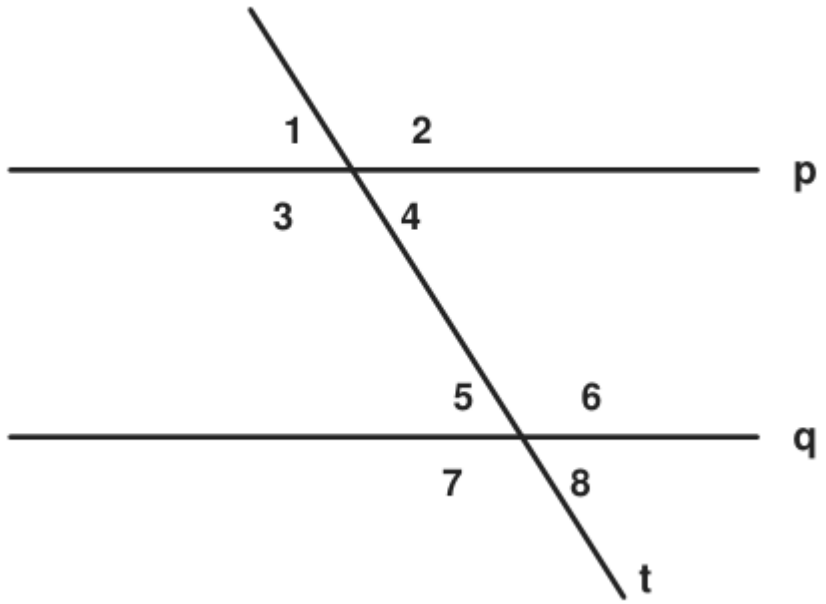


p is parallel to q

x = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q3
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Without knowing that p and q are parallel, one can conclude that angle 3 = angle 6.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q4
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.05 (local) · MP3

P is on the perpendicular bisector of segment AB. PA = 3x - 4 and PB = x + 10. What is PA?

PA = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q5
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.05 (local) · MP3

Set 44: Invariants and impossibility

Local outcome LOGIC.INVARIANT.02: Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

Puzzle set · genre: invariants · MP7, MP3

MAIN: Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

BACKWARD 1: Strategy: look for something that never changes

FORWARD 1: ADVANCED APPLICATION: Invariants with remainders

FORWARD 2: EXPERT CHALLENGE: Prove targets impossible

The numbers 1, 2, 3, ..., 10 are written on a board. A move erases any two numbers and writes their sum. After 9 moves one number is left. What is it?

A. It depends on the moves chosen

B. 55

C. 10

D. 45

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q1
Logic, Set 44: Invariants and impossibility · MAIN · Q1/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

The numbers 1, 2, ..., 10 are written on a board. A move erases two numbers and writes their difference (larger minus smaller). After 9 moves one number is left. It can be 0.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q2
Logic, Set 44: Invariants and impossibility · MAIN · Q2/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Seven coins lie in a row: H H T H T T T (H = heads, T = tails). A move turns over any two coins that are next to each other. What is the smallest number of heads that can ever be showing?

Heads: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q3
Logic, Set 44: Invariants and impossibility · MAIN · Q3/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Seven coins lie in a row: H H T H T T T. A move turns over any two coins that are next to each other. Give a number of heads (0 to 7) that can be showing after some moves.

Heads = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q4
Logic, Set 44: Invariants and impossibility · MAIN · Q4/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A robot starts at the point (0, 0). Each move takes it from (x, y) to (x + 1, y + 2) or to (x + 2, y + 1). Which point can it reach?

A. (5, 5)
B. (2, 3)
C. (4, 5)
D. (7, 0)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q5
Logic, Set 44: Invariants and impossibility · MAIN · Q5/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A counter shows 12. A move adds 4 or subtracts 6. Can it ever show 15?

A. No: 15 is not a multiple of 4.

B. Yes: after enough moves.

C. Yes: add 4, then subtract 1.

D. No: every move keeps the number even, and 15 is odd.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q1
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Three numbers on a board are 2, 4 and 9. A move adds 2 to any one of them. The total of the three numbers always stays odd.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q2
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A move erases two numbers a and b on a board and writes a + b. The board starts with 3, 5, 6 and 8. What number is left after 3 moves?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q3
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

If a quantity is the same after every allowed move, it can never change, however many moves are made.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q4
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A robot starts at (0, 0). Each move goes from (x, y) to (x + 2, y) or to (x, y + 2). Which point can it NEVER reach?

A. (0, 6)
B. (3, 4)
C. (2, 4)
D. (4, 0)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q5
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

The numbers 1, 2, ..., 10 are on a board. A move erases two numbers a and b and writes a + b + 1. After 9 moves one number is left. What is it?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F1-Q1
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A robot starts at (0, 0). Each move goes from (x, y) to (x + 1, y + 2) or to (x + 2, y + 1). It can reach (10, 10).

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F1-Q2
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Three numbers are on a board: 1, 2, 3. A move picks two of them, adds 1 to one and subtracts 1 from the other. Which quantity never changes?

A. The product of the three numbers

B. The number of odd numbers

C. The total of the three numbers

D. The largest number

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F1-Q3
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Ten coins all show heads. A move turns over any four coins (they need not be next to each other). Give a number of heads from 0 to 10 that can NEVER be showing.

Heads = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F1-Q4
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A counter shows 100. A move subtracts 7 or adds 3. What is the smallest number of moves that makes it show 0?

Moves: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F1-Q5
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

The numbers 1, 2, 3, 4 and 5 are on a board. A move erases two numbers a and b and writes a x b + a + b. After 4 moves one number is left. What is it?

Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q1
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A 4 by 4 grid of lights starts all off. A move switches every light in one whole row, or every light in one whole column. Some sequence of moves leaves exactly one light on.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q2
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A robot starts at (0, 0). Each move goes from (x, y) to (x + 3, y), to (x, y + 3), or to (x + 1, y + 1). Which point can it reach?

A. (2, 3)
B. (4, 0)
C. (1, 5)
D. (5, 2)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q3
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A 4 by 4 grid of lights starts all off. A move switches every light in one whole row, or every light in one whole column. Give a number of lit lights from 1 to 15 that some sequence of moves can leave.

Lit lights = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q4
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

The numbers 1, 2, 3, ..., 10 are on a board. A move erases two numbers a and b and writes a + b + 1. The last number left does not depend on which moves are made.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q5
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

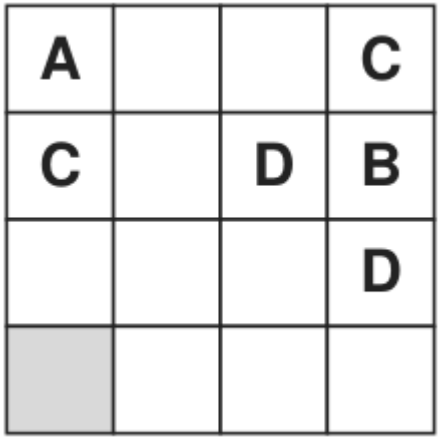
Set 45: Letter grids and word squares

Local outcome LOGIC.GRID.02: Fill and analyze letter grids under stated rules and a supplied word list

Puzzle set · genre: letter grids · MP1, MP7

MAIN: Fill and analyze letter grids under stated rules and a supplied word list
BACKWARD 1: Strategy: fill the most constrained cell first
FORWARD 1: ADVANCED APPLICATION: Word squares from a list
FORWARD 2: EXPERT CHALLENGE: Prove a grid is impossible or count it

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. The grid in the figure has six letters filled in. Which letter belongs in the shaded cell (bottom row, first column)?



Letter: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-M-Q1
Logic, Set 45: Letter grids and word squares · MAIN · Q1/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. The first row of a word square is TEN. Give the second and third rows.

Row 2 = _____

Row 3 = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-M-Q2
Logic, Set 45: Letter grids and word squares · MAIN · Q2/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

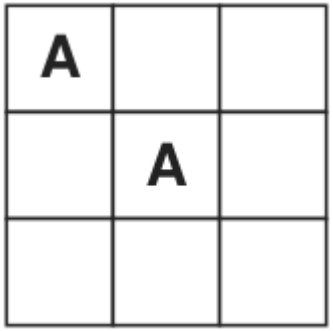
Rule: a 3 by 3 grid is filled with the letters A and B so that no two cells that share a side hold the same letter. The center cell holds A. Then all four corner cells hold the same letter.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-M-Q3
Logic, Set 45: Letter grids and word squares · MAIN · Q3/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. The top-left cell holds A and the center cell holds A. Which letter must go in the bottom-right cell?



A. A

B. B

C. It could be B or C

D. C

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-M-Q4
Logic, Set 45: Letter grids and word squares · MAIN · Q4/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. The first row is A B C. How many ways are there to complete the grid?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-M-Q5
Logic, Set 45: Letter grids and word squares · MAIN · Q5/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. A row reads A _ C. Which letter fills the gap?

Letter: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-B1-Q1
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. If column 1 holds A and B, its third cell must hold C.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-B1-Q2
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. A cell's row already holds A and D, and its column holds B. Which letter must go in the cell?

A. A
B. D
C. B
D. C

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-B1-Q3
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. A grid can have the row A B C directly above the row A C B.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-B1-Q4
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. Rows 1 and 2 are A B C and C A B. What is row 3? Type the three letters with no spaces.

Row 3: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-B1-Q5
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. The first row of a word square is ATE. Give the second and third rows.

Row 2 = _____

Row 3 = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q1
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. How many words on the list can be the first row of at least one word square?

Words: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q2
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

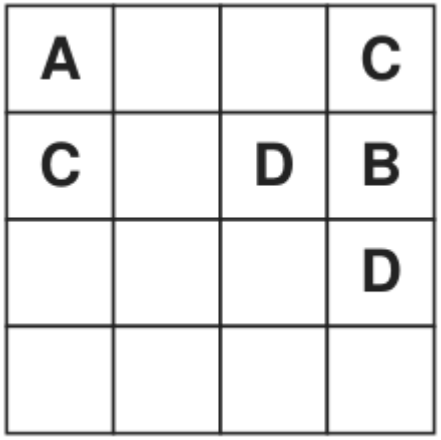
Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. A word square can start with the row ONE.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q3
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. The grid in the figure has six letters filled in. Type the complete top row (four letters, no spaces).



Row 1: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q4
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: a 2 by 3 grid is filled with A and B so that no two cells sharing a side hold the same letter. How many such grids are there?

A. 2

B. 4

C. 6

D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q5
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. How many ways are there to fill the whole grid if the first row is A B C D?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F2-Q1
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: a 3 by 3 grid is filled with A and B so that no two cells sharing a side hold the same letter. The top-left cell holds A and the top-right cell holds B.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F2-Q2
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. Which partly filled grid can NOT be completed?

A. Row 1 starts A, and row 2 starts B

B. Row 1 starts A B, and row 2 ends in C

C. The diagonal from top left to bottom right is A, A, A

D. Row 1 is A B C

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F2-Q3
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. Give the second and third rows of the word square whose first row is ROE.

Row 2 = _____

Row 3 = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F2-Q4
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. How many ways are there to fill an empty 3 by 3 grid?

Ways: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F2-Q5
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Set 46: Explanations and missing assumptions

Local outcome LOGIC.EVIDENCE.01: Identify the unstated assumption an argument needs and evaluate competing explanations

Formal logic set · genre: formal logic · MP3

MAIN: Identify the unstated assumption an argument needs and evaluate competing explanations
BACKWARD 1: Find the gap between premise and conclusion
FORWARD 1: ADVANCED APPLICATION: Weaken and strengthen explanations
FORWARD 2: EXPERT CHALLENGE: Judge explanations with care

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "The school's robot passed every test in the lab, so it will work at the competition." Which assumption does this argument need?

A. The robot failed some tests last year.

B. The robot was built by students.

C. Conditions at the competition are like those in the lab.

D. Other teams also tested their robots.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q1
Logic, Set 46: Explanations and missing assumptions · MAIN · Q1/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Mia's plant grew faster after she moved it to the window, so sunlight made it grow faster." Which assumption does this argument need?

A. Nothing else that affects growth changed when the plant was moved.

B. The window faces south.

C. Mia has other plants indoors.

D. Plants need water to grow.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q2
Logic, Set 46: Explanations and missing assumptions · MAIN · Q2/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Every member of the chess club is in grade 10. Jo is in grade 10. So Jo is in the chess club." This argument would become valid if we add the assumption "Everyone in grade 10 is in the chess club."

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q3
Logic, Set 46: Explanations and missing assumptions · MAIN · Q3/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

The cafeteria sold fewer sandwiches this week than last week. Which explanation is best supported if we also learn that the cafeteria was closed on two days this week?

A. Fewer students attend the school this year.

B. The cafeteria was open on fewer days to sell sandwiches.

C. Students stopped liking sandwiches.

D. The sandwich price went up.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q4
Logic, Set 46: Explanations and missing assumptions · MAIN · Q4/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "This bridge is safe because it has stood for 80 years." The argument needs the assumption that the bridge will be used the same way as it has been.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q5
Logic, Set 46: Explanations and missing assumptions · MAIN · Q5/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Sam is taller than Ana, so Sam is taller than Raj." Which assumption fills the gap?

A. Raj is taller than Ana.
B. Ana and Raj are friends.
C. Sam is the tallest in his class.
D. Ana is at least as tall as Raj.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q1
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. Every true statement about the topic of an argument is an assumption that the argument needs.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q2
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "The park was crowded on Saturday, so it will be crowded this Saturday." What is the gap?

A. From Saturday to Sunday
B. From the park to the city
C. From one past Saturday to a future Saturday
D. From crowded to empty

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q3
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. If an assumption an argument needs turns out to be false, the conclusion no longer follows from the premises.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q4
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "All squares are rectangles, so all rectangles are squares" needs no assumption to be valid.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q5
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

Claim: "Students who eat breakfast score higher on tests, so eating breakfast raises test scores." Which fact would most WEAKEN the claim?

A. Test scores rose last year.
B. Many students eat breakfast.
C. Breakfast foods are cheap.
D. Students who eat breakfast also sleep more, and sleep is linked to higher scores.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q1
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

Claim: "The new crosswalk signal reduced accidents at Elm Street." Which fact would most STRENGTHEN the claim?

A. The signal cost a lot of money.

B. Elm Street is near a school.

C. Accidents did not drop at similar streets that did not get the signal.

D. Drivers like the new signal.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q2
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "No reptiles have fur. This animal has fur. So this animal is not a reptile." This argument needs an unstated assumption to be valid.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q3
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Our team lost every game we played in the rain. It will rain on Friday. So we will lose Friday's game." Which assumption does the argument need?

A. It rains more in the spring.

B. The rain caused the losses, rather than something else about those games.

C. Friday's game is at home.

D. Our team has played in the rain before.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q4
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

Two explanations both fit the facts. Then no new fact can help decide between them.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q5
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

A town's ice-cream sales and its number of sunburns rise and fall together during the year. Which explanation is best?

A. Hot, sunny weather increases both.
B. Eating ice cream causes sunburn.
C. Sunburns make people buy ice cream.
D. The two numbers match by chance every year.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q1
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Ming read the book, so Ming knows how it ends." Which assumption is needed?

A. Ming liked the book.
B. Ming read the book all the way to the end.
C. The book has a surprise ending.
D. The book is long.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q2
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Every prize winner practiced daily. Kai practiced daily. So Kai won a prize." Adding "Only prize winners practiced daily" makes the argument valid.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q3
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

A survey of the chess club finds that 90% of members prefer quiet study rooms. A student concludes that 90% of the whole school prefers quiet study rooms. What assumption does this need?

A. The survey asked about study rooms.

B. The chess club has more than 10 members.

C. Chess club members have the same preferences as the whole school.

D. Quiet rooms are better for studying.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q4
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

If a needed assumption is shown to be false, the conclusion of the argument must be false.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q5
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

Set 47: Is there enough information?

Local outcome LOGIC.EVIDENCE.02: Decide whether the given information determines the answer, and what extra fact would

Formal logic set · genre: formal logic · MP1, MP3

MAIN: Decide whether the given information determines the answer, and what extra fact would

BACKWARD 1: Strategy: list what each clue leaves

FORWARD 1: ADVANCED APPLICATION: Which clue is enough?

FORWARD 2: EXPERT CHALLENGE: Design the missing clue

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Ana, Ben and Cal ran a race with no ties. Who finished first? Clue (1): Ana finished ahead of Ben. Clue (2): Cal finished last.

A. Neither clue alone is enough, but both together are.
B. Clue (2) alone is enough, but clue (1) alone is not.
C. Clue (1) alone is enough, but clue (2) alone is not.
D. Each clue alone is enough.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q1
Logic, Set 47: Is there enough information? · MAIN · Q1/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) x is a whole number from 1 to 20. What is x? Clue (1): x is a multiple of 7 and greater than 10. Clue (2): x is even.

A. Each clue alone is enough.

B. Neither clue alone is enough, but both together are.

C. Clue (2) alone is enough, but clue (1) alone is not.

D. Clue (1) alone is enough, but clue (2) alone is not.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q2
Logic, Set 47: Is there enough information? · MAIN · Q2/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Two whole numbers, each from 1 to 9, add up to 17. Knowing only this is enough to know which two numbers they are.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q3
Logic, Set 47: Is there enough information? · MAIN · Q3/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) A rectangle has whole-number side lengths. What is its area? Clue (1): its perimeter is 12. Clue (2): one side is twice as long as the other.

A. Clue (1) alone is enough, but clue (2) alone is not.

B. Neither clue alone is enough, but both together are.

C. Clue (2) alone is enough, but clue (1) alone is not.

D. Each clue alone is enough.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q4
Logic, Set 47: Is there enough information? · MAIN · Q4/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

x is a whole number from 1 to 50. You know that x is a multiple of 6 and also a multiple of 4. How many values of x are still possible?

Values: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q5
Logic, Set 47: Is there enough information? · MAIN · Q5/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

n is a whole number from 1 to 10, and n is a multiple of 4. How many values of n are possible?

Values: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-B1-Q1
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

n is a whole number from 1 to 10. Knowing that n is prime and even is enough to find n.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-B1-Q2
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Two whole numbers add to 10. Knowing only this is enough to find the larger number.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-B1-Q3
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

A clue is "enough" to answer a question when...

A. exactly one answer is left after using it.
B. at least one answer is left after using it.
C. it is true.
D. it rules out at least one answer.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-B1-Q4
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

n is a whole number from 1 to 20. Clue: n is a square number and n is odd. How many values of n are left?

Values: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-B1-Q5
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Four friends sit in a row of four seats numbered 1 to 4: Dee, Eli, Fay and Gus. Who sits in seat 1? Clue (1): Dee sits in seat 4. Clue (2): Eli sits directly left of Fay, and Gus sits directly left of Eli.

A. Clue (2) alone is enough, but clue (1) alone is not.

B. Clue (1) alone is enough, but clue (2) alone is not.

C. Neither clue alone is enough, but both together are.

D. Each clue alone is enough.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F1-Q1
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Pat is older than Quinn. How old is Quinn, in whole years? Clue (1): their ages add to 30. Clue (2): Pat is 4 years older than Quinn.

A. Each clue alone is enough.

B. Clue (2) alone is enough, but clue (1) alone is not.

C. Clue (1) alone is enough, but clue (2) alone is not.

D. Neither clue alone is enough, but both together are.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F1-Q2
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 30. What is n? Clue (1): n leaves remainder 1 when divided by 7. Clue (2): n is a perfect square.

A. Each clue alone is enough.

B. Clue (1) alone is enough, but clue (2) alone is not.

C. Neither clue alone is enough, but both together are.

D. Clue (2) alone is enough, but clue (1) alone is not.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F1-Q3
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

A bag holds only red and blue marbles, 12 in all. Knowing that there are 4 more red marbles than blue ones is enough to find the number of red marbles.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F1-Q4
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

x is a whole number from 1 to 20 and x is a multiple of 3. Give one extra clue as a single number: a value m from 2 to 10 such that knowing "x is also a multiple of m" leaves exactly one possible x.

m = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F1-Q5
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 20. What is n? Clue (1): n + 5 = 12. Clue (2): 2n = 14.

A. Clue (2) alone is enough, but clue (1) alone is not.

B. Clue (1) alone is enough, but clue (2) alone is not.

C. Neither clue alone is enough, but both together are.

D. Each clue alone is enough.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F2-Q1
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 20. What is n? Clue (1): n is odd. Clue (2): n is a multiple of 13.

A. Clue (2) alone is enough, but clue (1) alone is not.

B. Clue (1) alone is enough, but clue (2) alone is not.

C. Neither clue alone is enough, but both together are.

D. Each clue alone is enough.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F2-Q2
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Two whole numbers from 1 to 9 multiply to 12. Give a sum s such that also knowing "the two numbers add to s" tells you exactly which two numbers they are.

s = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F2-Q3
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Two whole numbers, each from 1 to 9, multiply to 36. Knowing only this is enough to know which two numbers they are.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F2-Q4
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Three different whole numbers from 1 to 9 add up to 23. How many different sets of three numbers are possible?

Sets: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F2-Q5
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Set 48: Digital displays

Local outcome LOGIC.SPATIAL.06: Reason about seven-segment digits: segments used, rotations and segment moves

Puzzle set · genre: seven-segment displays · MP7, MP6

MAIN: Reason about seven-segment digits: segments used, rotations and segment moves
BACKWARD 1: Read the segments of each digit
FORWARD 1: ADVANCED APPLICATION: Turned displays and segment counts
FORWARD 2: EXPERT CHALLENGE: Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many segments are lit when the display shows 2026?



Segments: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-M-Q1
Logic, Set 48: Digital displays · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Which digit uses exactly 3 lit segments?



A. 4
B. 7
C. 1
D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-M-Q2
Logic, Set 48: Digital displays · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. When the display showing 69 is turned upside down, it shows 69 again.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-M-Q3
Logic, Set 48: Digital displays · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Give a digit that looks exactly the same after the display is turned upside down.



Digit = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-M-Q4
Logic, Set 48: Digital displays · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Exactly one lit segment of the digit 2 is moved to an unlit position. Which digit can result?



A. 5
B. 7
C. 8
D. 3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-M-Q5
Logic, Set 48: Digital displays · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many segments does the digit 4 light?



Segments: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q1
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digit 8 lights all seven segments.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q2
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Which digit lights the fewest segments?



A. 4
B. 0
C. 1
D. 7

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q3
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digits 0 and 6 light the same number of segments.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q4
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digit 3 lights the middle segment and both left-hand segments.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q5
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display shows 1958. How many segments are lit?



Segments: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F1-Q1
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display shows 625. What does it show after being turned upside down?



A. 529

B. 526

C. 925

D. It does not show a number

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F1-Q2
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display showing 1001 still shows a number after it is turned upside down.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F1-Q3
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many two-digit numbers (10 to 99) still show a number after the display is turned upside down?



Numbers: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F1-Q4
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Give a digit d such that adding exactly one lit segment to d (moving nothing) makes the digit 8.



d = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F1-Q5
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Exactly one lit segment of the digit 3 is moved to an unlit position. Give a digit that can result.



Digit = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F2-Q1
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. You may light at most 12 segments in total. What is the largest whole number you can show? (Any number of digits; the first digit is not 0.)



Number: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F2-Q2
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Every digit can be turned into some other digit by moving exactly one lit segment.



A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F2-Q3
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. A clock shows hours 1 to 12 and minutes 00 to 59 (for example 9:58 uses the digits 9, 5, 8). Which time lights the most segments?



A. 9:58

B. 10:08

C. 12:59

D. 11:11

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F2-Q4
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many digits light more than 5 segments?



Digits: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F2-Q5
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Set 49: Common fallacies

Local outcome LOGIC.FALLACY.01: Recognize common informal fallacies in self-contained arguments

Formal logic set · genre: informal logic · MP3

MAIN: Recognize common informal fallacies in self-contained arguments
BACKWARD 1: Strategy: name what the argument does instead of giving reasons
FORWARD 1: ADVANCED APPLICATION: Straw men, slopes and popularity
FORWARD 2: EXPERT CHALLENGE: Is it really a fallacy?

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Hasty generalization: drawing a broad conclusion from too few cases. Circular reasoning: using the conclusion, in other words, as a reason for itself. Appeal to popularity: claiming something is true because many people believe it. Dana argues that the library should stay open later. Raj replies: "Dana is always late to class, so her opinion about library hours does not count." Which fallacy is this?

A. Circular reasoning
B. Hasty generalization
C. Appeal to popularity
D. Attacking the person

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q1
Logic, Set 49: Common fallacies · MAIN · Q1/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: False choice: presenting only two options when others are possible. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Straw man: attacking a distorted version of someone's argument instead of the real one. After, so because: concluding that A caused B only because B happened after A. "Either we cancel the field trip, or every student will get lost in the city." Which fallacy is this?

A. False choice
B. After, so because
C. Straw man
D. Slippery slope

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q2
Logic, Set 49: Common fallacies · MAIN · Q2/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. False choice: presenting only two options when others are possible. Circular reasoning: using the conclusion, in other words, as a reason for itself. "I met two rude people from that town, so everyone from that town is rude." Which fallacy is this?

A. Attacking the person
B. Circular reasoning
C. Hasty generalization
D. False choice

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q3
Logic, Set 49: Common fallacies · MAIN · Q3/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Circular reasoning: using the conclusion, in other words, as a reason for itself. "This app is the best because it is better than all the others" is an example of circular reasoning.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q4
Logic, Set 49: Common fallacies · MAIN · Q4/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: After, so because: concluding that A caused B only because B happened after A. Hasty generalization: drawing a broad conclusion from too few cases. Appeal to popularity: claiming something is true because many people believe it. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. "After I started wearing my lucky socks, our team won three games in a row. The socks made us win." Which fallacy is this?

A. Hasty generalization
B. After, so because
C. Slippery slope
D. Appeal to popularity

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q5
Logic, Set 49: Common fallacies · MAIN · Q5/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Appeal to popularity: claiming something is true because many people believe it. Straw man: attacking a distorted version of someone's argument instead of the real one. False choice: presenting only two options when others are possible. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Which definition describes "Millions of people watch this show, so it must be educational"?

A. Straw man
B. Appeal to popularity
C. Slippery slope
D. False choice

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q1
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. Concluding that all dogs are friendly after meeting three friendly dogs is a hasty generalization.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q2
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Saying "The bridge design is unsafe because the beams are too thin for the load" is attacking the person.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q3
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

An argument's reason is about who made a claim rather than about the claim. Which question best helps you notice this?

A. Is the reason about the claim, or about something else?
B. Do I agree with the conclusion?
C. Is the conclusion surprising?
D. Is the argument long?

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q4
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. "If we allow phones at lunch, then students will use them in class, then stop learning, then fail school" is a slippery slope.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q5
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Straw man: attacking a distorted version of someone's argument instead of the real one. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. False choice: presenting only two options when others are possible. Hasty generalization: drawing a broad conclusion from too few cases. Ava says the school should add one vegetarian meal option. Ben replies: "Ava wants to ban meat from the cafeteria completely, and that is unfair to everyone." Which fallacy is this?

A. Hasty generalization
B. Attacking the person
C. Straw man
D. False choice

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q1
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. False choice: presenting only two options when others are possible. After, so because: concluding that A caused B only because B happened after A. Circular reasoning: using the conclusion, in other words, as a reason for itself. "If we let one student retake the quiz, soon everyone will retake every quiz, and grades will mean nothing." Which fallacy is this?

A. Circular reasoning
B. After, so because
C. False choice
D. Slippery slope

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q2
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Appeal to popularity: claiming something is true because many people believe it. "Most dentists in a large survey recommend brushing twice a day, so brushing twice a day is likely a good idea" is an appeal to popularity.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q3
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Circular reasoning: using the conclusion, in other words, as a reason for itself. Appeal to popularity: claiming something is true because many people believe it. Straw man: attacking a distorted version of someone's argument instead of the real one. Hasty generalization: drawing a broad conclusion from too few cases. "The rule must be fair because it is the rule, and rules are fair." Which fallacy is this?

A. Hasty generalization
B. Appeal to popularity
C. Circular reasoning
D. Straw man

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q4
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: After, so because: concluding that A caused B only because B happened after A. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. False choice: presenting only two options when others are possible. "The new principal started in September, and in October the cafeteria food got worse. The principal ruined the food." Which fallacy is this?

A. Attacking the person
B. After, so because
C. False choice
D. Slippery slope

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q5
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Which reply is NOT attacking the person?

A. "Your numbers are from 2010, and the prices have changed since then."

B. "Nobody listens to you anyway."

C. "You are too young to understand prices."

D. "You only say that because you work for the store."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q1
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: False choice: presenting only two options when others are possible. "A whole number is either even or odd" is a false choice.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q2
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. After, so because: concluding that A caused B only because B happened after A. Straw man: attacking a distorted version of someone's argument instead of the real one. Appeal to popularity: claiming something is true because many people believe it. Which argument avoids all four fallacies defined above?

A. My two cousins liked the plan, so most voters like it.

B. Everyone online says the plan is good, so it is good.

C. Our team won after the coach changed shoes, so the shoes helped.

D. A random sample of 500 voters favored the plan, so most voters probably favor it.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q3
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Straw man: attacking a distorted version of someone's argument instead of the real one. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Appeal to popularity: claiming something is true because many people believe it. Circular reasoning: using the conclusion, in other words, as a reason for itself. Mo says homework should be shorter on weekends. Lee replies: "So you think students should never study at all? That would be terrible." Which fallacy is this?

A. Slippery slope
B. Straw man
C. Appeal to popularity
D. Circular reasoning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q4
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. An argument that predicts a chain of results is always a slippery slope, even when each step is supported by evidence.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q5
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Set 50: Weighing evidence

Local outcome LOGIC.EVIDENCE.03: Judge whether evidence supports a claim: samples, comparisons and correlation

Formal logic set · genre: informal logic · MP3, MP2

MAIN: Judge whether evidence supports a claim: samples, comparisons and correlation
BACKWARD 1: Compare rates, not counts
FORWARD 1: ADVANCED APPLICATION: Samples and comparisons
FORWARD 2: EXPERT CHALLENGE: Comparisons that reverse

To learn how often people in a town exercise, a student surveys 100 people leaving a gym. What is the main problem with this evidence?

A. People leaving a gym probably exercise more than the town as a whole.

B. The survey should have asked about diet too.

C. 100 people is always too few for a survey.

D. Surveys can never give useful evidence.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-M-Q1
Logic, Set 50: Weighing evidence · MAIN · Q1/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

School A: 45 of its 60 students passed a test. School B: 52 of its 80 students passed. What percent of School A's students passed?

Percent: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-M-Q2
Logic, Set 50: Weighing evidence · MAIN · Q2/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

In many towns, more ice-cream sales come with more swimming accidents. This shows that eating ice cream causes swimming accidents.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-M-Q3
Logic, Set 50: Weighing evidence · MAIN · Q3/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Which study design gives the best evidence that a tutoring program raises test scores?

A. Measure the scores of tutored students only.
B. Compare tutored students with students who chose not to sign up.
C. Randomly assign students to get tutoring or not, then compare the two groups' scores.
D. Ask tutored students whether they feel they improved.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-M-Q4
Logic, Set 50: Weighing evidence · MAIN · Q4/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A claim says that more than half of a school's students prefer a later start time. In a random sample of 40 students, n said they prefer it. Give a value of n that would support the claim.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-M-Q5
Logic, Set 50: Weighing evidence · MAIN · Q5/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

18 of 24 students in a class walk to school. What percent is that?

Percent: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q1
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Club X has 30 members and 12 play chess. Club Y has 10 members and 6 play chess. A larger share of Club Y plays chess.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q2
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Town P had 50 car accidents and Town Q had 30. What else do you need before saying Town P's roads are more dangerous?

A. The colors of the cars
B. Nothing; 50 is more than 30
C. The names of the drivers
D. How many people or cars each town has

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q3
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A survey of 200 people finds that 30 own a bike. What percent own a bike?

Percent: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q4
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

If Group A has more members with a trait than Group B, then a larger percent of Group A has the trait.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q5
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A website asks visitors to vote on whether the town needs a new park. Of 300 votes, 80% say yes. Why is this weak evidence about the whole town?

A. 80% is too high to be believed.

B. 300 votes is more than enough.

C. Websites cannot count votes.

D. People who choose to vote online may care more about parks than others.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F1-Q1
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Class A: 21 of 28 students finished the project on time. Class B: 24 of 30. What is the higher of the two on-time percents?

Percent: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F1-Q2
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A study finds that students who own more books tend to have higher reading scores. This alone proves that buying books raises reading scores.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F1-Q3
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

To test whether a new fertilizer helps tomato plants grow, which design is best?

A. Give fertilizer to a randomly chosen half of the plants and compare them with the other half.

B. Give fertilizer to all plants and measure them.

C. Ask a gardener whether the plants look bigger.

D. Use the fertilizer only on the healthiest plants.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F1-Q4
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A coin is flipped 50 times to test whether it is fair. Give a number of heads h from 0 to 50 that would be strong evidence the coin favors heads. Use this test: h is at least 35.

h = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F1-Q5
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Two clinics treat a condition. Clinic A treats 100 mild cases (90 recover) and 20 severe cases (8 recover). Clinic B treats 20 mild cases (19 recover) and 100 severe cases (50 recover). What percent of all Clinic A patients recover, to the nearest whole percent?

Percent: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F2-Q1
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

In the clinics above, Clinic A has the higher overall recovery rate (about 82% to 58%). Which statement is true?

A. Clinic A is better for both mild and severe cases.

B. Clinic B has the higher recovery rate for mild cases AND for severe cases.

C. Clinic A must be the better clinic.

D. The two clinics are equally good for mild cases.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F2-Q2
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

In the clinics above, comparing only the overall recovery rates gives a fair comparison of the clinics.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F2-Q3
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A test for a rare condition is positive for 95 of every 100 people who have it and for 5 of every 100 people who do not. In a town of 10,000 people, 100 have the condition. About how many people in the town test positive in all?

People: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F2-Q4
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

In the town above, 590 people test positive and 95 of them have the condition. What fraction of the positive tests are correct?

A. About 5%
B. About 50%
C. About 16%
D. About 95%

REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F2-Q5
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Set 51: Best possible: constrained optimization

Local outcome LOGIC.OPTIMIZE.01: Find the best value under stated constraints and show nothing better is possible

Puzzle set · genre: optimization · MP1, MP7

MAIN: Find the best value under stated constraints and show nothing better is possible
BACKWARD 1: Strategy: find a good answer, then prove nothing beats it
FORWARD 1: ADVANCED APPLICATION: When the obvious choice is not best
FORWARD 2: EXPERT CHALLENGE: Prove the best value

Two whole numbers add up to 20. What is the largest possible product?

Product: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-M-Q1
Logic, Set 51: Best possible: constrained optimization · MAIN · Q1/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

A rectangular pen uses exactly 24 meters of fence, and its sides are whole numbers of meters. What is the largest possible area, in square meters?

Area: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-M-Q2
Logic, Set 51: Best possible: constrained optimization · MAIN · Q2/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

A 12-hour digital clock shows times from 1:00 to 12:59. Which time has the largest product of its digits?

A. 8:59
B. 9:59
C. 11:59
D. 12:59

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-M-Q3
Logic, Set 51: Best possible: constrained optimization · MAIN · Q3/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Using coins worth 1, 5, 10 and 25 cents, the fewest coins that make exactly 30 cents is 2.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-M-Q4
Logic, Set 51: Best possible: constrained optimization · MAIN · Q4/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Whole numbers x and y are each 0 or more, with x + 2y at most 10 and x at most 6. Make x + 3y as large as possible. Give x and y.

x = _____

y = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-M-Q5
Logic, Set 51: Best possible: constrained optimization · MAIN · Q5/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Two whole numbers add to 14. What is the largest possible product?

Product: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-B1-Q1
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

To show that a value is the largest possible, it is enough to find one example that reaches it.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-B1-Q2
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Which numbers from the list 4, 9, 2, 7, 5 give the largest sum if you may choose any two?

A. 4 and 9
B. 9 and 5
C. 9 and 7
D. 7 and 5

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-B1-Q3
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Using coins worth 1, 5 and 10 cents, what is the fewest coins that make 27 cents?

Coins: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-B1-Q4
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

A rectangle with whole-number sides and perimeter 20 can have area 25.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-B1-Q5
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Coins are worth 1, 4 and 6 cents. What is the fewest coins that make exactly 8 cents?

Coins: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F1-Q1
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Whole numbers add up to 10 (you choose how many numbers). What is the largest possible product?

Product: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F1-Q2
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Coins are worth 1, 4 and 6 cents. Always taking the largest coin that fits gives the fewest coins for every amount.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F1-Q3
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

A backpack can hold at most 10 kilograms. Items: tent 6 kg, value 8; stove 4 kg, value 5; food 5 kg, value 6; lamp 1 kg, value 3. Which choice has the greatest total value?

A. Food, stove, lamp

B. Food, lamp

C. Tent, stove

D. Tent, lamp

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F1-Q4
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Two whole numbers are each at least 1 and their product is 36. Make their sum as small as possible. Give the two numbers.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F1-Q5
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Whole numbers add up to 20 (you choose how many numbers). What is the largest possible product?

Product: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F2-Q1
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Coins are worth 1, 7 and 10 cents. What is the fewest coins that make exactly 14 cents?

Coins: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F2-Q2
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Among rectangles with whole-number sides and area 36, the 6 by 6 square has the smallest perimeter.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F2-Q3
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Whole numbers a, b and c are each at least 1 and add up to 12. Make the product a x b x c as large as possible. Give a, b and c.

a = _____

b = _____

c = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F2-Q4
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Use each of the digits 1, 2, 3 and 4 once to make two 2-digit numbers. Which pair has the largest product?

A. 31 x 24
B. 43 x 21
C. 42 x 31
D. 41 x 32

REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F2-Q5
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Set 52: Patterns from a stated rule family

Local outcome LOGIC.PATTERN.01: Identify the rule in a stated family that fits a sequence, and continue it

Puzzle set · genre: sequences · MP8, MP7

MAIN: Identify the rule in a stated family that fits a sequence, and continue it
BACKWARD 1: Strategy: write the differences
FORWARD 1: ADVANCED APPLICATION: Rules for letters and far terms
FORWARD 2: EXPERT CHALLENGE: Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 3, 7, 11, 15, ...?

Next term: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-M-Q1
Logic, Set 52: Patterns from a stated rule family · MAIN · Q1/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Which rule fits 2, 6, 18, 54, ...?

A. Multiply by 3 each time

B. Add 4, then 12, then 36

C. Add 4 each time

D. Multiply by 2 each time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-M-Q2
Logic, Set 52: Patterns from a stated rule family · MAIN · Q2/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 1, 2, 4, 7, 11, ...?

Next term: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-M-Q3
Logic, Set 52: Patterns from a stated rule family · MAIN · Q3/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 5, 10, 20, 35, ... follows rule (3).

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-M-Q4
Logic, Set 52: Patterns from a stated rule family · MAIN · Q4/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. The letters C, F, I, L follow rule (1) by their numbers. Give the next letter.

Next letter = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-M-Q5
Logic, Set 52: Patterns from a stated rule family · MAIN · Q5/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Write the differences of 2, 5, 9, 14. What is the last difference?

Difference: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-B1-Q1
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 10, 7, 4, 1, ... follows rule (1).

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-B1-Q2
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Which rule fits 1, 4, 16, 64, ...?

A. None of the rules
B. Rule (2)
C. Rule (3)
D. Rule (1)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-B1-Q3
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 50, 45, 40, 35, ...?

Next term: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-B1-Q4
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 3, 6, 12, 24, ... follows rule (1).

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-B1-Q5
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. The letters B, D, G, K follow rule (3) by their numbers. Give the next letter.

Next letter = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F1-Q1
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 10th term of 4, 7, 10, 13, ...?

10th term: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F1-Q2
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Knowing only the first two terms, 3 and 6, is enough to know the third term.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F1-Q3
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The first three terms are 1, 2, 4. Which statement is true?

A. Only rule (3) fits, so the next term is 7.

B. Rule (1) fits, so the next term is 6.

C. Rule (2) and rule (3) both fit, giving 8 or 7 next.

D. Only rule (2) fits, so the next term is 8.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F1-Q4
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 8th term of 2, 6, 18, 54, ...?

8th term: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F1-Q5
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence is 5, 9, 13, 17, ... Give the 25th term.

25th term = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F2-Q1
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). In the sequence 3, 6, 12, 24, ..., which term is the first one greater than 1000? Give its position (1 for the first term).

Position: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F2-Q2
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 10th term of 1, 3, 6, 10, ...?

10th term: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F2-Q3
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). A sequence that follows rule (1) with a positive amount added will eventually pass every number.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F2-Q4
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. Which letter sequence follows rule (2) by the letters' numbers?

A. A, B, D, G
B. A, C, E, G
C. B, C, E, H
D. A, B, D, H

REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F2-Q5
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Set 53: Rearranging letters

Local outcome LOGIC.WORD.03: Find anagrams and letter-bank words within a supplied word list and letter rules

Puzzle set · genre: anagrams · MP7, MP1

MAIN: Find anagrams and letter-bank words within a supplied word list and letter rules
BACKWARD 1: Strategy: sort the letters
FORWARD 1: ADVANCED APPLICATION: Letter banks and joined parts
FORWARD 2: EXPERT CHALLENGE: Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. Which word on the list is an anagram of LEMON?

A. SOLEMN
B. LANE
C. TONE
D. MELON

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q1
Logic, Set 53: Rearranging letters · MAIN · Q1/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. Give a word on the list that is an anagram of STARE.

Word = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q2
Logic, Set 53: Rearranging letters · MAIN · Q2/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. LISTEN, SILENT, ENLIST and TINSEL are all anagrams of one another.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q3
Logic, Set 53: Rearranging letters · MAIN · Q3/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. How many words on the list can be spelled using the letters of PLANET, each letter at most once?

Words: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q4
Logic, Set 53: Rearranging letters · MAIN · Q4/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Read the letters STOPAN from right to left. Which word on the list appears as a block of neighboring letters in that backward reading?

A. SPOT
B. POTS
C. PAN
D. STOP

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q5
Logic, Set 53: Rearranging letters · MAIN · Q5/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Write the letters of NOTE in alphabetical order (no spaces).

Letters: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-B1-Q1
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

An anagram uses exactly the same letters, each as many times, in a different order. STONE and NOTES are anagrams.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-B1-Q2
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

An anagram uses exactly the same letters, each as many times, in a different order. TASTE and STATE are anagrams.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-B1-Q3
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

An anagram uses exactly the same letters, each as many times, in a different order. Which word is NOT an anagram of SPOT?

A. POTS
B. STOP
C. SPOTS
D. TOPS

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-B1-Q4
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

An anagram uses exactly the same letters, each as many times, in a different order. BREAD and BEARD are anagrams.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-B1-Q5
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Join one part from the first group to one part from the second group to make a word on the list. First group: PL, ST, HE. Second group: ART, ANET, ONE. Give the word.

Word = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q1
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. How many words on the list are anagrams of EARTH (not counting EARTH itself)?

Words: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q2
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. You may use the letters S, T, O, P as often as you like. Which is the longest word on the list you can spell?

A. All of SPOT, STOP, POTS, TOPS and POST tie at 4 letters

B. SPOTS

C. STONE

D. STOP is the only longest word

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q3
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. NIGHT and THING are anagrams.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q4
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. Group the list words into anagram families (words with the same sorted letters). How many families have exactly 4 or more words?

Families: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q5
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. Spaces do not count. Which pair of list words, written together, is an anagram of the phrase PAST TEEN?

A. STAR PEN
B. NEAT POTS
C. TASTE PEN
D. PETAL NET

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q1
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Every word on the list that is also a list word when read backward has 3 or 4 letters.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q2
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Give a word on the list that is also a word on the list when read backward.

Word = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q3
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. How many words on the list can be spelled using the letters of STONE, each letter at most once?

Words: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q4
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. How many words on the list use exactly the letters of NOTES, counting NOTES itself?

Words: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q5
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Set 54: Weighings and the least information needed

Local outcome LOGIC.SEARCH.02: Plan weighings or questions and find the least number that always identifies the answer

Puzzle set · genre: balance puzzles · MP1, MP3

MAIN: Plan weighings or questions and find the least number that always identifies the answer
BACKWARD 1: Strategy: each weighing has three outcomes
FORWARD 1: ADVANCED APPLICATION: Bigger groups and comparisons
FORWARD 2: EXPERT CHALLENGE: Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 9 coins and one is heavier. What is the least number of weighings that always finds the heavy coin?

Weighings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q1
Logic, Set 54: Weighings and the least information needed · MAIN · Q1/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 9 coins and one is heavier. Which first weighing lets you always finish in the fewest total weighings?

A. 3 coins against 3 coins
B. 1 coin against 1 coin
C. 2 coins against 2 coins
D. 4 coins against 4 coins

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q2
Logic, Set 54: Weighings and the least information needed · MAIN · Q2/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 3 weighings you can always find the heavier coin among 27 coins.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q3
Logic, Set 54: Weighings and the least information needed · MAIN · Q3/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. What is the largest number of coins n for which 2 weighings always find the one heavier coin?

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q4
Logic, Set 54: Weighings and the least information needed · MAIN · Q4/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 10 coins and one is heavier. What is the least number of weighings that always finds the heavy coin?

Weighings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q5
Logic, Set 54: Weighings and the least information needed · MAIN · Q5/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

How many different outcomes can one weighing on a balance scale have?

Outcomes: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-B1-Q1
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 3 coins and one heavier, one weighing always finds the heavy coin.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-B1-Q2
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 5 coins and one is heavier. You weigh 2 against 2 and they balance. What do you know?

A. Nothing yet.
B. The fifth coin is the heavy one.
C. You must weigh again to know.
D. One of the four weighed coins is heavy.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-B1-Q3
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

How many outcome patterns do 2 weighings have in all?

Patterns: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-B1-Q4
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 4 coins and one heavier, one weighing always finds the heavy coin.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-B1-Q5
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. You may use 4 weighings. What is the largest number of coins among which you can always find the one heavier coin?

Coins: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F1-Q1
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 6 coins and one is heavier. You weigh 2 against 2 and the left side goes down. What should the next weighing be?

A. The two coins left off against each other

B. All six coins again, 3 against 3

C. One left-side coin against a right-side coin

D. One left-side coin against the other left-side coin

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F1-Q2
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

You have 3 weights of different sizes and compare them two at a time on a balance. Finding the heaviest always takes at least 2 weighings.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F1-Q3
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

You have 8 weights of different sizes and compare them two at a time on a balance. What is the least number of weighings that always finds the heaviest?

Weighings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F1-Q4
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Give a number of coins n from 20 to 30 for which 3 weighings always find the one heavier coin.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F1-Q5
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 7 coins and one is LIGHTER (not heavier). What is the least number of weighings that always finds it?

Weighings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F2-Q1
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Why can 2 weighings NOT always find the heavy coin among 10?

A. Two weighings give only 9 outcome patterns, but there are 10 possible heavy coins.

B. You cannot put 5 coins on each side.

C. 10 is an even number.

D. The first weighing must use all 10 coins.

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F2-Q2
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 4 coins and one heavier, 2 weighings always find the heavy coin.

A. True

B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F2-Q3
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Give a number of coins n from 20 to 90 for which 3 weighings are NOT enough but 4 weighings are enough to always find the one heavier coin.

n = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F2-Q4
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

You have 4 weights of different sizes and compare them two at a time on a balance. What is the least number of weighings that always finds both the heaviest and the lightest?

Weighings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F2-Q5
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3